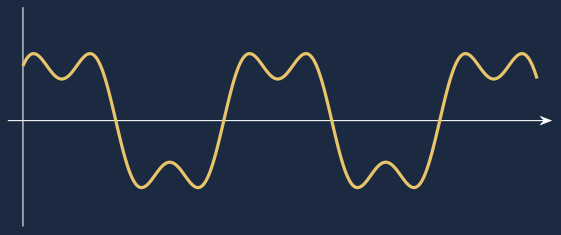
BUET • CSE 219 • EEE 211

Signals and Linear Systems

Topic-wise questions with complete worked solutions



16	169	6	3
topic sections	exam questions	exam sessions	course codes

CSE 219, EEE 211/301 and CSE 311 term finals, sessions 2018–2019 to 2023–2024. Anik Sir's part (signals, systems, convolution, Fourier series and transform) comes first, followed by Ashraf Sir's part (sampling, DFT/FFT, Laplace and z-transform).

Every question on the content of the course is organised topic-wise (most recent session first) and followed by a step-by-step worked solution. Diagrams are redrawn in vector graphics.

Solutions checked numerically in Python

Contents

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How to read this book

2099–2100 • Q0(a) Course • TF page MARKS 10

A question card. The coral badge on the right shows the printed marks. Figures from the original paper are redrawn in vector graphics inside the card.

SOLUTION

The solution panel always follows its question: derivations, intermediate tables and short explanations. Key results are highlighted:

Answer. like this.

REPEATED ×2 **Same question in 2 papers — stacked below, solved once.**

Paper A Q1 MARKS 12

A question asked in a recent session. . .

Paper B Q4 MARKS 15

. . . and asked again in an earlier session.

SOLUTION

One shared solution covers every stacked card.

NOTE Gold boxes flag a misprint, an illegible part of the scanned paper, or an assumption made in the solution.

Conventions

- **Scope:** 2018–19 to 2023–24. Course codes: **CSE 219** (current), **EEE 211/301** (the earlier course on the same content; "EEE p.N" is the page in the EEE paper collection), **CSE 311** (Data Communication; only the signals/Fourier/sampling parts are included). CSE 219 2023–24 pages are in the "22 batch" file.
- **Ordering:** within a topic, the most recent session first.
- **Notation:** $\mathcal{F}$, $\mathcal{L}$, $\mathcal{Z}$ are the Fourier, Laplace and z transforms; $r(t) = t\,u(t)$; $\operatorname{sinc} x = \sin(\pi x)/(\pi x)$ unless a solution says otherwise; $W_N = e^{-j2\pi/N}$.
- **Syllabus:** Anik Sir excludes Parseval's theorem and slides marked $(*)$; Parseval questions are still solved here because they were asked, and are marked in a note.
- **Figures:** some scanned figures are partly illegible; the reading used is stated in a note before the solution.
- **Last chapter:** state-space and mechanical-analogue questions are not on either teacher's slides; they are listed but not solved.

01

Signal Energy, Power and Periodicity

ANIK SIR • LECTURES 1–2

2023–2024 • Q3(a) CSE 219 • 22 batch p.2 MARKS 15

Suppose $x[n] = (3-4j)\, e^{j\frac{13\pi}{7}n}$ is a complex exponential discrete-time signal.

(i) Show whether the real and imaginary parts of $x[n]$ are periodic or not. If periodic, determine the fundamental period.

(ii) Justify carefully whether the phase shift in each part can be interpreted as an integer time shift of a cosine sequence.

SOLUTION

Write $3-4j = 5e^{-j\theta}$ with $\theta = \tan^{-1}\frac{4}{3} = 0.9273$ rad. Then

$$x[n] = 5\, e^{j(\frac{13\pi}{7}n-\theta)}, \qquad \operatorname{Re} x[n] = 5\cos\Big(\tfrac{13\pi}{7}n-\theta\Big), \quad \operatorname{Im} x[n] = 5\sin\Big(\tfrac{13\pi}{7}n-\theta\Big) = 5\cos\Big(\tfrac{13\pi}{7}n-\theta-\tfrac{\pi}{2}\Big).$$

(i) Periodicity. A sequence $\cos(\omega_0 n+\varphi)$ is periodic iff $\omega_0 N = 2\pi m$ for integers N, m, i.e. $2\pi/\omega_0 = N/m$ is rational; the phase plays no role. Here $\omega_0 = 13\pi/7$ and

$$\frac{2\pi}{\omega_0} = \frac{14}{13}\ \text{(rational)} \implies N_0 = 14 \quad (m=13).$$

Both the real and the imaginary part are periodic with fundamental period $N_0 = 14$; after 14 samples the angle has advanced by $13\cdot 2\pi$.

(ii) Phase as an integer shift. Shifting $\cos(\omega_0 n)$ right by an integer n_0 gives $\cos(\omega_0 n-\omega_0 n_0)$, so the phase obtainable by an integer shift is $\omega_0 n_0 = \frac{13\pi}{7}n_0$, which (mod 2π) is always an integer multiple of $\pi/7$. The real part has phase $\theta = 0.9273$ rad $= 2.066\,\frac{\pi}{7}$, not an integer multiple of $\pi/7$ ($\cos\theta = 3/5$, and θ/π is irrational). The imaginary part has the further offset $\pi/2$, which is $3.5\,\frac{\pi}{7}$ — again not a multiple. Hence *neither* phase can be produced by an integer time shift. They are shifts of the underlying continuous-time cosine by a fraction of a sample, $t_0 = \theta/\omega_0 \approx 0.1264$ samples.

Answer. (i) Both parts are periodic, $N_0 = 14$. (ii) No: $\theta = \tan^{-1}\frac{4}{3}$ is not an integer multiple of $\pi/7$, so the phase is not an integer shift (it is a non-integer shift of ≈ 0.126 sample).

2022–2023 • Q5(c) CSE 219 • TF p.3 MARKS 5

Is the signal $x[n] = 2\cos\big(\frac{16\pi n}{63}+\frac{\pi}{3}\big)$ a periodic signal? Explain your answer.

SOLUTION

$\omega_0 = \frac{16\pi}{63}$, so $\frac{2\pi}{\omega_0} = \frac{63}{8} = \frac{N}{m}$ is rational with $N = 63$, $m = 8$ in lowest terms. The sequence repeats after 63 samples (the angle advances by $16\pi = 8 \cdot 2\pi$); the phase $\pi/3$ does not affect the period. A smaller N would require $\frac{16\pi}{63} N \in 2\pi\mathbb{Z}$, i.e. $8N/63 \in \mathbb{Z}$, so $63 \mid N$.

Answer. Yes, periodic with fundamental period $N_0 = 63$.

2021–2022 • Q1(a) EEE 211 • EEE p.12 MARKS 10

Is the following signal periodic? If so, what is the period?

$$x(t) = \sqrt{e^{j5\pi t}} + \sin^2\left(3t - \frac{\pi}{4}\right).$$

SOLUTION

$$\sqrt{e^{j5\pi t}} = e^{j5\pi t/2} \ \text{(principal root)}, \qquad \sin^2\left(3t - \tfrac{\pi}{4}\right) = \tfrac{1}{2} - \tfrac{1}{2}\cos\left(6t - \tfrac{\pi}{2}\right).$$

Periods: $T_1 = \dfrac{2\pi}{5\pi/2} = \dfrac{4}{5}$; the constant has no period; $T_2 = \dfrac{2\pi}{6} = \dfrac{\pi}{3}$. The sum is periodic only if T_1/T_2 is rational:

$$\frac{T_1}{T_2} = \frac{4/5}{\pi/3} = \frac{12}{5\pi} \ \text{(irrational)}.$$

Answer. Not periodic.

2021–2022 • Q1(b) EEE 211 • EEE p.12 MARKS 10

Determine whether the signal $e^{-3t}\sin\left(\frac{2\pi}{3}t - 4\right)u(t)$ is a power or an energy signal. Justify your answer.

SOLUTION

Let $\omega = 2\pi/3$. Since $\sin^2 \le 1$,

$$E = \int_0^\infty e^{-6t}\sin^2(\omega t - 4)\,\mathrm{d}t \le \int_0^\infty e^{-6t}\,\mathrm{d}t = \frac{1}{6} < \infty,$$

so the signal has finite energy and therefore zero average power. The exact value follows from $\sin^2\theta = \frac{1}{2}(1 - \cos 2\theta)$:

$$E = \frac{1}{12} - \frac{1}{2}\,\mathrm{Re}\left\{\frac{e^{-j8}}{6 - j2\omega}\right\} = \frac{1}{12} - \frac{6\cos 8 + 2\omega\sin 8}{2(36 + 4\omega^2)} = 0.08333 - 0.03055 = 0.0528.$$

Answer. Energy signal ($E \approx 0.0528$, $P = 0$).

2020–2021 • Q5(b) EEE 211 • EEE p.20 MARKS 10

Analytically determine whether the following signals are power or energy signals: (i) $u(t-6) - u(-t+8)$, and (ii) $2r(t-4)u(t)$.

SOLUTION

(i) $u(t-6)=1$ for $t>6$ and $u(8-t)=1$ for $t<8$. Hence

$$x(t)=u(t-6)-u(8-t)=\begin{cases}-1, & t<6,\\ 0, & 6<t<8,\\ +1, & t>8.\end{cases}$$

The energy is infinite, but the average power is finite:

$$P_\infty=\lim_{T\to\infty}\frac{1}{2T}\Big[\int_{-T}^{6}1\,\mathrm{d}t+\int_{8}^{T}1\,\mathrm{d}t\Big]=\lim\frac{(T+6)+(T-8)}{2T}=1.$$

(ii) $2r(t-4)u(t)=2(t-4)$ for $t\geq 4$ and 0 otherwise: a ramp that grows without bound. $E=\int_4^\infty 4(t-4)^2\,\mathrm{d}t=\infty$ and $P=\lim\frac{1}{2T}\int_4^T 4(t-4)^2\,\mathrm{d}t=\lim\frac{4(T-4)^3}{6T}=\infty$.

Answer. (i) Power signal, $P=1$. (ii) Neither energy nor power signal.

2018–2019 • Q6(a) EEE 211 • EEE p.30 MARKS 15

Determine whether the following signals are power signals, energy signals or neither. Justify the answer. (i) $x_1(t)=\exp\big[j(2\pi t+\pi/4)\big]+\exp[-2|t|]$; (ii) $x_2(t)=\big[u(\cos t)-\delta(t-k\pi)\big]^{0.5}$, where $k=\pm1,\pm3,\pm5,\ldots,\infty$.

NOTE The impulses of x_2 sit at the odd multiples of π, where $\cos t=-1$ and $u(\cos t)=0$, so the bracket is negative there. The solution takes $|x_2|^2$ as $u(\cos t)$ plus the magnitudes of the impulse terms; the printed signal is not a classical function.

SOLUTION

(i) $|x_1|^2=1+e^{-4|t|}+2e^{-2|t|}\cos(2\pi t+\pi/4)$. The last two terms are absolutely integrable, so they add nothing to $\frac{1}{2T}\int_{-T}^{T}$ as $T\to\infty$:

$$P=1,\qquad E=\infty.$$

(ii) $u(\cos t)$ is a $0/1$ square wave with period 2π (it equals 1 for $|t|<\pi/2$ within a period). The impulses repeat every 2π as well, so $|x_2|^2$ is periodic with period 2π and $E=\infty$. Over one period $(-\pi,\pi]$ the energy is π (from the square wave) plus 1 (the unit impulse at $t=\pi$ in magnitude), so

$$P=\frac{1}{2\pi}(\pi+1)=\frac{1}{2}+\frac{1}{2\pi}\approx 0.659.$$

Answer. (i) Power signal, $P=1$. (ii) Periodic with infinite energy and finite average power: a power signal (value $\frac{1}{2}+\frac{1}{2\pi}$ under the stated reading).

02

Signal Transformations, Even–Odd Parts and Elementary Signals

ANIK SIR • LECTURES 1–2

2023–2024 • Q1(a) CSE 219 • 22 batch p.1 MARKS 15

Let $x(t) = \begin{cases} 1-|t|, & |t| \le 1, \\ 0, & \text{otherwise.} \end{cases}$ Sketch the signal $y(t) = x\big(-2(t-1)\big) + x\big(\frac{t+2}{2}\big)$. Clearly show the effect of each signal operation on the two component signals before obtaining the final sketch of $y(t)$. Label both axes properly.

SOLUTION

Recipe for $x(\alpha t+\beta)$: shift by β, reverse if $\alpha<0$, then scale by $|\alpha|$. A feature of x at $t=a$ appears at $t=(a-\beta)/\alpha$.

Component 1: $x(-2(t-1)) = x(-2t+2)$, $\alpha=-2$, $\beta=2$.

(i) Shift left by 2: $x(t+2)$ — triangle centred at $t=-2$.

(ii) Reverse: $x(-t+2)$ — triangle centred at $t=+2$.

(iii) Compress by 2: $x(-2t+2)$ — triangle centred at $t=1$, half-width $\frac{1}{2}$.

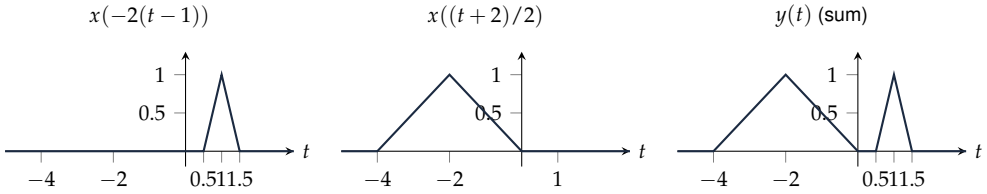
Check: $a=0\mapsto t=1$, $a=\pm1\mapsto t=\frac{1}{2},\frac{3}{2}$.

Component 2: $x\big(\frac{t+2}{2}\big) = x\big(\frac{t}{2}+1\big)$, $\alpha=\frac{1}{2}$, $\beta=1$.

(i) Shift left by 1: $x(t+1)$ — centred at $t=-1$.

(ii) Stretch by 2: $x\big(\frac{t}{2}+1\big)$ — centred at $t=-2$, half-width 2.

Check: $a=0\mapsto t=-2$, $a=\pm1\mapsto t=0,-4$.



The two triangles do not overlap (the first starts at $t=\frac{1}{2}$, the second ends at $t=0$), so

$$y(t) = \begin{cases} \frac{t+4}{2}, & -4 \le t \le -2, \\ -\frac{t}{2}, & -2 < t \le 0, \\ 2t-1, & \frac{1}{2} \le t \le 1, \\ 3-2t, & 1 < t \le \frac{3}{2}, \\ 0, & \text{otherwise.} \end{cases}$$

Answer. $y(t)$ is the sum of a triangle of height 1 centred at $t=-2$ (base from -4 to 0) and a narrower triangle of height 1 centred at $t=1$ (base from $\frac{1}{2}$ to $\frac{3}{2}$).

2023–2024 • Q1(a) EEE 211 • EEE p.1 MARKS 11

Consider the signal $x(t)$ shown in the figure (a triangle on $[-2,0]$ with peak 1 at $t=-1$, then a ramp up to 2 at $t=2$, flat at 2 until $t=3$, and a line down to 0 at $t=4$). (i) Draw $x(-3t+2)$, $x'(t)$ and $x''(t)$. (ii) Express $x'(t)$ using elementary signals such as $\delta(t)$, $u(t)$, $r(t)$.

SOLUTION

Piecewise: $x=t+2$ on $[-2,-1]$; $-t$ on $[-1,0]$; t on $[0,2]$; 2 on $[2,3]$; $8-2t$ on $[3,4]$; 0 otherwise. The signal is continuous.

(i) $x(-3t+2)$**.** $\alpha=-3$, $\beta=2$: shift left by 2, reverse, compress by 3. A feature at a moves to $t=(2-a)/3$:

$$a=4,3,2,0,-1,-2 \longmapsto t=-\tfrac{2}{3},-\tfrac{1}{3},0,\tfrac{2}{3},1,\tfrac{4}{3}, \qquad \text{values } 0,2,2,0,1,0.$$

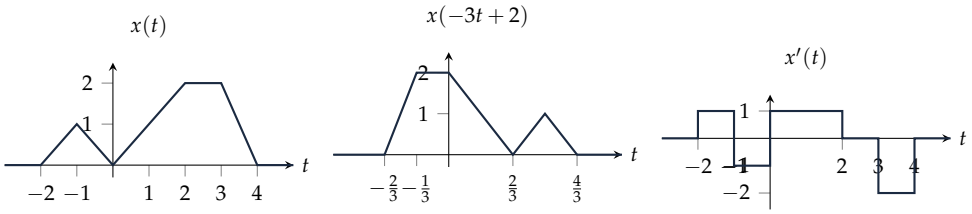
$$x(-3t+2)=\begin{cases} 6t+4, & -\frac{2}{3}\le t\le -\frac{1}{3},\\ 2, & -\frac{1}{3}\le t\le 0,\\ 2-3t, & 0\le t\le \frac{2}{3},\\ 3t-2, & \frac{2}{3}\le t\le 1,\\ 4-3t, & 1\le t\le \frac{4}{3}. \end{cases}$$

Derivatives. Since x is continuous there are no impulses in x'; slopes are $+1,-1,+1,0,-2$ on the five pieces:

$$x'(t)=\begin{cases} 1, & -2<t<-1\\ -1, & -1<t<0\\ 1, & 0<t<2\\ 0, & 2<t<3\\ -2, & 3<t<4 \end{cases}$$

The slope changes by $+2$ at $t=-1$, $+2$ at 0, -1 at 2, -2 at 3, $+2$ at 4 (and $+1$ at -2, from slope 0 to 1), so

$$x''(t)=\delta(t+2)-2\delta(t+1)+2\delta(t)-\delta(t-2)-2\delta(t-3)+2\delta(t-4).$$



(ii) $x'(t)=u(t+2)-2u(t+1)+2u(t)-u(t-2)-2u(t-3)+2u(t-4)$, equivalently $x(t)=r(t+2)-2r(t+1)+2r(t)-r(t-2)-2r(t-3)+2r(t-4)$.

Answer. $x'=u(t+2)-2u(t+1)+2u(t)-u(t-2)-2u(t-3)+2u(t-4)$; $x''=\delta(t+2)-2\delta(t+1)+2\delta(t)-\delta(t-2)-2\delta(t-3)+2\delta(t-4)$.

2022–2023 • Q5(a) CSE 219 • TF p.2 MARKS 15

A signal $f(t)$ is defined as $f(t) = \begin{cases} t+1, & -2 \le t \le 0, \\ -\frac{t}{2}+1, & 0 < t \le 2, \\ 0, & \text{elsewhere.} \end{cases}$ (i) Draw the graph of $f(t)$. (ii) Determine the even–odd decomposition of $f(t)$; apply the definitions of both the even and the odd function. (iii) Draw the graphs of the even and odd functions of the decomposition.

SOLUTION

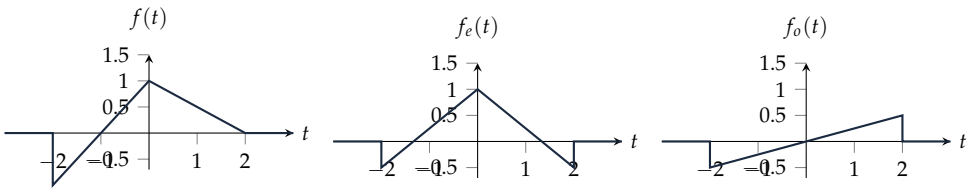
$f(-t) = -t+1$ for $0 \le t \le 2$ and $\frac{t}{2}+1$ for $-2 \le t < 0$.

$$f_e(t) = \tfrac{1}{2}[f(t)+f(-t)], \qquad f_o(t) = \tfrac{1}{2}[f(t)-f(-t)].$$

$0 < t \le 2$: $f_e = \frac{1}{2}\big[(1-\frac{t}{2})+(1-t)\big] = 1-\frac{3t}{4}$, $\quad f_o = \frac{1}{2}\big[(1-\frac{t}{2})-(1-t)\big] = \frac{t}{4}$.
$-2 \le t < 0$: $f_e = \frac{1}{2}\big[(t+1)+(\frac{t}{2}+1)\big] = 1+\frac{3t}{4}$, $\quad f_o = \frac{1}{2}\big[(t+1)-(\frac{t}{2}+1)\big] = \frac{t}{4}$.
Hence

$$f_e(t) = 1-\tfrac{3}{4}|t|, \qquad f_o(t) = \tfrac{t}{4}, \qquad |t| \le 2 \quad (0 \text{ elsewhere}).$$

Check: $f_e(t) = f_e(-t)$ (a "tent"), $f_o(-t) = -f_o(t)$ (a straight line through the origin), and at $t=1$: $\frac{1}{4}+\frac{1}{4} = \frac{1}{2} = f(1)$; at $t=-1$: $\frac{1}{4}-\frac{1}{4} = 0 = f(-1)$.



NOTE $f(t)$ as printed is $f(-2) = -1$ and $f(2) = 0$; the end values give the jumps of f_e and f_o at $|t| = 2$ (the drawing shows the jump values).

Answer. $f_e(t) = 1-\frac{3}{4}|t|$ and $f_o(t) = \frac{t}{4}$ for $|t| \le 2$.

2022–2023 • Q7(a)–(c) CSE 219 • TF p.4 MARKS 23

The graph of a signal $x(t)$ is shown in Figure 2 (it is 0 for $t < -2$ or $t > 2$). It rises linearly from -1 at $t = -2$ to 0 at $t = -1$, jumps to 1 and stays at 1 up to $t = 0$, then falls linearly to 0 at $t = 2$. (a) Write down the piece-wise definition of $x(t)$. (b) Determine $x(-2t+2)$ using the properties of signals, explicitly mentioning the properties used, and draw its graph. (c) Determine $x\big(-\frac{t}{2}+1\big)$ in the same way and draw its graph.

SOLUTION

(a) $x(t) = \begin{cases} t+1, & -2 \le t < -1, \\ 1, & -1 \le t \le 0, \\ 1-\frac{t}{2}, & 0 < t \le 2, \\ 0, & \text{otherwise.} \end{cases}$

(b) $x(-2t+2)$ ($\alpha = -2$, $\beta = 2$): time shift, time reversal, time scaling.

(i) Time shift left by 2: $g_1(t) = x(t+2) = t+3$ on $[-4,-3)$; 1 on $[-3,-2]$; $-\frac{t}{2}$ on $(-2,0]$.

(ii) Time reversal: $g_2(t) = g_1(-t) = x(-t+2) = \frac{t}{2}$ on $[0,2)$; 1 on $[2,3]$; $3-t$ on $(3,4]$.

(iii) Time compression by 2: $g_2(2t) = x(-2t+2)$:

$$x(-2t+2) = \begin{cases} t, & 0 \le t < 1, \\ 1, & 1 \le t \le \frac{3}{2}, \\ 3-2t, & \frac{3}{2} < t \le 2, \\ 0, & \text{otherwise.} \end{cases}$$

Check: $t = 2 \Rightarrow x(-2) = -1$ and $3-4 = -1$; $t = 1 \Rightarrow x(0) = 1$.

(c) $x(-\frac{t}{2}+1)$ $(\alpha = -\frac{1}{2},\ \beta = 1)$: shift left by 1, reverse, stretch by 2.

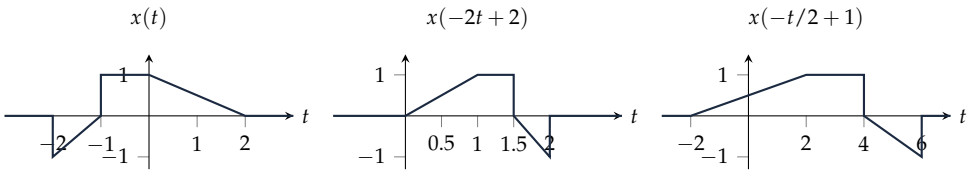
(i) $x(t+1) = t+2$ on $[-3,-2)$; 1 on $[-2,-1]$; $\frac{1-t}{2}$ on $(-1,1]$.

(ii) $x(-t+1) = \frac{1+t}{2}$ on $[-1,1)$; 1 on $[1,2]$; $2-t$ on $(2,3]$.

(iii) Stretch by 2:

$$x(-\tfrac{t}{2}+1) = \begin{cases} \frac{1}{2}+\frac{t}{4}, & -2 \le t < 2, \\ 1, & 2 \le t \le 4, \\ 2-\frac{t}{2}, & 4 < t \le 6, \\ 0, & \text{otherwise.} \end{cases}$$

Check: $t = 6 \Rightarrow x(-2) = -1$ and $2-3 = -1$; $t = -2 \Rightarrow x(2) = 0$.



Answer. $x(-2t+2) = t$ $(0 \le t < 1)$, 1 $(1 \le t \le \frac{3}{2})$, $3-2t$ $(\frac{3}{2} < t \le 2)$; $x(-\frac{t}{2}+1) = \frac{1}{2}+\frac{t}{4}$ $(-2 \le t < 2)$, 1 $(2 \le t \le 4)$, $2-\frac{t}{2}$ $(4 < t \le 6)$.

2022–2023 • Q7(d) CSE 219 • TF p.4 MARKS 7

Discrete signals have many properties. Suppose you have an image that you are using as an example of a discrete signal. Mention the specific signal properties that could be useful in accomplishing each task (no details needed): image compression, blurring the image, sharpening the image, flipping the image, zooming in or out, determining the image contrast, detecting edges in the image.

SOLUTION

Task	Signal property / operation
Compression	Frequency-domain representation (Fourier/DCT): keep large low-frequency coefficients, drop small high-frequency ones
Blurring	Convolution with a low-pass (averaging) kernel; removes high frequencies
Sharpening	High-pass filtering / adding back the difference (derivative) image; boosts high frequencies
Flipping	Time (index) reversal $x[-n]$ (mirror about the axis)
Zooming	Time (index) scaling $x[\alpha n]$ with up-/down-sampling and interpolation
Contrast	Amplitude scaling (and shifting) of pixel values; the spread (variance/power) of the values measures it
Edge detection	First difference $x[n] - x[n-1]$ (derivative): large values where the signal jumps; high-frequency content

Answer. Compression: frequency content; blur/sharpen: convolution (low/high-pass); flip: reversal; zoom: scaling; contrast: amplitude scaling/power; edges: difference/derivative.

2022–2023 • Q5(c) EEE 211 • EEE p.10 MARKS 10

For the signal $x(t)$ shown in the figure ($5t$ on $(0,1)$, then -6 on $(1,2)$, then $5(t-2)$ on $(2,3)$, zero elsewhere), write the signal as a linear combination of elementary signals. Sketch $\frac{\mathrm{d}x(t)}{\mathrm{d}t}$ with proper labels on the x- and y-axes.

SOLUTION

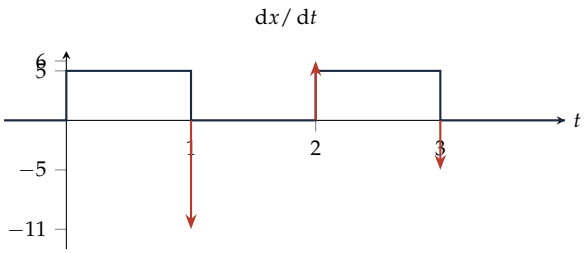
Scan from left to right, adding (new slope $-$ old slope) $r(t-a)$ and (jump) $u(t-a)$ at every break point.

t	event	term
0	slope $0 \to 5$	$5r(t)$
1	slope $5 \to 0$; jump $5 \to -6$ (-11)	$-5r(t-1) - 11u(t-1)$
2	jump $-6 \to 0$ $(+6)$; slope $0 \to 5$	$6u(t-2) + 5r(t-2)$
3	slope $5 \to 0$; jump $5 \to 0$ (-5)	$-5r(t-3) - 5u(t-3)$

$$x(t) = 5r(t) - 5r(t-1) - 11u(t-1) + 6u(t-2) + 5r(t-2) - 5r(t-3) - 5u(t-3).$$

Derivative (replace $r \to u$, $u \to \delta$):

$$\frac{\mathrm{d}x}{\mathrm{d}t} = 5\big[u(t) - u(t-1)\big] + 5\big[u(t-2) - u(t-3)\big] - 11\delta(t-1) + 6\delta(t-2) - 5\delta(t-3).$$



Answer. $x = 5r(t) - 5r(t-1) - 11u(t-1) + 6u(t-2) + 5r(t-2) - 5r(t-3) - 5u(t-3)$; x' has height-5 rectangles on $(0,1)$ and $(2,3)$ and impulses of strength -11, $+6$, -5 at $t = 1, 2, 3$.

2022–2023 • Q6(a) EEE 211 • EEE p.10 MARKS 20

Check whether the following signals can be used as a mathematical model of a delta function: (i) $p_1(t) = \lim_{\epsilon \to 0} \left(\epsilon\pi \cosh \frac{t}{\epsilon}\right)^{-1}$; (ii) $p_2(t) = \lim_{\epsilon \to 0} \frac{1}{\pi} \frac{\sin \epsilon t}{t}$.

SOLUTION

A family $p_\epsilon(t)$ models $\delta(t)$ if (1) its area is 1 for every ϵ, (2) $p_\epsilon(t) \to 0$ for every $t \neq 0$, and (3) its peak grows without bound (the area concentrates at $t = 0$).

(i) Area (substitute $u = t/\epsilon$): $\displaystyle\int \frac{\mathrm{d}t}{\epsilon\pi \cosh(t/\epsilon)} = \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{\mathrm{d}u}{\cosh u} = \frac{1}{\pi} \cdot \pi = 1$. For $t \neq 0$: $\cosh(t/\epsilon) \approx \frac{1}{2} e^{|t|/\epsilon}$, so $p_1 \approx \frac{2}{\epsilon\pi} e^{-|t|/\epsilon} \to 0$. Peak: $p_1(0) = \frac{1}{\epsilon\pi} \to \infty$. All three conditions hold.

(ii) Area: $\int \frac{\sin \epsilon t}{\pi t}\,\mathrm{d}t = 1$ for every $\epsilon > 0$. But for fixed t, $\frac{\sin \epsilon t}{\pi t} \to 0$ as $\epsilon \to 0$ (the peak value $\epsilon/\pi \to 0$): the unit area is spread over a main lobe of width $2\pi/\epsilon \to \infty$. The function becomes flat, not narrow. Condition (3)

fails, so as $\epsilon \to 0$ it is *not* a delta model. (The correct family is $\frac{\sin(t/\epsilon)}{\pi t}$, i.e. $\epsilon \to \infty$ in the printed form.)

Answer. (i) Yes, p_1 is a valid model of $\delta(t)$. (ii) No, not as $\epsilon \to 0$ (it becomes a delta only as $\epsilon \to \infty$).

2021–2022 • Q1(c) EEE 211 • EEE p.12 MARKS 15

Given the signal $x(t) = \delta(t-1) + u(t)u(t-3) + u(\cos t)$, compute $y(t) = \int_{-\infty}^{t} x(\tau)\,\mathrm{d}\tau$ and sketch $y(t/2)$.

NOTE The lower limit is partly illegible. Because $u(\cos t)$ equals 1 for half of every period, the integral from $-\infty$ diverges; the solution integrates from $t = 0$, which fixes y up to an additive constant.

SOLUTION

$u(t)u(t-3) = u(t-3)$. The three parts integrate separately:

$$\int \delta(\tau-1)\,\mathrm{d}\tau = u(t-1), \qquad \int u(\tau-3)\,\mathrm{d}\tau = r(t-3), \qquad w(t) = \int_0^t u(\cos\tau)\,\mathrm{d}\tau.$$

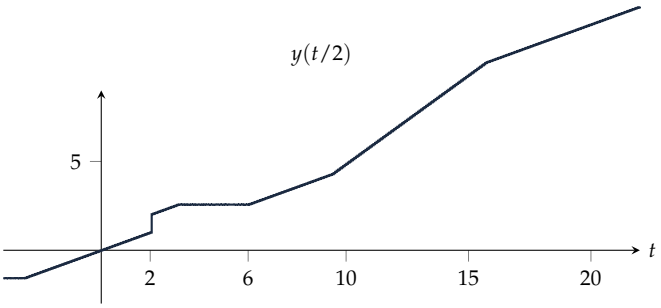
Write $u(\cos\tau) = \frac{1}{2} + \frac{1}{2}\operatorname{sgn}(\cos\tau)$. Since $\frac{\mathrm{d}}{\mathrm{d}\tau}\arcsin(\sin\tau) = \operatorname{sgn}(\cos\tau)$, we have $\int_0^t \operatorname{sgn}(\cos\tau)\,\mathrm{d}\tau = \arcsin(\sin t)$ and

$$w(t) = \frac{t}{2} + \frac{1}{2}\arcsin(\sin t) = \begin{cases} t, & |t| \le \frac{\pi}{2}, \\ \frac{\pi}{2}, & \frac{\pi}{2} \le t \le \frac{3\pi}{2}, \end{cases}$$ then repeats with slope 1 for half of each period.

Therefore

$$y(t) = u(t-1) + r(t-3) + \frac{t}{2} + \frac{1}{2}\arcsin(\sin t), \qquad y\Big(\frac{t}{2}\Big) = u\Big(\frac{t}{2}-1\Big) + r\Big(\frac{t}{2}-3\Big) + \frac{t}{4} + \frac{1}{2}\arcsin\Big(\sin\frac{t}{2}\Big).$$

$y(t/2)$ is $y(t)$ stretched by 2: the break points $t = 1, 3, \frac{\pi}{2}, \frac{3\pi}{2}, \ldots$ move to $t = 2, 6, \pi, 3\pi, \ldots$ (jump of 1 at $t = 2$, extra ramp of slope $\frac{1}{2}$ from $t = 6$, a staircase-ramp of average slope $\frac{1}{4}$).



Answer. $y(t) = u(t-1) + r(t-3) + \frac{t}{2} + \frac{1}{2}\arcsin(\sin t)$ (integrating $u(\cos t)$ from 0); $y(t/2)$ is its 2-times-stretched version.

2020–2021 • Q5(a) EEE 211 • EEE p.19 MARKS 25

Three signals $x_1(t), x_2(t), x_3(t)$ are shown in the figure. x_1 ramps from 0 at $t = 0$ to 2 at $t = 2$, stays 2 until $t = 6$ and drops to 0. x_2 ramps from 0 at $t = -4$ to 2 at $t = -2$, stays 2 until $t = 2$, equals 3 on $(2, 4)$ and is 0 afterwards. x_3 rises from 0 at $t = -3$ to 4 at $t = -2$, stays 4 until $t = 0$, drops to 1 on $(0, 2)$ and falls to 0 at $t = 3$. (i) Express $x_1(t)$ and $x_2(t)$ in terms of the ramp $r(t)$ and unit step $u(t)$. (10) (ii) Draw $x_1(t+2)\,r(t-2)$ and $x_2(t-2)\,u(-2t)$. (10) (iii) Express $x_3(t)$ in terms of $x_1(t)$. (5)

SOLUTION

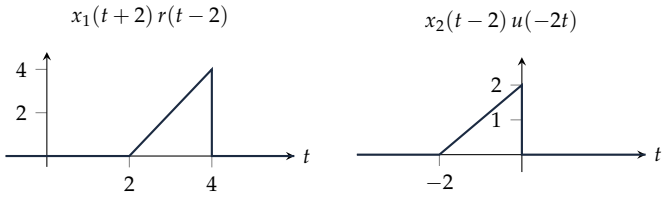
(i) x_1: slope 1 from $t=0$, slope returns to 0 at $t=2$, jump -2 at $t=6$:

$$x_1(t) = r(t) - r(t-2) - 2u(t-6).$$

x_2: slope 1 from $t=-4$ to -2, jump $+1$ at $t=2$, jump -3 at $t=4$:

$$x_2(t) = r(t+4) - r(t+2) + u(t-2) - 3u(t-4).$$

(ii) $x_1(t+2)$ is the ramp from $(-2,0)$ to $(0,2)$ then 2 until $t=4$; $r(t-2) = (t-2)u(t-2)$. For $2 \le t \le 4$ the product is $2(t-2)$; it is 0 elsewhere. $x_2(t-2)$ is the ramp on $(-2,0)$, then 2 on $(0,4)$, 3 on $(4,6)$. $u(-2t) = u(-t)$ keeps only $t<0$: the product is $t+2$ on $[-2,0)$ and 0 elsewhere.



(iii) The left part (rise from 0 to 4 over one unit, then 4 for two units) is $2x_1(2t+6)$ (compress by 2, shift left, scale by 2): ramp for $2t+6 \in [0,2] \Leftrightarrow t \in [-3,-2]$, flat 4 for $t \in [-2,0]$. The right part (1 for two units, then a ramp down over one unit) is $\frac{1}{2}x_1(6-2t)$ (reversed): flat 1 for $t \in [0,2]$, ramp down on $[2,3]$. Hence

$$x_3(t) = 2x_1(2t+6) + \tfrac{1}{2}x_1(6-2t).$$

Check: $x_3(-1) = 2\cdot 2\cdot 1 = 4$; $x_3(2.5) = \frac{1}{2}\cdot 1 = \frac{1}{2}$; $x_3(-2.5) = 2\cdot 1 = 2$ (mid-ramp).

Answer. $x_1 = r(t) - r(t-2) - 2u(t-6)$, $x_2 = r(t+4) - r(t+2) + u(t-2) - 3u(t-4)$, $x_3(t) = 2x_1(2t+6) + \frac{1}{2}x_1(6-2t)$.

2019–2020 • Q2(b) CSE 311 • TF p.15 **MARKS 11**

Prove that a unit step signal can be derived from a unit impulse signal and vice versa.

SOLUTION

Definition. The unit impulse is the limit of a pulse $\delta_\Delta(t)$ of width Δ and height $1/\Delta$ (area 1 for every Δ) as $\Delta \to 0$: $\delta(t) = 0$ for $t \neq 0$ and $\int \delta\,\mathrm{d}t = 1$.

Step from impulse. Integrate the impulse up to time t:

$$u(t) = \int_{-\infty}^{t} \delta(\tau)\,\mathrm{d}\tau = \begin{cases} 0, & t<0 \text{ (the window } (-\infty,t] \text{ misses the impulse)},\\ 1, & t>0 \text{ (the window contains it, area } 1). \end{cases}$$

Impulse from step. Replace the jump of u by a ramp $u_\Delta(t)$ rising from 0 to 1 over $[0,\Delta]$. Its derivative is

$$\frac{\mathrm{d}u_\Delta}{\mathrm{d}t} = \begin{cases} 1/\Delta, & 0<t<\Delta,\\ 0, & \text{otherwise} \end{cases} = \delta_\Delta(t),$$

a pulse of unit area. As $\Delta \to 0$, $u_\Delta \to u$ and $\delta_\Delta \to \delta$, so $\delta(t) = \dfrac{\mathrm{d}u(t)}{\mathrm{d}t}$ (in the generalised sense: zero away from the jump, an impulse of area equal to the jump height at $t=0$).
In discrete time the same relations are exact: $\delta[n] = u[n] - u[n-1]$ and $u[n] = \sum_{m=-\infty}^{n} \delta[m] = \sum_{k=0}^{\infty} \delta[n-k]$.

Answer. $u(t) = \int_{-\infty}^{t} \delta(\tau)\,\mathrm{d}\tau$ and $\delta(t) = \dfrac{\mathrm{d}u}{\mathrm{d}t}$.

2019–2020 • Q5(a) EEE 211 • EEE p.24 MARKS 15

Describe the signal $x(t)$ in the figure by a single expression valid for all t. Sketch the time-compressed version of $x(t)$ by a factor of 3. (The figure shows $x = 2$ on $(-1.5, 0)$, then $x = 2e^{-t/2}$ on $(0, 3)$, and 0 elsewhere.)

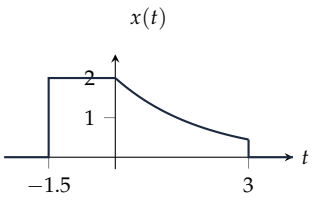
SOLUTION

Use steps to switch each piece on and off:

$$x(t) = 2\big[u(t+1.5) - u(t)\big] + 2e^{-t/2}\big[u(t) - u(t-3)\big] \quad \text{(valid for all } t\text{)}.$$

Time compression by 3 replaces t by $3t$, so every break point is divided by 3:

$$x(3t) = 2\big[u(t+0.5) - u(t)\big] + 2e^{-3t/2}\big[u(t) - u(t-1)\big].$$



Answer. $x(t) = 2[u(t+1.5) - u(t)] + 2e^{-t/2}[u(t) - u(t-3)]$; $x(3t) = 2[u(t+0.5) - u(t)] + 2e^{-3t/2}[u(t) - u(t-1)]$.

2018–2019 • Q6(b) EEE 211 • EEE p.30 MARKS 20

The voltage waveform $v(t)$ is shown in the figure: it is 0 for $t < -1$, jumps to -2 at $t = -1$, rises linearly through $(0, 0)$ to $(1, 2)$, stays 2 until $t = 2$, rises linearly to $(4, 3)$, drops to 1 at $t = 4$, stays 1 until $t = 5$, falls linearly through $(6, 0)$ to $(7, -1)$ and jumps back to 0 at $t = 7$. (i) Express $v(t)$ in terms of unit step and unit ramp functions. (ii) Compute the derivative of $v(t)$ and sketch its waveform. (iii) Evaluate and sketch the even and odd components of the derivative of $v(t)$.

NOTE The scanned label at $t = 2$ reads "(2,1)" while the curve is drawn at height 2; the solution follows the drawing ($v(2) = 2$, slope $\frac{1}{2}$ on $(2, 4)$).

SOLUTION

Slopes and jumps. Slopes: 2 on $(-1, 1)$; 0 on $(1, 2)$; $\frac{1}{2}$ on $(2, 4)$; 0 on $(4, 5)$; -1 on $(5, 7)$. Jumps: -2 at $t = -1$, -2 at $t = 4$, $+1$ at $t = 7$.

(i) Add (slope change) $r(t-a)$ and (jump) $u(t-a)$ at each break point:

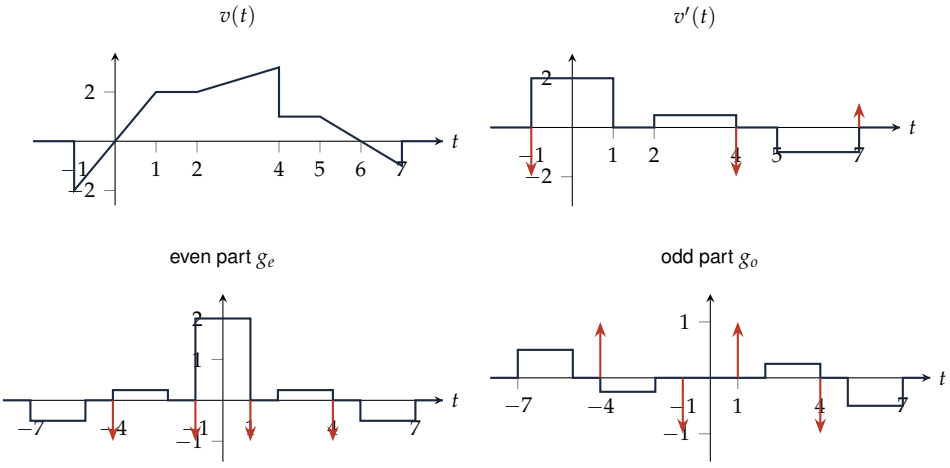
$$v(t) = -2u(t+1) + 2r(t+1) - 2r(t-1) + \tfrac{1}{2}r(t-2) - 2u(t-4) - \tfrac{1}{2}r(t-4) - r(t-5) + r(t-7) + u(t-7).$$

Check: $v(3) = -2 + 8 - 4 + \frac{1}{2} = 2.5$ (line from $(2, 2)$ with slope $\frac{1}{2}$); $v(6) = -2 + 14 - 10 + 2 - 2 - 1 - 1 = 0$.

(ii) $\dfrac{dv}{dt} = 2\,[u(t+1) - u(t-1)] + \frac{1}{2}[u(t-2) - u(t-4)] - [u(t-5) - u(t-7)] - 2\delta(t+1) - 2\delta(t-4) + \delta(t-7)$.

(iii) Let $g = v'$ and $g(-t)$ its reflection. $g_e = \frac{1}{2}[g(t) + g(-t)]$, $g_o = \frac{1}{2}[g(t) - g(-t)]$:

interval / location	g_e	g_o
$\lvert t\rvert < 1$ (slope part)	2	0
$2 < t < 4$ $(-4 < t < -2)$	$\frac{1}{4}$ $(\frac{1}{4})$	$\frac{1}{4}$ $(-\frac{1}{4})$
$5 < t < 7$ $(-7 < t < -5)$	$-\frac{1}{2}$ $(-\frac{1}{2})$	$-\frac{1}{2}$ $(+\frac{1}{2})$
$\delta(t+1)$, $\delta(t-1)$	-1, -1	-1, $+1$
$\delta(t+4)$, $\delta(t-4)$	-1, -1	$+1$, -1
$\delta(t+7)$, $\delta(t-7)$	$+\frac{1}{2}$, $+\frac{1}{2}$	$-\frac{1}{2}$, $+\frac{1}{2}$



(The even part has impulses $-\delta(t\pm1)$, $-\delta(t\pm4)$, $\frac{1}{2}\delta(t\pm7)$; the odd part has $-\delta(t+1)+\delta(t-1)$, $\delta(t+4)-\delta(t-4)$, $-\frac{1}{2}\delta(t+7)+\frac{1}{2}\delta(t-7)$.)

Answer. $v = -2u(t+1)+2r(t+1)-2r(t-1)+\frac{1}{2}r(t-2)-2u(t-4)-\frac{1}{2}r(t-4)-r(t-5)+r(t-7)+u(t-7)$; v' = slopes $2, 0, \frac{1}{2}, 0, -1$ plus impulses $-2, -2, +1$ at $t = -1, 4, 7$.

03

System Properties and Responses of LTI Systems

ANIK SIR • LECTURE 3

REPEATED ×2 **Same question in 2 papers — stacked below, solved once.**

2023–2024 • Q1(b) EEE 211 • EEE p.1 MARKS 7

Consider a system with input $x(t)$ and output $y(t)$ described by the differential equation $\frac{dy(t)}{dt} + t^2 y(t) = (2t+3)x(t)$. Determine whether the system is (i) linear; (ii) time invariant.

2022–2023 • Q8(b) EEE 211 • EEE p.11 MARKS 15

Consider a system with input $x(t)$ and output $y(t)$ described by $\frac{dy(t)}{dt} + t^2 y(t) = (2t+3)x(t)$. Comment on whether the system is (i) linear; (ii) time invariant.

SOLUTION

Assume the system is initially relaxed ($y = 0$ before the input starts).

(i) Linearity. Let $x_1 \mapsto y_1$ and $x_2 \mapsto y_2$, i.e. $y_i' + t^2 y_i = (2t+3)x_i$. For $x = ax_1 + bx_2$, try $y = ay_1 + by_2$:

$$y' + t^2 y = a(y_1' + t^2 y_1) + b(y_2' + t^2 y_2) = (2t+3)(ax_1 + bx_2) = (2t+3)x.$$

Superposition holds (both sides are first order in x, y and no constant term appears), so the system is *linear*.

(ii) Time invariance. Let y be the response to $x(t)$. Delay the input: $x_d(t) = x(t - t_0)$. If the system were time invariant the response would be $y(t - t_0)$, which satisfies

$$y'(t - t_0) + (t - t_0)^2 y(t - t_0) = \big(2(t - t_0) + 3\big)x(t - t_0).$$

The actual system requires $y_d' + t^2 y_d = (2t+3)x(t - t_0)$. The coefficients t^2 and $2t + 3$ differ from $(t - t_0)^2$ and $2(t - t_0) + 3$ whenever $t_0 \neq 0$, so $y(t - t_0)$ is not the response to $x(t - t_0)$.

Answer. Linear, but *not* time invariant (the coefficients depend explicitly on t).

2022–2023 • Q5(b) EEE 211 • EEE p.9–10 MARKS 15

The input $x(t)$ and output $y(t)$ of an LTI system are shown in the figure: $x(t) = |t|$ for $|t| \leq 1$ (a "V" with value 1 at $t = \pm 1$) and $y(t) = 1$ for $|t| \leq 1$ (a unit rectangle). Find the outputs of this system for the inputs (i) $x(t - \frac{1}{2}) - x(t + \frac{1}{2})$ and (ii) $\frac{dx(t)}{dt}$. Sketch both outputs with proper axis labels.

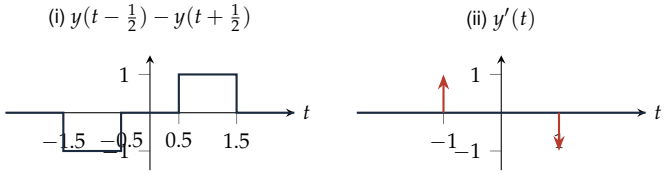
SOLUTION

Linearity and time invariance give: $x(t-t_0)\mapsto y(t-t_0)$ and $\alpha x_1+\beta x_2\mapsto \alpha y_1+\beta y_2$. Differentiation is a limit of differences of shifted inputs, so $\dfrac{\mathrm{d}x}{\mathrm{d}t}\mapsto\dfrac{\mathrm{d}y}{\mathrm{d}t}$ for an LTI system.

(i) Output $=y(t-\tfrac12)-y(t+\tfrac12)$, where $y(t-\tfrac12)=1$ on $(-\tfrac12,\tfrac32)$ and $y(t+\tfrac12)=1$ on $(-\tfrac32,\tfrac12)$:

$$\text{output}=\begin{cases}-1, & -\tfrac32<t<-\tfrac12,\\ 0, & -\tfrac12<t<\tfrac12,\\ +1, & \tfrac12<t<\tfrac32,\\ 0, & \text{otherwise.}\end{cases}$$

(ii) $y(t)=u(t+1)-u(t-1)$, so $\dfrac{\mathrm{d}y}{\mathrm{d}t}=\delta(t+1)-\delta(t-1)$.



Answer. (i) -1 on $(-\tfrac32,-\tfrac12)$, 0 on $(-\tfrac12,\tfrac12)$, $+1$ on $(\tfrac12,\tfrac32)$. (ii) $\delta(t+1)-\delta(t-1)$.

2021–2022 • Q2(b) EEE 211 • EEE p.12 **MARKS 18**

Let h be the impulse response of an LTI system, and let x_1, x_2, y_1, y_2 denote functions such that $y_1=x_1*h$ and $y_2=x_2*h$. The functions x_1 and x_2 are shown in the figure (x_1 is a triangle on $(0,2)$ with peak 1 at $t=1$; x_2 is a trapezoid on $(-2,2)$ with plateau height 2). *[the remaining part of the wording is cut off in the scan]*

NOTE The question text after the figure is not legible, so the exact task cannot be recovered. The solution gives the standard method for this setting.

SOLUTION

In this setting the data usually include a given $y_1(t)$ and ask for $y_2(t)$. Method:

(i) Write $x_2(t)$ as a linear combination of shifted/scaled copies of $x_1(t)$ (a trapezoid is a sum of delayed triangles, or $x_2=\sum_k a_k x_1(t-t_k)$).

(ii) By linearity and time invariance, $y_2(t)=\sum_k a_k y_1(t-t_k)$.

(iii) Otherwise differentiate the signals: x_i' consists of rectangles/impulses and $y_i'=x_i'*h$, so h-dependent parts cancel when comparing x_1 and x_2.

Answer. Not solvable without the illegible wording; use $y_2=\sum a_k y_1(t-t_k)$ after decomposing x_2 into shifted copies of x_1.

2020–2021 • Q6(a) EEE 211 • EEE p.20 **MARKS 15**

Consider a continuous-time system with input–output relationship $y(t)=t\,x(t+2)$. Determine whether the system is (i) linear, (ii) causal, (iii) time-variant, (iv) memoryless and (v) stable. You will have to use time-domain techniques.

SOLUTION

(i) *Linear.* $S\{ax_1+bx_2\}=t\,[ax_1(t+2)+bx_2(t+2)]=ay_1+by_2$ — yes.

(ii) *Causal.* $y(t)$ uses $x(t+2)$, a value two seconds in the future — not causal.

(iii) *Time-variant.* $S\{x(t-t_0)\} = t\,x(t-t_0+2)$ but $y(t-t_0) = (t-t_0)\,x(t-t_0+2)$. These differ $(t \neq t-t_0)$ — time-variant.

(iv) *Memoryless.* The output at t depends on $x(t+2)$, not on $x(t)$ — has memory.

(v) *Stable.* Take the bounded input $x = 1$: $y(t) = t$, unbounded — not BIBO stable.

Answer. Linear; not causal; time-variant; with memory; unstable.

2019–2020 • Q1 EEE 211 • EEE p.22 MARKS 35

A continuous-time system is given by $y(t) = \frac{1}{3}\int_{-\infty}^{\infty} x(\tau)\big[\delta(t-\tau) + \delta(t-\tau+t_0) + \delta(t-\tau-t_0)\big]\,\mathrm{d}\tau$. (i) Explain what this system does. (ii) Is the system BIBO stable? (iii) Linear? (iv) Memoryless? (v) Causal? (vi) Time invariant? Justify each answer.

SOLUTION

Sifting each impulse ($\int x(\tau)\delta(t-\tau+a)\,\mathrm{d}\tau = x(t+a)$):

$$y(t) = \tfrac{1}{3}\big[x(t) + x(t+t_0) + x(t-t_0)\big], \qquad h(t) = \tfrac{1}{3}\big[\delta(t) + \delta(t+t_0) + \delta(t-t_0)\big].$$

(i) It averages the signal over the three instants $t-t_0$, t, $t+t_0$: a three-point moving average (a smoothing low-pass filter).

(ii) *BIBO stable:* $|y| \le \frac{1}{3}(B+B+B) = B$; equivalently $\int|h| = 1 < \infty$.

(iii) *Linear:* a weighted sum of shifted inputs — yes.

(iv) *Memoryless:* no, it uses $x(t \pm t_0)$.

(v) *Causal:* no (for $t_0 \neq 0$): the term $x(t+t_0)$ needs a future value; $h \neq 0$ at $t = -t_0 < 0$.

(vi) *Time invariant:* yes — the system is the convolution with a fixed h; delaying x by T delays every term by T.

Answer. A 3-point moving average; BIBO stable, linear, time invariant, but with memory and non-causal.

2018–2019 • Q7(a) EEE 211 • EEE p.30 MARKS 20

A linear system with impulse response $h(t)$ has the input–output pair shown in the figure: $x(t) = 1$ on $(1,2)$ gives $y(t)$ that rises linearly from 0 at $t = 1$ to 1 at $t = 2$, stays 1 until $t = 3$ and is 0 afterwards. Determine whether the system is causal, time-invariant and memoryless. Also find the output of the system for the input $x_1(t) = u(t) + u(t-1) - 2u(t-2)$.

NOTE The system is treated as LTI (it is described by an impulse response $h(t)$). A single input–output pair cannot prove time invariance by itself.

SOLUTION

Here $x(t) = u(t-1) - u(t-2)$ and $y(t) = r(t-1) - r(t-2) - u(t-3)$.

Causal. The output is zero for $t < 1$, exactly where the input is zero: nothing appears before the cause. Solving $x * h = y$ for the LTI case, $x' = \delta(t-1) - \delta(t-2)$ and $y' = u(t-1) - u(t-2) - \delta(t-3)$ give $h(t) - h(t-1) = u(t) - u(t-1) - \delta(t-2)$, so $h(t) = u(t) - \sum_{k=2}^{\infty}\delta(t-k)$, which is zero for $t < 0$: *causal*.

Memoryless. No. A memoryless system would give a scaled copy of x (a rectangle on $(1,2)$), but y extends to $t = 3$ and is not a rectangle.

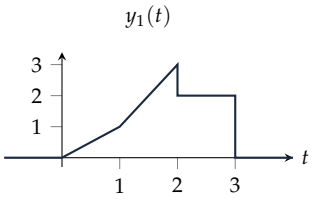
Time invariance. Assumed (LTI).

Output for x_1. Write $x_1 = [u(t) - u(t-1)] + 2[u(t-1) - u(t-2)] = x(t+1) + 2x(t)$. By linearity and

time invariance $y_1(t) = y(t+1) + 2y(t)$:

$$y(t+1) = \begin{cases} t, & 0 \le t \le 1 \\ 1, & 1 \le t \le 2 \\ 0, & \text{else} \end{cases} \qquad 2y(t) = \begin{cases} 2(t-1), & 1 \le t \le 2 \\ 2, & 2 \le t \le 3 \\ 0, & \text{else} \end{cases}$$

$$y_1(t) = \begin{cases} t, & 0 \le t \le 1, \\ 2t-1, & 1 \le t \le 2, \\ 2, & 2 < t \le 3, \\ 0, & \text{otherwise.} \end{cases}$$



Answer. Causal, not memoryless; $y_1 = t$ $(0 \le t \le 1)$, $2t-1$ $(1 \le t \le 2)$, 2 $(2 < t \le 3)$, 0 otherwise.

04

Convolution and Interconnected Systems

ANIK SIR • LECTURES 3–4

2023–2024 • Q1(b) CSE 219 • 22 batch p.1 MARKS 15

Consider the combination of discrete-time LTI systems in Figure 1: the input $x[n]$ goes through h_1 and h_2 in parallel, their outputs are added and passed through h_3 to give $y[n]$. The impulse responses are $h_1[n] = \delta[n] - \delta[n-2]$, $h_2[n] = u[n-1]$, $h_3[n] = \delta[n] - \delta[n-1]$. Suppose the input is $x[n] = u[n]$. (i) Express the equivalent impulse response $h_{eq}[n]$ of the overall system. (ii) Infer the output $y[n]$ for the given input.

SOLUTION

Parallel branches add; the cascade convolves: $h_{eq} = (h_1 + h_2) * h_3$.

$$h_2 * h_3 = u[n-1] - u[n-2] = \delta[n-1],$$
$$h_1 * h_3 = \delta[n] - \delta[n-1] - \delta[n-2] + \delta[n-3].$$

$$h_{eq}[n] = h_1 * h_3 + h_2 * h_3 = \delta[n] - \delta[n-2] + \delta[n-3].$$

For $x = u[n]$: $y = u * h_{eq} = u[n] - u[n-2] + u[n-3]$, i.e.

n	<0	0	1	2	3	4	≥ 5
$y[n]$	0	1	1	0	1	1	1

Answer. $h_{eq}[n] = \delta[n] - \delta[n-2] + \delta[n-3]$; $y[n] = u[n] - u[n-2] + u[n-3] = \{1, 1, 0, 1, 1, 1, \dots\}$.

2023–2024 • Q1(c) EEE 211 • EEE p.1 MARKS 17

Consider the system in the figure: $x(t)$ feeds three parallel paths — $h_1(t)$; $h_2(t)$ followed by $h_3(t)$; and $h_4(t)$ — whose outputs are added and passed through $h_5(t)$ to give $y(t)$. Given $h_1 = e^{-2t}u(t)$, $h_2 = e^{-2t}u(t)$, $h_3 = e^{-t}u(t)$, $h_4 = \delta(t)$, $h_5 = e^{-3t}u(t)$. Using time-domain techniques, (i) determine whether the system is BIBO stable; (ii) determine the output $y(t)$ for the input $x(t) = e^{-t}u(t)$.

SOLUTION

Useful pair: $e^{-at}u * e^{-bt}u = \dfrac{e^{-bt} - e^{-at}}{a-b}u(t)$ for $a \neq b$.

Middle path: $h_2 * h_3 = e^{-2t}u * e^{-t}u = (e^{-t} - e^{-2t})u(t)$.

Sum of the three paths: $g = h_1 + h_2 * h_3 + h_4 = e^{-2t}u + (e^{-t} - e^{-2t})u + \delta(t) = \delta(t) + e^{-t}u(t)$.

Overall: $h_{eq} = g * h_5 = e^{-3t}u + e^{-t}u * e^{-3t}u = e^{-3t}u + \tfrac{1}{2}(e^{-t} - e^{-3t})u$, i.e.

$$h_{eq}(t) = \tfrac{1}{2}\big(e^{-t} + e^{-3t}\big)u(t).$$

(i) h_{eq} is causal and $\int_0^\infty |h_{eq}|\,\mathrm{d}t = \tfrac{1}{2}(1 + \tfrac{1}{3}) = \tfrac{2}{3} < \infty$: BIBO stable.
(ii) $y = e^{-t}u * h_{eq} = \tfrac{1}{2}\,e^{-t}u * e^{-t}u + \tfrac{1}{2}\,e^{-t}u * e^{-3t}u = \tfrac{1}{2}te^{-t}u + \tfrac{1}{4}(e^{-t} - e^{-3t})u$.

Answer. (i) Stable ($\int |h_{eq}| = \tfrac{2}{3}$). (ii) $y(t) = \big[\tfrac{1}{2}te^{-t} + \tfrac{1}{4}e^{-t} - \tfrac{1}{4}e^{-3t}\big]u(t)$.

2023–2024 • Q2(a) EEE 211 • EEE p.2 MARKS 18

Graphically determine the convolution of the pair of signals in the figure: $x(t) = 1$ for $1 < t < 2$ (and 0 elsewhere), and $h(t)$ a triangle of height 1 on $(-1, 1)$ with peak at $t = 0$.

SOLUTION

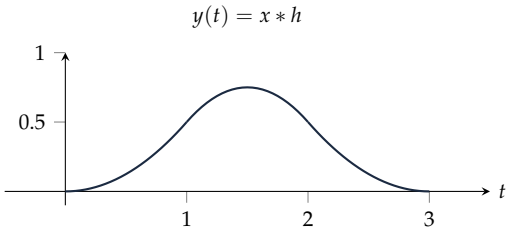
Since x is a rectangle on $(1,2)$, $y(t) = \int h(\tau)\,x(t-\tau)\,\mathrm{d}\tau = \int_{t-2}^{t-1} h(s)\,\mathrm{d}s = H(t-1) - H(t-2)$, where $H(s) = \int_{-\infty}^{s} h$ is the running integral of the triangle:

$$H(s) = \begin{cases} 0, & s < -1, \\ \frac{(s+1)^2}{2}, & -1 \le s \le 0, \\ 1 - \frac{(1-s)^2}{2}, & 0 \le s \le 1, \\ 1, & s > 1. \end{cases}$$

Slide the flipped rectangle (width 1) along τ; the interval table is

interval	overlap	$y(t)$
$t < 0$	none	0
$0 \le t \le 1$	right part of h enters	$\dfrac{t^2}{2}$
$1 \le t \le 2$	window $(t-2, t-1)$ covers the peak	$-t^2 + 3t - \dfrac{3}{2}$
$2 \le t \le 3$	left part of h leaves	$\dfrac{(3-t)^2}{2}$
$t > 3$	none	0

Checks: continuity at $t = 1$ ($\tfrac{1}{2}$), $t = 2$ ($\tfrac{1}{2}$), peak $y(1.5) = \tfrac{3}{4}$, and area $\int y = (1)(1) = 1$.



Answer. $y = \frac{t^2}{2}$ $(0 \le t \le 1)$, $-t^2 + 3t - \frac{3}{2}$ $(1 \le t \le 2)$, $\frac{(3-t)^2}{2}$ $(2 \le t \le 3)$, 0 otherwise.

2022–2023 • Q8(a) CSE 219 • TF p.5 MARKS 20

Convolve $x[n] = 3^n u[-n-1] + \left(\tfrac{1}{3}\right)^n u[n]$ with $h[n] = \left(\tfrac{1}{4}\right)^n u[n+3]$.

SOLUTION

Split the two-sided input (convolution distributes): $x = x_1 + x_2$, $x_1 = 3^n u[-n-1]$, $x_2 = (\frac{1}{3})^n u[n]$. The kernel $h[n-k] = (\frac{1}{4})^{n-k}$ is nonzero for $k \le n+3$.

$y_1 = x_1 * h$. $x_1[k] \neq 0$ for $k \le -1$, so $k \le \min(-1, n+3)$ and $y_1[n] = 4^{-n}\sum_k 12^k$:

- $n \le -4$ (upper limit $n+3$): $y_1[n] = 4^{-n}\dfrac{12^{n+3}}{1-\frac{1}{12}} = \dfrac{12}{11}4^{-n}12^{n+3} = \dfrac{12^4}{11}3^n = \dfrac{256}{11}3^{n+4}$.
- $n \ge -3$ (upper limit -1): $y_1[n] = 4^{-n}\dfrac{12^{-1}}{1-\frac{1}{12}} = \dfrac{4^{-n}}{11}$.

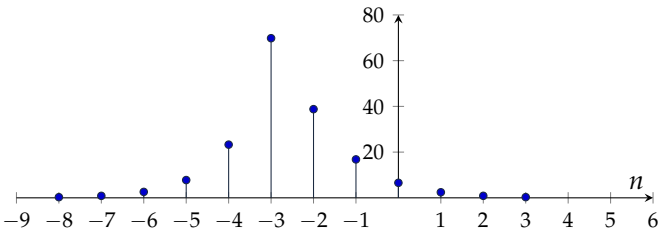
$y_2 = x_2 * h$. $x_2[k] \neq 0$ for $k \ge 0$, so $0 \le k \le n+3$ (needs $n \ge -3$):

$$y_2[n] = 4^{-n}\sum_{k=0}^{n+3}\left(\tfrac{4}{3}\right)^k = 4^{-n}\,\frac{(4/3)^{n+4}-1}{1/3} = 256\cdot 3^{-(n+3)} - 3\cdot 4^{-n}.$$

Total.

$$y[n] = \begin{cases} \dfrac{256}{11}3^{n+4}, & n \le -4, \\ 256\cdot 3^{-(n+3)} - \dfrac{32}{11}4^{-n}, & n \ge -3. \end{cases}$$

For $n \ge -3$: $y_1 + y_2 = \frac{4^{-n}}{11} + 256\cdot 3^{-(n+3)} - 3\cdot 4^{-n}$ and $\frac{1}{11} - 3 = -\frac{32}{11}$. Values: $y[-4] = 23.27$, $y[-3] = 69.82$ (peak), $y[-2] = 38.79$, $y[-1] = 16.81$, $y[0] = 6.57$, $y[1] = 2.43$.



Answer. $y[n] = \frac{256}{11}3^{n+4}$ for $n \le -4$; $y[n] = 256\cdot 3^{-(n+3)} - \frac{32}{11}4^{-n}$ for $n \ge -3$.

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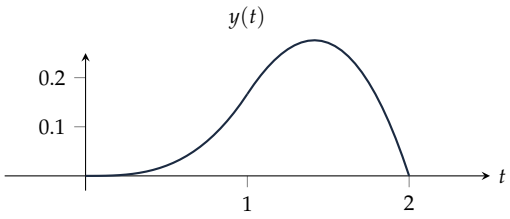
Consider an LTI system where $x(t) = h(t) = t\,[u(t) - u(t-1)]$. Find the expression of the output $y(t)$ of the system for all t.

SOLUTION

Both signals are the ramp t on $(0,1)$. $y(t) = \int x(\tau)h(t-\tau)\,\mathrm{d}\tau$ with $0 < \tau < 1$ and $0 < t-\tau < 1$, i.e. $\max(0, t-1) < \tau < \min(1, t)$.

- $0 \le t \le 1$ (limits 0 to t): $y = \int_0^t \tau(t-\tau)\,\mathrm{d}\tau = \dfrac{t^3}{6}$.
- $1 \le t \le 2$ (limits $t-1$ to 1): $y = \int_{t-1}^1 \tau(t-\tau)\,\mathrm{d}\tau = -\dfrac{t^3}{6} + t - \dfrac{2}{3}$.
- otherwise 0.

Check: both formulas give $\frac{1}{6}$ at $t = 1$; the second gives 0 at $t = 2$; the area of y is $\frac{1}{2}\cdot\frac{1}{2} = \frac{1}{4}$.



Answer. $y(t) = \frac{t^3}{6}$ for $0 \le t \le 1$; $-\frac{t^3}{6} + t - \frac{2}{3}$ for $1 \le t \le 2$; 0 otherwise.

2021–2022 • Q2(a) EEE 211 • EEE p.12 **MARKS 17**

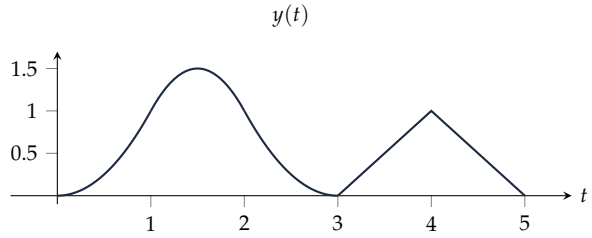
The input $x(t)$ and the impulse response $h(t)$ of an LTI system are shown in the figure: $x(t)$ is a triangle on $(0,2)$ with peak 1 at $t = 1$; $h(t)$ is a rectangle of height 2 on $(0,1)$ together with a unit impulse at $t = 3$. Compute the output $y(t)$ using the graphical method. Also write the analytical expression of $y(t)$.

SOLUTION

$h(t) = 2[u(t) - u(t-1)] + \delta(t-3)$, so $y = 2\int_{t-1}^{t} x(s)\,\mathrm{d}s + x(t-3)$. The running integral of the triangle is $X(s) = \frac{s^2}{2}$ $(0 \le s \le 1)$, $1 - \frac{(2-s)^2}{2}$ $(1 \le s \le 2)$, 1 $(s \ge 2)$.

interval	rectangular part $2[X(t) - X(t-1)]$
$0 \le t \le 1$	t^2
$1 \le t \le 2$	$2 - (2-t)^2 - (t-1)^2 = -2t^2 + 6t - 3$
$2 \le t \le 3$	$(3-t)^2$

The impulse produces the delayed triangle $x(t-3)$: $t-3$ on $(3,4)$ and $5-t$ on $(4,5)$.



Answer. $y = t^2$ $(0 \le t \le 1)$; $-2t^2 + 6t - 3$ $(1 \le t \le 2)$; $(3-t)^2$ $(2 \le t \le 3)$; then the triangle $x(t-3)$ on $(3,5)$.

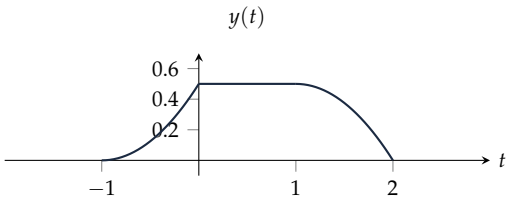
2020–2021 • Q6(b) EEE 211 • EEE p.20 **MARKS 10**

Using graphical convolution, determine the output $y(t)$ of an LTI system if the input is $x(t) = \mathrm{rect}(t/2)$ and the system impulse response is $h(t) = r(t)\,u(-t+1)$. Also write the analytical expression of $y(t)$ and draw it.

SOLUTION

$x = 1$ on $(-1,1)$ and $h(t) = t$ on $(0,1)$. With x a rectangle, $y(t) = \int_{t-1}^{t+1} h(s)\,\mathrm{d}s$.

interval	overlap of $(t-1, t+1)$ with $(0,1)$	$y(t)$
$-1 \le t \le 0$	$(0, t+1)$	$\frac{(t+1)^2}{2}$
$0 \le t \le 1$	the whole $(0,1)$	$\frac{1}{2}$
$1 \le t \le 2$	$(t-1, 1)$	$\frac{1}{2} - \frac{(t-1)^2}{2}$
else	none	0



Answer. $y = \frac{(t+1)^2}{2}$ $(-1 \le t \le 0)$, $\frac{1}{2}$ $(0 \le t \le 1)$, $\frac{1-(t-1)^2}{2}$ $(1 \le t \le 2)$, 0 otherwise.

2019–2020 • Q2 EEE 211 • EEE p.22–23 MARKS 35

Suppose an LTI system has impulse response $h(t)$ and input $x(t)$. The figure shows $x(t)$ (a rectangle of height 1 on $(0,2)$) and $h(t+1)$ (a triangle of height 1 on $(-1,1)$). (i) Is the system causal? (ii) Use the graphical convolution technique to determine $y(t) = x(t) * h(t)$. (iii) Compare $x(t)$ and $y(t)$, and explain what this system does. (iv) Sketch the output of the system for a different input $z(t) = x(t) - x(t-4)$.

SOLUTION

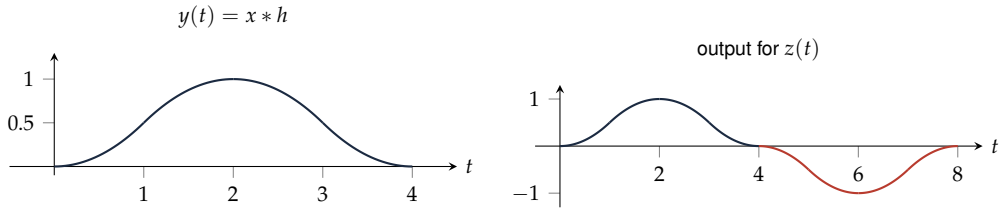
(i) $h(t+1)$ is a triangle on $(-1,1)$, so $h(t)$ is a triangle on $(0,2)$ with peak at $t=1$: $h(t) = 0$ for $t < 0$ — *causal*.

(ii) x is a rectangle on $(0,2)$: $y(t) = \int_{t-2}^{t} h(s)\,\mathrm{d}s = H(t) - H(t-2)$ with $H(s) = \frac{s^2}{2}$ $(0 \le s \le 1)$, $1 - \frac{(2-s)^2}{2}$ $(1 \le s \le 2)$, 1 $(s \ge 2)$:

$$y(t) = \begin{cases} \frac{t^2}{2}, & 0 \le t \le 1, \\ 1 - \frac{(2-t)^2}{2}, & 1 \le t \le 2, \\ 1 - \frac{(t-2)^2}{2}, & 2 \le t \le 3, \\ \frac{(4-t)^2}{2}, & 3 \le t \le 4, \\ 0, & \text{otherwise.} \end{cases}$$

(iii) y is a smoothed copy of x: the sharp edges become S-shaped curves, the duration grows from 2 to 4 (sum of the durations), the peak is $y(2) = 1$ and the area is preserved ($\int y = \int x \int h = 2 \cdot 1 = 2$). The system is a causal smoothing (low-pass) filter: a triangular-weighted moving average over 2 seconds with a delay of 1.

(iv) $z = x(t) - x(t-4)$, so the output is $y(t) - y(t-4)$: the smooth pulse on $(0,4)$ followed by its negative on $(4,8)$.



Answer. Causal; y is a 4-second smooth pulse (peak 1 at $t = 2$), i.e. a smoothing low-pass system; for z the output is $y(t) - y(t-4)$.

2018–2019 • Q7(b) EEE 211 • EEE p.31 MARKS 15

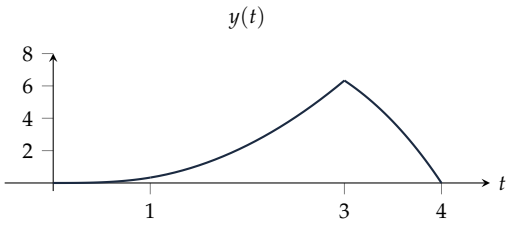
Let $h(t) = u(t) - u(t-1)$ and $x(t) = t^2\,[u(t) - u(t-3)]$. Using graphical convolution compute the output $y(t)$ and write its analytical expression.

SOLUTION

h is a rectangle of width 1, so $y(t) = \int_{t-1}^{t} x(s)\,\mathrm{d}s$ with $x = s^2$ on $(0,3)$:

interval	window $(t-1,t)$	$y(t)$
$0 \le t \le 1$	$(0,t)$	$\dfrac{t^3}{3}$
$1 \le t \le 3$	inside $(0,3)$	$\dfrac{t^3-(t-1)^3}{3} = t^2 - t + \dfrac{1}{3}$
$3 \le t \le 4$	$(t-1,3)$	$\dfrac{27-(t-1)^3}{3}$
else	none	0

Check: continuity at $t = 1$ ($\frac{1}{3}$) and $t = 3$ ($\frac{19}{3}$), $y(4) = 0$, area $= 9 \cdot 1 = 9$.



Answer. $y = \frac{t^3}{3}$ $(0 \le t \le 1)$; $t^2 - t + \frac{1}{3}$ $(1 \le t \le 3)$; $\frac{27-(t-1)^3}{3}$ $(3 \le t \le 4)$; 0 otherwise.

2018–2019 • Q4(b) CSE 311 • TF p.18 MARKS 17

Prove that for an input signal $x(t)$, the output of a linear time-invariant system with impulse response $h(t)$ is given by $y(t) = x(t) * h(t)$. Use *only* time-domain variables and equations.

SOLUTION

Let S denote the system and $h(t) = S\{\delta(t)\}$ (definition of the impulse response).

(i) *Sifting.* Any input is a continuum of impulses: $x(t) = \displaystyle\int_{-\infty}^{\infty} x(\tau)\,\delta(t-\tau)\,\mathrm{d}\tau$.

(ii) *Time invariance.* $S\{\delta(t-\tau)\} = h(t-\tau)$.

(iii) *Homogeneity.* The weight $x(\tau)\,\mathrm{d}\tau$ does not depend on t, so $S\{x(\tau)\,\mathrm{d}\tau\,\delta(t-\tau)\} = x(\tau)\,\mathrm{d}\tau\,h(t-\tau)$.

(iv) *Additivity (the integral is a limit of sums).*

$$y(t) = S\{x(t)\} = \int_{-\infty}^{\infty} x(\tau)\,h(t-\tau)\,\mathrm{d}\tau = (x*h)(t).$$

Only linearity (steps 3–4) and time invariance (step 2) were used, so the result holds for every LTI system. (Discrete time: $x[n] = \sum_k x[k]\delta[n-k] \Rightarrow y[n] = \sum_k x[k]h[n-k]$.)

Answer. $y(t) = \int x(\tau)h(t-\tau)\,\mathrm{d}\tau = x * h$, from sifting, time invariance and linearity.

05

Causality, Stability and Impulse Response

ANIK SIR • LECTURE 4

2023–2024 • Q3(b) CSE 219 • 22 batch p.2 MARKS 15

The following are the impulse responses of continuous-time LTI systems. Determine whether each system is causal and/or stable. Justify your answers. (i) $h(t) = e^{-4t}u(3-t)$ (ii) $h(t) = e^{-6|t|}$ (iii) $h(t) = te^{-t}u(t)$

SOLUTION

Causal $\Leftrightarrow h(t) = 0$ for $t < 0$. Stable $\Leftrightarrow \int |h|\,\mathrm{d}t < \infty$.

(i) $u(3-t) = 1$ for $t \le 3$, so $h \neq 0$ for all $t \le 3$, in particular for $t < 0$: *not causal*. $\int_{-\infty}^{3} e^{-4t}\,\mathrm{d}t = \infty$ (the exponential grows as $t \to -\infty$): *not stable*.

(ii) $h \neq 0$ for $t < 0$: *not causal*. $\int e^{-6|t|}\,\mathrm{d}t = 2\int_0^\infty e^{-6t}\,\mathrm{d}t = \frac{1}{3} < \infty$: *stable*.

(iii) $h = 0$ for $t < 0$: *causal*. $\int_0^\infty te^{-t}\,\mathrm{d}t = 1 < \infty$: *stable*.

	causal	stable
(i) $e^{-4t}u(3-t)$	no	no
(ii) $e^{-6\lvert t\rvert}$	no	yes
(iii) $te^{-t}u(t)$	yes	yes

Answer. (i) neither; (ii) stable but non-causal; (iii) causal and stable.

2023–2024 • Q2(b) EEE 211 • EEE p.2 MARKS 17

Using a time-domain technique, find the impulse response of the LC circuit shown in the figure (input voltage $x(t)$ applied through the series inductor L to the capacitor C, output $y(t)$ the capacitor voltage).

SOLUTION

KVL: $x = L\dfrac{\mathrm{d}i}{\mathrm{d}t} + y$, with $i = C\dfrac{\mathrm{d}y}{\mathrm{d}t}$, so

$$LC\,y'' + y = x, \qquad \omega_0 = \frac{1}{\sqrt{LC}}.$$

For $x = \delta(t)$ the system is initially relaxed, so $h(t) = 0$ for $t < 0$. For $t > 0$: $LC\,h'' + h = 0 \Rightarrow h = A\cos\omega_0 t + B\sin\omega_0 t$. Integrating the equation over $(0^-, 0^+)$ gives the jump conditions

$$h(0^+) = 0, \qquad LC\,h'(0^+) = 1 \;\Rightarrow\; h'(0^+) = \omega_0^2.$$

Hence $A = 0$ and $B\omega_0 = \omega_0^2 \Rightarrow B = \omega_0$.

Answer. $h(t) = \dfrac{1}{\sqrt{LC}} \sin\!\left(\dfrac{t}{\sqrt{LC}}\right) u(t)$ — an undamped oscillation (marginally stable: $\int |h| = \infty$).

2022–2023 • Q8(b) CSE 219 • TF p.5 MARKS 10

Determine whether the continuous-time LTI systems with the following impulse responses are causal and/or stable: $h_1(t) = e^{2t}u(-2-t), \quad h_2(t) = te^{-t}u(t)$.

SOLUTION

h_1: $u(-2-t) = 1 \Leftrightarrow t \le -2$, so $h_1 \neq 0$ for $t \le -2 < 0$: *non-causal*. $\int_{-\infty}^{-2} e^{2t}\,\mathrm{d}t = \frac{1}{2}e^{-4} < \infty$: *stable*.

h_2: zero for $t < 0$: *causal*; $\int_0^{\infty} te^{-t}\,\mathrm{d}t = 1 < \infty$: *stable*.

Answer. h_1: stable but not causal. h_2: causal and stable.

2020–2021 • Q6(c) EEE 211 • EEE p.20 MARKS 10

Consider a series RC circuit as an LTI system with an input voltage $x(t)$. The voltage across the resistor, $v_R(t)$, is the output. Assume the circuit is initially relaxed. Using time-domain analysis, determine the impulse response $h(t)$ of the circuit.

SOLUTION

Let v_C be the capacitor voltage: $x = v_R + v_C$, $i = C\dot{v}_C = v_R/R$, so $RC\,\dot{v}_C + v_C = x$ and $v_R = x - v_C$.

For $x = \delta(t)$: $\dot{v}_C + \dfrac{v_C}{RC} = \dfrac{1}{RC}\delta(t)$ with $v_C(0^-) = 0$, so v_C jumps to $1/RC$ and decays:

$$h_C(t) = \frac{1}{RC} e^{-t/RC} u(t).$$

Hence

$$h_R(t) = \delta(t) - h_C(t) = \delta(t) - \frac{1}{RC} e^{-t/RC} u(t).$$

The resistor passes the impulse instantly, then the discharging capacitor drives a negative current: the circuit is a first-order high-pass filter. $\int |h_R|\,\mathrm{d}t = 1 + 1 = 2 < \infty$ (stable), $h_R = 0$ for $t < 0$ (causal).

Answer. $h(t) = \delta(t) - \dfrac{1}{RC} e^{-t/RC} u(t)$.

06

Orthogonality and Fourier Series Coefficients

ANIK SIR • LECTURES 5–6

Reminder: $x(t) = \sum a_k e^{jk\omega_0 t},\ a_k = \frac{1}{T}\int_T x(t)e^{-jk\omega_0 t}\,dt,\ \omega_0 = 2\pi/T,\ a_0 = \text{average value}.$

2023–2024 • Q4(a)–(d) CSE 219 • 22 batch p.2 MARKS 30

Let $x(t)$ be the odd square wave $x(t) = A$ for $0 < t < \frac{T_0}{2}$ and $x(t) = -A$ for $-\frac{T_0}{2} < t < 0$, with $x(t+T_0) = x(t)$. Define $y(t) = x\left(t - \frac{T_0}{6}\right)$. (a) Show that $y(t)$ still has half-wave symmetry. (b) Find the complex Fourier series coefficients Y_k from those of $x(t)$. (c) Draw the magnitude and phase spectra of Y_k (up to $k = 5$ is sufficient). (d) Compare the trigonometric Fourier series of $x(t)$ and $y(t)$.

SOLUTION

Here $\omega_0 = 2\pi/T_0$.

Coefficients of x.

$$X_k = \frac{1}{T_0}\left[\int_0^{T_0/2} Ae^{-jk\omega_0 t}\,dt - \int_{-T_0/2}^{0} Ae^{-jk\omega_0 t}\,dt\right] = \frac{A\left(1-(-1)^k\right)}{jk\pi} = \begin{cases} \dfrac{2A}{jk\pi}, & k \text{ odd}, \\ 0, & k \text{ even}. \end{cases}$$

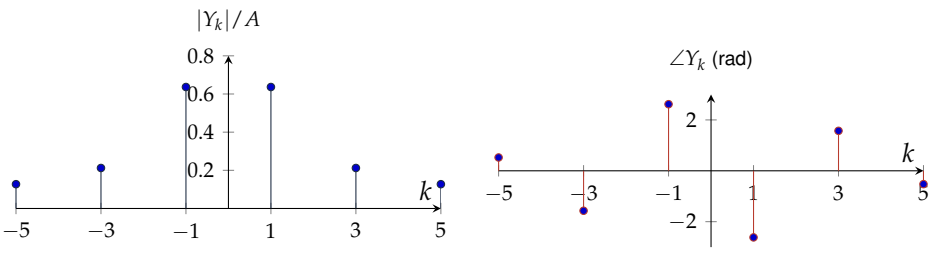
(a) Half-wave symmetry means $f(t + \frac{T_0}{2}) = -f(t)$. For y: $y(t+\frac{T_0}{2}) = x\left(t+\frac{T_0}{2}-\frac{T_0}{6}\right) = -x\left(t-\frac{T_0}{6}\right) = -y(t)$, because x itself has this property. A time shift never destroys half-wave symmetry. Equivalently $Y_k = 0$ for even k.

(b) Time-shift property: $Y_k = X_k e^{-jk\omega_0 T_0/6} = X_k e^{-jk\pi/3}$,

$$Y_k = \frac{2A}{jk\pi}e^{-jk\pi/3} \quad (k \text{ odd}), \qquad Y_k = 0 \quad (k \text{ even}).$$

(c) $|Y_k| = \dfrac{2A}{\pi|k|}$ and, for $k > 0$ odd, $\angle Y_k = -\dfrac{\pi}{2} - \dfrac{k\pi}{3}$ (the phase is odd in k: $Y_{-k} = Y_k^*$):

k	±1	±2	±3	±4	±5	0
$\lvert Y_k\rvert/A$	0.637	0	0.212	0	0.127	0
$\angle Y_k$ $(k>0)$	$-\frac{5\pi}{6}$	–	$\frac{\pi}{2}$	–	$-\frac{\pi}{6}$	–



(d) Real form: $x(t) = \sum_{k\ \text{odd}>0} \dfrac{4A}{\pi k}\sin(k\omega_0 t)$ and, with the shift,

$$y(t) = \sum_{k\ \text{odd}>0} \frac{4A}{\pi k}\sin\!\Big(k\omega_0 t - \frac{k\pi}{3}\Big).$$

Both contain only odd harmonics with identical amplitudes $\frac{4A}{\pi k}$; the shift changes only the phase, by $-k\pi/3$ for harmonic k (linear in k: every harmonic is delayed by the same $T_0/6$).

Answer. $Y_k = \dfrac{2A}{jk\pi}e^{-jk\pi/3}$ for odd k (0 for even k); $|Y_k| = \frac{2A}{\pi|k|}$; amplitudes unchanged, phases shifted by $-k\pi/3$.

2023–2024 • Q7(a) EEE 211 • EEE p.4–5 MARKS 10

Show that the three functions in the figure are pair-wise orthogonal over the interval $(-1,2)$, and calculate the value of A that makes these signals orthonormal to each other. ($\phi_1 = -A$ on $(-1,0)$, $+A$ on $(0,1)$; $\phi_2 = A$ on $(-1,1)$; $\phi_3 = 1$ on $(1,2)$.)

SOLUTION

$$\int \phi_1\phi_2\,\mathrm{d}t = \int_{-1}^{0}(-A)(A)\,\mathrm{d}t + \int_{0}^{1}(A)(A)\,\mathrm{d}t = -A^2 + A^2 = 0.$$

ϕ_3 is nonzero only on $(1,2)$ where $\phi_1 = \phi_2 = 0$, so $\int \phi_1\phi_3 = \int \phi_2\phi_3 = 0$: the set is pairwise orthogonal. Energies: $E_1 = \int \phi_1^2 = 2A^2$, $E_2 = 2A^2$, $E_3 = 1$. Orthonormal requires $E_i = 1$:

$$2A^2 = 1 \;\Rightarrow\; A = \frac{1}{\sqrt{2}} \quad (\phi_3 \text{ already has unit energy}).$$

Answer. Pairwise orthogonal; orthonormal for $A = 1/\sqrt{2}$.

2022–2023 • Q5(a) EEE 211 • EEE p.9 MARKS 10

A periodic signal $x(t)$ is represented by the trigonometric Fourier series $x(t) = 2 + 3\cos 2t + 4\sin 2t + 2\sin(3t + 30^\circ) - \cos(7t + 150^\circ)$. Express this series as a compact trigonometric Fourier series $x(t) = A_0 + \sum A_n \cos\left(\frac{2\pi nt}{T} + \phi_n\right)$. Sketch the amplitude spectrum with proper axis labels.

SOLUTION

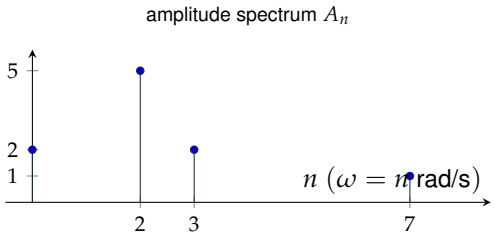
Convert each term to a cosine with phase:

$$\begin{aligned} 3\cos 2t + 4\sin 2t &= 5\cos(2t - 53.13^\circ), \qquad (\sqrt{3^2+4^2} = 5,\ \tan^{-1}\tfrac{4}{3} = 53.13^\circ)\\ 2\sin(3t + 30^\circ) &= 2\cos(3t + 30^\circ - 90^\circ) = 2\cos(3t - 60^\circ),\\ -\cos(7t + 150^\circ) &= \cos(7t + 150^\circ - 180^\circ) = \cos(7t - 30^\circ). \end{aligned}$$

$$x(t) = 2 + 5\cos(2t - 53.13^\circ) + 2\cos(3t - 60^\circ) + \cos(7t - 30^\circ).$$

Fundamental: $\gcd(2,3,7) = 1$ rad/s, so $T = 2\pi$ and $2\pi n/T = n$: the nonzero harmonics are $n = 2, 3, 7$.

n	0	2	3	7
A_n	2	5	2	1
ϕ_n	–	-53.13°	-60°	-30°



Answer. $x = 2 + 5\cos(2t - 53.13°) + 2\cos(3t - 60°) + \cos(7t - 30°)$; $A_0 = 2, A_2 = 5, A_3 = 2, A_7 = 1$.

2022–2023 • Q7(a) EEE 211 • EEE p.11 MARKS 20

For the signal $x(t)$ shown in the figure, find the expression of the exponential Fourier series. (The figure shows a periodic signal of period 2 with cusp-shaped peaks of height 1 at the odd integers and smooth zeros at the even integers.)

NOTE The scanned figure is small. It is read as $x(t) = (1 - |t - 1|)^2$ on $0 \le t \le 2$ repeated with period $T = 2$ (peak 1 at $t = 1$, value 0 with zero slope at $t = 0, 2$). The method is unchanged for another arc shape.

SOLUTION

$T = 2$, $\omega_0 = \pi$. Put $s = t - 1$: one period is the even function $g(s) = (1 - |s|)^2$ on $|s| \le 1$, and $x(t) = g(t - 1)$, so $a_k = (-1)^k a_k^g$.

$$a_0 = \int_0^1 (1 - s)^2 \, ds = \frac{1}{3}, \qquad a_k^g = \frac{1}{2}\int_{-1}^{1} g(s)\cos k\pi s \, ds = \int_0^1 (1 - s)^2 \cos(k\pi s) \, ds.$$

Integrate by parts twice (with $\sin k\pi = 0$): $\int_0^1 (1 - s)^2 \cos(\kappa s) \, ds = \dfrac{2}{\kappa^2}$, $\kappa = k\pi$. Hence

$$a_k = (-1)^k \frac{2}{\pi^2 k^2} \ (k \neq 0), \qquad x(t) = \frac{1}{3} + \sum_{k \neq 0} \frac{2(-1)^k}{\pi^2 k^2} e^{jk\pi t}.$$

(Check $a_1 = -2/\pi^2 = -0.203$ against numerical integration.)

Answer. $a_0 = \frac{1}{3}$, $a_k = \dfrac{2(-1)^k}{\pi^2 k^2}$ $(k \neq 0)$, i.e. $x(t) = \frac{1}{3} + \sum_{k \neq 0} \frac{2(-1)^k}{\pi^2 k^2} e^{jk\pi t}$.

2022–2023 • Q5(d) CSE 219 • TF p.3 MARKS 5

Determine whether the signal $x(t)$ is real-valued and even. Here $x(t)$ is a continuous-time periodic signal presented in its Fourier series representation: $x(t) = \sum_{k=0}^{100} \left(\frac{1}{2}\right)^k e^{jk\frac{2\pi}{50}t}$.

SOLUTION

The Fourier coefficients are $a_k = (\frac{1}{2})^k$ for $0 \le k \le 100$ and $a_k = 0$ otherwise (with $\omega_0 = 2\pi/50$).

- Real $x(t)$ requires $a_{-k} = a_k^*$. For $k \ge 1$: $a_{-k} = 0$ but $a_k^* = (\frac{1}{2})^k \neq 0$. *Not real.*
- Even $x(t)$ requires $a_{-k} = a_k$. Again $a_{-k} = 0 \neq a_k$. *Not even.*

(The signal contains only positive-frequency exponentials, so $x(t)$ is complex.)

Answer. $x(t)$ is neither real-valued nor even.

2022–2023 • Q8(a) EEE 211 • EEE p.11 **MARKS 20**

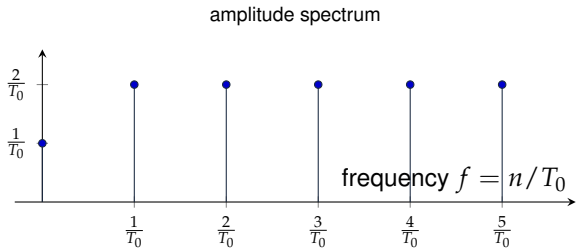
Find the analytical expression of the trigonometric Fourier series (in the form $\delta_{T_0}(t) = A_0 + \sum A_n \cos\left(\frac{2\pi nt}{T} + \phi\right)$) for the impulse train $\delta_{T_0}(t) = \sum_m \delta(t - mT_0)$ depicted in the figure. Sketch the amplitude spectrum with proper x- and y-axis labels.

SOLUTION

Take the period $(-\frac{T_0}{2}, \frac{T_0}{2})$, which contains one impulse at $t = 0$. By sifting,

$$a_k = \frac{1}{T_0}\int_{-T_0/2}^{T_0/2} \delta(t) e^{-jk\omega_0 t}\,\mathrm{d}t = \frac{1}{T_0} \quad \text{for every } k, \qquad \omega_0 = \frac{2\pi}{T_0}.$$

Real coefficients pair up ($a_{-n} = a_n$): $\delta_{T_0}(t) = \dfrac{1}{T_0} + \dfrac{2}{T_0}\sum_{n=1}^{\infty} \cos\left(\dfrac{2\pi nt}{T_0}\right)$, so $A_0 = \frac{1}{T_0}$, $A_n = \frac{2}{T_0}$, $\phi = 0$.



Answer. $\delta_{T_0}(t) = \frac{1}{T_0} + \frac{2}{T_0}\sum_{n\geq 1}\cos\frac{2\pi nt}{T_0}$; $A_0 = \frac{1}{T_0}$, $A_n = \frac{2}{T_0}$ (flat, equal-height lines).

2021–2022 • Q3(b) EEE 211 • EEE p.13 **MARKS 15**

Show that the three functions ψ_1, ψ_2, ψ_3 shown in the figure are pairwise orthogonal over $(-2, 2)$. How can you make these functions an orthonormal set? Express $x(t) = 1$ for $0 \leq t \leq 2$ (0 otherwise) in terms of the orthonormal set. Validate Parseval's relation in this case. ($\psi_1 = -A$ on $(-2,-1)$, A on $(-1,1)$, $-A$ on $(1,2)$; $\psi_2 = -A$ on $(-2,0)$, A on $(0,2)$; $\psi_3 = -A$ on $(-2,2)$.)

SOLUTION

Orthogonality.

$$\int \psi_1\psi_2 = A^2\left[(+1)\cdot 1 + (-1)\cdot 1 + (+1)\cdot 1 + (-1)\cdot 1\right] = 0 \quad \text{(intervals } (-2,-1),(-1,0),(0,1),(1,2)),$$

$$\int \psi_1\psi_3 = -A\int \psi_1 = -A\,(-A + 2A - A) = 0, \qquad \int \psi_2\psi_3 = -A\int \psi_2 = -A(-2A + 2A) = 0.$$

Orthonormalisation. Each ψ_i has energy $E_i = 4A^2$, so $\phi_i = \psi_i/\sqrt{E_i} = \psi_i/(2A)$; with $A = \frac{1}{2}$ the ψ_i themselves are orthonormal.

Expansion of x. With $A = \frac{1}{2}$: $c_i = \int_0^2 x\,\phi_i\,\mathrm{d}t$:

$$c_1 = \int_0^2 \phi_1 = \tfrac{1}{2}(1) + (-\tfrac{1}{2})(1) = 0, \qquad c_2 = \int_0^2 \tfrac{1}{2} = 1, \qquad c_3 = \int_0^2 (-\tfrac{1}{2}) = -1.$$

$$x(t) = \phi_2(t) - \phi_3(t) \quad \text{(check on } (0,2):\ \tfrac{1}{2} + \tfrac{1}{2} = 1;\ \text{on } (-2,0):\ -\tfrac{1}{2} + \tfrac{1}{2} = 0).$$

Parseval. $\int x^2\,\mathrm{d}t = 2$ and $\sum c_i^2 = 0 + 1 + 1 = 2$: equal, so Parseval's relation holds.

Answer. Orthogonal; orthonormal for $A = \frac{1}{2}$; $x = \phi_2 - \phi_3$ ($c_1 = 0$, $c_2 = 1$, $c_3 = -1$); $\int x^2 = 2 = \sum c_i^2$.

2020–2021 • Q2(b) — CSE 311 CSE 311 • TF p.12 MARKS 17

Find the exponential Fourier series coefficients of $\sin 2\pi t$ and $\cos 2\pi t$. Do you need to integrate over a full cycle to obtain them? Explain.

SOLUTION

Use Euler's formula; no integration is needed ($\omega_0 = 2\pi$, $T = 1$).

$$\cos 2\pi t = \tfrac{1}{2}e^{j2\pi t} + \tfrac{1}{2}e^{-j2\pi t} \;\Rightarrow\; a_{\pm 1} = \tfrac{1}{2}\ (\text{real}),$$

$$\sin 2\pi t = \tfrac{1}{2j}e^{j2\pi t} - \tfrac{1}{2j}e^{-j2\pi t} \;\Rightarrow\; a_1 = \tfrac{1}{2j} = -\tfrac{j}{2},\ \ a_{-1} = \tfrac{j}{2}\ (\text{imaginary}).$$

All other coefficients are zero. The defining integral $a_k = \int_0^1 x(t)e^{-j2\pi kt}\,\mathrm{d}t$ needs a full cycle in general. But for symmetric signals the cosine part (even x, real a_k) or the sine part (odd x, imaginary a_k) can be computed from half a cycle: $a_k = 2\int_0^{1/2} x\cos k\omega_0 t\,\mathrm{d}t$ (even) and $a_k = -2j\int_0^{1/2} x\sin k\omega_0 t\,\mathrm{d}t$ (odd).

Answer. $\cos 2\pi t$: $a_{\pm 1} = \frac{1}{2}$; $\sin 2\pi t$: $a_{\pm 1} = \mp\frac{j}{2}$. By inspection no integral is needed; with symmetry, half a cycle suffices.

2020–2021 • Q3(b) — CSE 311 CSE 311 • TF p.12 MARKS 11

State the Dirichlet conditions. Are they necessary for the existence of a Fourier series? Give an example.

SOLUTION

A periodic $x(t)$ has a Fourier series converging to $x(t)$ at every point of continuity (and to the mid-value at jumps) if, over one period: (1) $\int_T |x(t)|\,\mathrm{d}t < \infty$ (absolutely integrable); (2) x has a finite number of maxima and minima; (3) x has a finite number of finite discontinuities.
These conditions are *sufficient, not necessary*. Violating them does not forbid a useful representation: $\cos\omega_0 t$ is fine, and an impulse train violates condition 1 yet has $a_k = 1/T$ in the generalised sense. Examples that genuinely fail: $\tan(2\pi t)$ (not absolutely integrable) and $\sin(2\pi/t)$ (infinitely many extrema near $t = 0$).

Answer. Dirichlet conditions: absolutely integrable, finitely many extrema and finite jumps per period — sufficient but not necessary.

2020–2021 • Q7(a) EEE 211 • EEE p.20 MARKS 20

For the periodic signals in the figure ($x(t)$ a triangle wave with peak 1 at $t = 0, \pm 4, \ldots$ and valley -1 at $t = \pm 2, \ldots$; $y(t)$ a rectangular wave equal to 4 on $(0,2)$, $(4,6), \ldots$ and 0 elsewhere), first compute the exponential Fourier series coefficients of $x(t)$. Then, using the properties of Fourier series, determine the Fourier series coefficients of $y(t)$. Also draw the amplitude spectra of $x(t)$ and $y(t)$.

SOLUTION

$T = 4$, $\omega_0 = \pi/2$. One period of x: $x = 1 - t$ on $(0,2)$... equivalently $x = -1\cdot$ slope: $x' = -1$ on $(0,2)$, $x' = +1$ on $(2,4)$.
Second derivative. The slope changes by -2 at $t = 0$ and by $+2$ at $t = 2$ (period $(0,4)$ counting $t = 0$): $x'' = -2\delta(t) + 2\delta(t-2)$ per period. Its coefficients: $\frac{1}{4}\left[-2 + 2e^{-jk\pi}\right] = \frac{(-1)^k - 1}{2}$. With $(jk\omega_0)^2 a_k = -\frac{k^2\pi^2}{4}a_k$:

$$a_0 = 0, \qquad a_k = \frac{4}{\pi^2 k^2}\ (k \text{ odd}), \qquad a_k = 0\ (k \text{ even}).$$

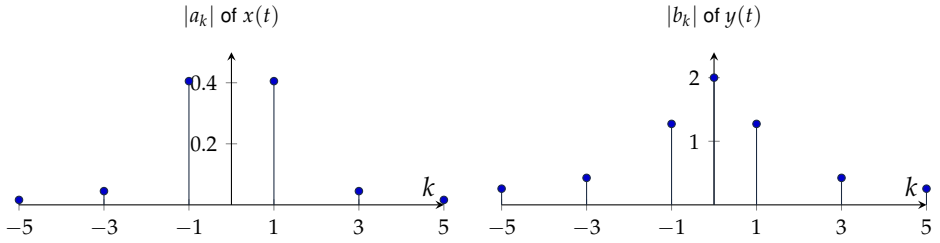
Coefficients of y. Compare: $y = 4$ on $(0,2)$, 0 on $(2,4)$, and $x' = -1$ on $(0,2)$, $+1$ on $(2,4)$. Hence

$$y(t) = 2 - 2\,x'(t).$$

Differentiation property: $x' \leftrightarrow jk\omega_0 a_k$. So $b_0 = 2$ and

$$b_k = -2jk\frac{\pi}{2}\cdot\frac{4}{\pi^2k^2} = -\frac{4j}{\pi k} \quad (k \text{ odd}), \qquad b_k = 0 \ (k \text{ even},\ k \neq 0).$$

Direct check: $b_k = \frac{1}{4}\int_0^2 4e^{-jk\pi t/2}\,\mathrm{d}t = \frac{2(1-(-1)^k)}{jk\pi} = -\frac{4j}{k\pi}$ for odd k (checked).



Answer. x: $a_0 = 0$, $a_k = \frac{4}{\pi^2k^2}$ (odd k). $y = 2 - 2x'$: $b_0 = 2$, $b_k = -\frac{4j}{\pi k}$ (odd k), 0 for even $k \neq 0$.

2019–2020 • Q7(a) EEE 211 • EEE p.25 MARKS 15

Find the exponential Fourier series for the two impulse trains shown in the figure: (a) unit impulses at every integer $t = n$; (b) impulses of strength $+1$ at odd integers and -1 at even integers.

SOLUTION

(a) $T = 1$, $\omega_0 = 2\pi$. One impulse per period: $a_k = \frac{1}{1}\int_{-1/2}^{1/2}\delta(t)e^{-jk2\pi t}\,\mathrm{d}t = 1$ for all k:

$$x_1(t) = \sum_{k=-\infty}^{\infty} e^{j2\pi kt}.$$

(b) The pattern $-1, +1$ repeats with period $T = 2$ ($\omega_0 = \pi$). Over one period $(-\frac{1}{2}, \frac{3}{2})$: $-\delta(t) + \delta(t-1)$:

$$a_k = \tfrac{1}{2}\left[-1 + e^{-jk\pi}\right] = \tfrac{1}{2}\left[(-1)^k - 1\right] = \begin{cases} -1, & k \text{ odd}, \\ 0, & k \text{ even}, \end{cases} \qquad x_2(t) = -\sum_{k\ \text{odd}} e^{jk\pi t} = -2\sum_{k=1,3,\ldots}\cos k\pi t.$$

Answer. (a) $a_k = 1$ for all k ($T = 1$). (b) $T = 2$: $a_k = -1$ for odd k, 0 for even k.

2019–2020 • Q1(a) — CSE 311 CSE 311 • TF p.15 MARKS 17

Justify why the signal $\cos 2\pi f_0 t$ (odd roll) / $\sin 2\pi f_0 t$ (even roll) can be represented using a Fourier series. Show whether the coefficients of the exponential Fourier series of your assigned signal are real or imaginary.

SOLUTION

Why a Fourier series exists. The signal is periodic with $T_0 = 1/f_0$, bounded, absolutely integrable over a period ($\int_{T_0}|x| < \infty$), has a finite number of extrema (one maximum and one minimum per period) and no discontinuities — the Dirichlet conditions hold. It is even simpler: it is already a sum of harmonic exponentials.

Coefficients.

$$\cos 2\pi f_0 t = \tfrac{1}{2}e^{j2\pi f_0 t} + \tfrac{1}{2}e^{-j2\pi f_0 t} \Rightarrow a_{\pm 1} = \tfrac{1}{2} \text{ (real; the signal is even)},$$

$$\sin 2\pi f_0 t = \tfrac{1}{2j}e^{j2\pi f_0 t} - \tfrac{1}{2j}e^{-j2\pi f_0 t} \Rightarrow a_{\pm 1} = \mp\tfrac{j}{2} \text{ (purely imaginary; the signal is odd)}.$$

In general $a_k = \frac{1}{T}\int x\cos k\omega_0 t\,\mathrm{d}t - \frac{j}{T}\int x\sin k\omega_0 t\,\mathrm{d}t$: even signals have only the real part, odd signals only the imaginary part.

Answer. Cosine: $a_{\pm 1} = \frac{1}{2}$ (real). Sine: $a_{\pm 1} = \mp \frac{j}{2}$ (imaginary). Both satisfy the Dirichlet conditions.

2019–2020 • Q8(b) EEE 211 • EEE p.26 MARKS 20

The trigonometric Fourier series of the periodic signal $x(t)$ with period T can be written as $x(t) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos \frac{2n\pi t}{T} + b_n \sin \frac{2n\pi t}{T}\right)$. For the triangular signal shown in the figure, calculate a_0, a_n and b_n.

NOTE The figure is read as a triangular wave of period $T = 2$ and amplitude $\pm A$: zero crossings at the integers, peak $+A$ at $t = \frac{1}{2}$, valley $-A$ at $t = -\frac{1}{2}$ (an odd function). The same method applies to other placements of the peaks.

SOLUTION

$T = 2$, $\omega_0 = \pi$. The waveform is odd: $x = 2At$ for $|t| \le \frac{1}{2}$ and $x = 2A(1-t)$ for $\frac{1}{2} \le t \le \frac{3}{2}$. Hence $a_0 = 0$ (zero average) and $a_n = 0$ (odd symmetry).

$$b_n = \frac{2}{T}\int_{-1}^{1} x \sin n\pi t \, dt = 2\Big[\int_0^{1/2} 2At \sin n\pi t \, dt + \int_{1/2}^{1} 2A(1-t)\sin n\pi t \, dt\Big].$$

Using $\int_0^{1/2} t \sin n\pi t \, dt = \dfrac{\sin(n\pi/2)}{n^2\pi^2} - \dfrac{\cos(n\pi/2)}{2n\pi}$ and the symmetric second integral, the cosine terms cancel and

$$b_n = \frac{8A}{n^2\pi^2}\sin\frac{n\pi}{2} = \begin{cases} \dfrac{8A(-1)^{(n-1)/2}}{\pi^2 n^2}, & n \text{ odd}, \\ 0, & n \text{ even}. \end{cases}$$

$$x(t) = \frac{8A}{\pi^2}\Big[\sin \pi t - \tfrac{1}{9}\sin 3\pi t + \tfrac{1}{25}\sin 5\pi t - \cdots\Big].$$

(Numerical check: $b_1 = 8A/\pi^2$.)

Answer. $a_0 = 0$, $a_n = 0$, $b_n = \dfrac{8A}{\pi^2 n^2}\sin\dfrac{n\pi}{2}$ (nonzero for odd n only).

2018–2019 • Q1(a) EEE 211 • EEE p.27 MARKS 10

What is the condition for two signals to be orthogonal? Are e^{j2t} and e^{j4t} orthogonal over the interval $[0, \pi]$?

SOLUTION

Two (complex) signals ϕ_1, ϕ_2 are orthogonal over (t_1, t_2) if $\int_{t_1}^{t_2} \phi_1(t)\,\phi_2^*(t)\,dt = 0$ (for real signals $\int \phi_1\phi_2 = 0$).

$$\int_0^{\pi} e^{j2t}\left(e^{j4t}\right)^* dt = \int_0^{\pi} e^{-j2t}\,dt = \frac{1 - e^{-j2\pi}}{2j} = 0.$$

Each has energy $\int_0^{\pi} 1\,dt = \pi$, so $\frac{1}{\sqrt{\pi}}e^{j2t}$, $\frac{1}{\sqrt{\pi}}e^{j4t}$ is an orthonormal pair.

Answer. Orthogonality: $\int \phi_1\phi_2^*\,dt = 0$. Yes, e^{j2t} and e^{j4t} are orthogonal on $[0, \pi]$.

2018–2019 • Q1(b) EEE 211 • EEE p.27 MARKS 10

Consider the signal $x(t) = 2 + \frac{1}{2}\cos(t + 45^\circ) + 2\cos 3t - \sin(4t + 30^\circ)$. Find its exponential Fourier series and draw the magnitude and phase spectra.

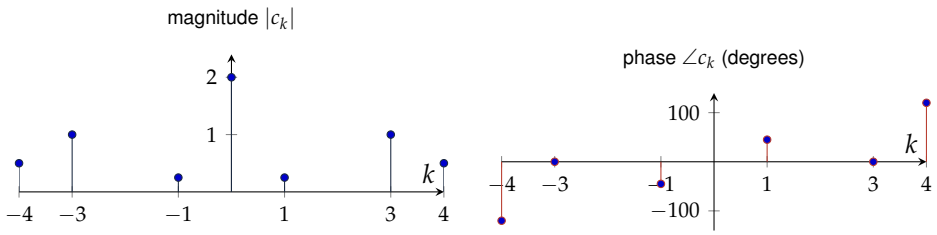
SOLUTION

Fundamental: $\omega_0 = \gcd(1,3,4) = 1$. Write $-\sin(4t+30^\circ) = \cos(4t+120^\circ)$ and $A\cos(\omega t+\theta) = \frac{A}{2}e^{j\theta}e^{j\omega t} + \frac{A}{2}e^{-j\theta}e^{-j\omega t}$:

$$c_0 = 2, \quad c_{\pm1} = \tfrac{1}{4}e^{\pm j45^\circ}, \quad c_{\pm3} = 1, \quad c_{\pm4} = \tfrac{1}{2}e^{\pm j120^\circ}, \quad c_{\pm2} = 0.$$

$$x(t) = 2 + \tfrac{1}{4}e^{j45^\circ}e^{jt} + \tfrac{1}{4}e^{-j45^\circ}e^{-jt} + e^{j3t} + e^{-j3t} + \tfrac{1}{2}e^{j120^\circ}e^{j4t} + \tfrac{1}{2}e^{-j120^\circ}e^{-j4t}.$$

k	0	±1	±3	±4	others
$\lvert c_k\rvert$	2	$\frac{1}{4}$	1	$\frac{1}{2}$	0
$\angle c_k$	0	$\pm45^\circ$	0	$\pm120^\circ$	–



Answer. $c_0 = 2$, $c_{\pm1} = \frac{1}{4}e^{\pm j45^\circ}$, $c_{\pm3} = 1$, $c_{\pm4} = \frac{1}{2}e^{\pm j120^\circ}$.

REPEATED ×2 Same question in 2 papers — stacked below, solved once.

2022–2023 • Q1(a) — CSE 311 CSE 311 • TF p.6 **MARKS 17**

Show that the coefficients of the exponential Fourier series of an even-symmetric periodic signal are real and those of an odd-symmetric periodic signal are imaginary (illustrate with $\cos 2\pi f_0 t$ and $\sin 2\pi f_0 t$). Also show that the coefficients can be computed by integrating the signal over the half cycle only.

2018–2019 • Q2(b) — CSE 311 CSE 311 • TF p.18 **MARKS 15**

Show that the coefficients of the exponential Fourier series of an even symmetric and an odd symmetric periodic signal are real and imaginary, respectively. Also show that the coefficients can be calculated by integrating the signal over the half cycle only.

SOLUTION

Split the defining integral over a symmetric period $(-\frac{T}{2}, \frac{T}{2})$ using $e^{-jk\omega_0 t} = \cos k\omega_0 t - j\sin k\omega_0 t$:

$$a_k = \frac{1}{T}\int_{-T/2}^{T/2} x(t)\cos k\omega_0 t\,\mathrm{d}t - \frac{j}{T}\int_{-T/2}^{T/2} x(t)\sin k\omega_0 t\,\mathrm{d}t.$$

Even signal $(x(-t) = x(t))$: $x\sin k\omega_0 t$ is odd, so its integral over the symmetric interval vanishes; $x\cos k\omega_0 t$ is even:

$$a_k = \frac{2}{T}\int_0^{T/2} x(t)\cos k\omega_0 t\,\mathrm{d}t \quad \text{(real, and } a_{-k} = a_k\text{)}.$$

Odd signal $(x(-t) = -x(t))$: $x\cos k\omega_0 t$ is odd and vanishes; $x\sin k\omega_0 t$ is even:

$$a_k = -\frac{2j}{T}\int_0^{T/2} x(t)\sin k\omega_0 t\,\mathrm{d}t \quad \text{(imaginary, } a_{-k} = -a_k,\ a_0 = 0\text{)}.$$

In both cases only the half cycle $(0, T/2)$ is integrated. Examples: $\cos 2\pi f_0 t$ gives $a_{\pm1} = \frac{1}{2}$ (real); $\sin 2\pi f_0 t$ gives $a_{\pm1} = \mp\frac{j}{2}$ (imaginary).

Answer. Even $\Rightarrow a_k = \frac{2}{T}\int_0^{T/2} x\cos k\omega_0 t\,\mathrm{d}t$ (real); odd $\Rightarrow a_k = -\frac{2j}{T}\int_0^{T/2} x\sin k\omega_0 t\,\mathrm{d}t$ (imaginary).

07

Fourier Series through LTI Systems

ANIK SIR • LECTURES 5–7

Method: if $x = \sum a_k e^{jk\omega_0 t}$ then $y = \sum a_k H(jk\omega_0) e^{jk\omega_0 t}$. Find $H(j\omega)$, evaluate it at each harmonic, multiply.

2020–2021 • Q7(b) EEE 211 • EEE p.20 MARKS 15

A system is described by $y'' + 3y' + 2y = x' + x$. For the input $x(t) = 2\cos t + 6\cos 3t$, find the output $y(t)$ and its average power.

SOLUTION

Frequency response. $H(j\omega) = \dfrac{j\omega + 1}{(j\omega)^2 + 3j\omega + 2} = \dfrac{j\omega + 1}{(j\omega + 1)(j\omega + 2)} = \dfrac{1}{2 + j\omega}.$

Each sinusoid. $A\cos(\omega t) \to A|H|\cos(\omega t + \angle H)$.

ω	$H(j\omega)$	$\lvert H\rvert$	$\angle H$
1	$\frac{1}{2+j}$	0.4472	-26.57°
3	$\frac{1}{2+3j}$	0.2774	-56.31°

$$y(t) = 0.894\cos(t - 26.57^\circ) + 1.664\cos(3t - 56.31^\circ).$$

Average power of a sum of sinusoids at distinct frequencies is $\sum \frac{A^2}{2}$:

$$P_y = \tfrac{1}{2}(0.894^2 + 1.664^2) = \tfrac{1}{2}\left(\tfrac{4}{5} + \tfrac{36}{13}\right) = 1.785.$$

Answer. $y(t) = 0.894\cos(t - 26.57^\circ) + 1.664\cos(3t - 56.31^\circ)$; $P_y \approx 1.785$.

2019–2020 • Q5(b) EEE 211 • EEE p.24 MARKS 20

A full-wave rectified signal $x(t) = |\sin t|$ is applied to an RC low-pass circuit with $R = 15\,\Omega$, $C = \frac{1}{5}$ F (output across the capacitor). Find $H(\omega)$ and the output $y(t)$.

SOLUTION

Circuit. $RC = 3$ s, so $H(j\omega) = \dfrac{1}{1 + j3\omega}.$

Input series. $|\sin t|$ has period π, $\omega_0 = 2$.

$$a_0 = \frac{1}{\pi}\int_0^{\pi} \sin t\,dt = \frac{2}{\pi}, \qquad |\sin t| = \frac{2}{\pi} - \frac{4}{\pi}\sum_{k=1}^{\infty} \frac{\cos 2kt}{4k^2-1}.$$

Output. At $\omega = 2k$: $H = \dfrac{1}{1+j6k}$, $|H| = \dfrac{1}{\sqrt{1+36k^2}}$, phase $-\tan^{-1} 6k$. DC passes with $H(0) = 1$:

$$y(t) = \frac{2}{\pi} - \frac{4}{\pi}\sum_{k=1}^{\infty} \frac{\cos\!\left(2kt - \tan^{-1} 6k\right)}{(4k^2-1)\sqrt{1+36k^2}}.$$

First terms: $y \approx 0.637 - 0.0698\cos(2t - 80.5^\circ) - 0.0070\cos(4t - 87.6^\circ) - \cdots$ (the RC filter has smoothed the ripple to about ± 0.07).

Answer. $H(j\omega) = \frac{1}{1+j3\omega}$; $y(t) = \frac{2}{\pi} - \frac{4}{\pi}\sum \frac{\cos(2kt - \tan^{-1} 6k)}{(4k^2-1)\sqrt{1+36k^2}}$.

2018–2019 • Q1(c) EEE 211 • EEE p.27 MARKS 15

A triangular wave of period $T = 4$ and amplitude $A = 4$ is applied to a series $R = 100\,\Omega$, $L = 0.1$ H circuit. Find the power dissipated in R.

NOTE The waveform is read as the zero-mean triangle swinging between -4 and $+4$ (period 4, $\omega_0 = \pi/2$) with the current as the response. If the figure shows a different offset, only a_0 changes.

SOLUTION

Zero-mean triangle of peak A and period T: $|a_k| = \frac{4A}{\pi^2 k^2}$ for odd k (here $\frac{16}{\pi^2 k^2}$). Voltage across the series circuit: $I_k = \dfrac{a_k}{R + jk\omega_0 L}$, and $P = \sum |I_k|^2 R$. With $k\omega_0 L = 0.05\pi k \ll R$ the inductor reactance is tiny (0.157 for $k = 1$ against 100):

$$|I_1| = \frac{1.621}{\sqrt{100^2 + 0.157^2}} \approx 0.01621, \qquad P \approx R\sum_{k\ \text{odd}} \frac{|a_k|^2}{R^2} = \frac{1}{R}\sum_{k\ \text{odd}} |a_k|^2.$$

By Parseval for the triangle, $\sum |a_k|^2 = \frac{1}{T}\int_T x^2\,dt = \frac{A^2}{3} = \frac{16}{3}$, so $P \approx \dfrac{16/3}{100} = 0.0533$ W.

Answer. $P_R \approx \dfrac{A^2/3}{R} = 0.0533$ W (the inductor barely affects the power at these frequencies).

08

Fourier Transform: Pairs and Properties

ANIK SIR • LECTURES 8–10

$X(\omega)=\int x(t)e^{-j\omega t}\,dt,\ x(t)=\frac{1}{2\pi}\int X(\omega)e^{j\omega t}\,d\omega$. Key pairs: $e^{-at}u(t)\leftrightarrow\frac{1}{a+j\omega},\ 1\leftrightarrow 2\pi\delta(\omega),\ \delta(t)\leftrightarrow 1$.

2023–2024 • Q2(a) CSE 219 • 22 batch p.2 MARKS 10

Find the Fourier transform of $x_1(t)=e^{-at}u(t)$, and then of $x_2(t)=e^{-a|t|}$ $(a>0)$.

SOLUTION

$$X_1(\omega)=\int_0^\infty e^{-(a+j\omega)t}\,dt=\frac{1}{a+j\omega}.$$

Write $x_2(t)=e^{-at}u(t)+e^{at}u(-t)=x_1(t)+x_1(-t)$. Time reversal gives $x_1(-t)\leftrightarrow X_1(-\omega)=\frac{1}{a-j\omega}$:

$$X_2(\omega)=\frac{1}{a+j\omega}+\frac{1}{a-j\omega}=\frac{2a}{a^2+\omega^2}.$$

Answer. $X_1=\dfrac{1}{a+j\omega};\ X_2=\dfrac{2a}{a^2+\omega^2}$ (real and even, as x_2 is real and even).

2022–2023 • Q6(a)–(b) CSE 219 • TF p.3 MARKS 15

Let $f(t)=e^{-at}u(t)$, $a>0$. Define $f(-t)$, and find the Fourier transforms of $f(t)$ and $f(-t)$.

SOLUTION

$f(-t)=e^{at}u(-t)$: it equals e^{at} for $t<0$ and 0 for $t>0$ (the mirror image of f).

$$F(\omega)=\frac{1}{a+j\omega},\qquad \mathcal{F}\{f(-t)\}=\int_{-\infty}^0 e^{(a-j\omega)t}\,dt=\frac{1}{a-j\omega}=F(-\omega).$$

Answer. $F(\omega)=\frac{1}{a+j\omega};\ \mathcal{F}\{f(-t)\}=\frac{1}{a-j\omega}$.

2022–2023 • Q6(c)–(d) CSE 219 • TF p.3 MARKS 15

Write $\operatorname{sign}(t)$ in terms of $f(t)$ and $f(-t)$, and find its Fourier transform using the properties of the Fourier transform.

SOLUTION

$\operatorname{sign}(t) = \lim_{a\to 0}\big[f(t) - f(-t)\big]$ since $e^{-a|t|} \to 1$ and the two halves carry opposite signs. By linearity,

$$\mathcal{F}\{\operatorname{sign} t\} = \lim_{a\to 0}\left[\frac{1}{a+j\omega} - \frac{1}{a-j\omega}\right] = \lim_{a\to 0}\frac{-2j\omega}{a^2+\omega^2} = \frac{-2j}{\omega} = \frac{2}{j\omega}.$$

(Check: sign is real and odd $\Rightarrow$ purely imaginary, odd transform.) With $u(t) = \frac{1}{2}[1 + \operatorname{sign} t]$ this gives $u(t) \leftrightarrow \pi\delta(\omega) + \frac{1}{j\omega}$.

Answer. $\operatorname{sign}(t) \leftrightarrow \dfrac{2}{j\omega}$.

2019–2020 • Q1(b) — CSE 311 CSE 311 • TF p.15 MARKS 18

Find the Fourier transform of $\cos 2\pi f_0 t$ and $\sin 2\pi f_0 t$. Are the Dirichlet conditions necessary for the existence of the Fourier transform?

SOLUTION

Use $\mathcal{F}\{e^{j2\pi f_0 t}\} = \delta(f - f_0)$ (from $\int e^{-j2\pi(f-f_0)t}\,\mathrm{d}t = \delta(f - f_0)$) and Euler's formula:

$$\cos 2\pi f_0 t \;\leftrightarrow\; \tfrac{1}{2}\big[\delta(f-f_0) + \delta(f+f_0)\big], \qquad \sin 2\pi f_0 t \;\leftrightarrow\; \tfrac{1}{2j}\big[\delta(f-f_0) - \delta(f+f_0)\big].$$

Neither signal is absolutely integrable, so the Dirichlet condition $\int |x|\,\mathrm{d}t < \infty$ fails; yet the transform exists as impulses. The conditions are *sufficient, not necessary*: the transform of a periodic or constant signal exists in the generalised (distribution) sense.

Answer. cos: impulses of weight $\frac{1}{2}$ at $\pm f_0$ (real, even). sin: $\pm\frac{1}{2j}$ at $\pm f_0$ (imaginary, odd). Dirichlet conditions are sufficient only.

2019–2020 • Q2(a) — CSE 311 CSE 311 • TF p.15 MARKS 24

Show that a time delay of a signal corresponds to a linear phase shift in its spectrum. What happens if a voice signal loses the linear phase in its spectrum?

SOLUTION

$$\mathcal{F}\{x(t-t_0)\} = \int x(t-t_0)e^{-j\omega t}\,\mathrm{d}t \overset{s=t-t_0}{=} e^{-j\omega t_0}\int x(s)e^{-j\omega s}\,\mathrm{d}s = X(\omega)e^{-j\omega t_0}.$$

The magnitude is unchanged, and the phase gets the extra term $-\omega t_0$, a straight line in ω with slope $-t_0$ (the group delay).

Loss of linear phase. Suppose the phase is no longer proportional to ω (for instance a phase that is constant or curved). Then different frequency components are delayed by different amounts, so the waveform is distorted in shape: sharp onsets smear and the peaks move. For a voice signal the ear is fairly insensitive to modest phase distortion, so intelligibility is largely retained but the waveform changes; for images and data pulses the effect is severe.

Answer. $x(t-t_0) \;\leftrightarrow\; X(\omega)e^{-j\omega t_0}$. Without linear phase the components arrive at different times (phase/dispersion distortion), though speech is mostly still intelligible.

2023–2024 • Q6(c) EEE 211 • EEE p.4 MARKS 10

Find the Fourier transform of the unit step $u(t)$. Prove the integration property of the Fourier transform.

SOLUTION

Step. $u(t) = \frac{1}{2} + \frac{1}{2}\operatorname{sign}(t)$, with $1 \leftrightarrow 2\pi\delta(\omega)$ and $\operatorname{sign} \leftrightarrow \frac{2}{j\omega}$:

$$U(\omega) = \pi\delta(\omega) + \frac{1}{j\omega}.$$

Integration property. $y(t) = \int_{-\infty}^{t} x(\tau)\,\mathrm{d}\tau = x(t) * u(t)$. Convolution in time is multiplication in frequency:

$$Y(\omega) = X(\omega)U(\omega) = \frac{X(\omega)}{j\omega} + \pi X(0)\delta(\omega),$$

because $X(\omega)\delta(\omega) = X(0)\delta(\omega)$. If $X(0) = 0$ the second term vanishes.

Answer. $u(t) \leftrightarrow \pi\delta(\omega) + \frac{1}{j\omega}$; $\int_{-\infty}^{t} x \leftrightarrow \frac{X(\omega)}{j\omega} + \pi X(0)\delta(\omega)$.

2020–2021 • Q1(a) EEE 211 • EEE p.17 MARKS 15

Let $X(\omega) = \dfrac{j\omega}{\omega^2 - 3j\omega - 2}$ be the Fourier transform of $x(t)$. Find the Fourier transform of $\dfrac{\mathrm{d}x}{\mathrm{d}t}\cos t$.

SOLUTION

Factor the denominator: $\omega^2 - 3j\omega - 2 = -\left[(j\omega)^2 + 3j\omega + 2\right] = -(j\omega+1)(j\omega+2)$, so $X = -\dfrac{j\omega}{(1+j\omega)(2+j\omega)}$.

Differentiation: $x' \leftrightarrow Z(\omega) = j\omega X(\omega) = \dfrac{\omega^2}{(1+j\omega)(2+j\omega)}$ (since $-(j\omega)^2 = \omega^2$).

Modulation: $z(t)\cos t \leftrightarrow \frac{1}{2}\left[Z(\omega-1) + Z(\omega+1)\right]$.

$$\mathcal{F}\{x'\cos t\} = \frac{1}{2}\left[\frac{(\omega-1)^2}{(1+j(\omega-1))(2+j(\omega-1))} + \frac{(\omega+1)^2}{(1+j(\omega+1))(2+j(\omega+1))}\right].$$

Answer. $\frac{1}{2}\left[Z(\omega-1) + Z(\omega+1)\right]$ with $Z(\omega) = \dfrac{\omega^2}{(1+j\omega)(2+j\omega)}$.

2019–2020 • Q6(b) EEE 211 • EEE p.25 MARKS 20

State and prove the duality property of the Fourier transform.

SOLUTION

Statement. If $x(t) \leftrightarrow X(\omega)$ then $X(t) \leftrightarrow 2\pi\, x(-\omega)$.
Proof. From the inverse transform, $x(t) = \frac{1}{2\pi}\int X(\omega)e^{j\omega t}\,\mathrm{d}\omega$. Replace $t \to -t$:

$$2\pi x(-t) = \int_{-\infty}^{\infty} X(\omega)e^{-j\omega t}\,\mathrm{d}\omega.$$

Now rename the variables, $t \leftrightarrow \omega$: $2\pi x(-\omega) = \int X(t)e^{-j\omega t}\,\mathrm{d}t = \mathcal{F}\{X(t)\}$. ■
Example. $\delta(t) \leftrightarrow 1$ gives $1 \leftrightarrow 2\pi\delta(-\omega) = 2\pi\delta(\omega)$; rect $\leftrightarrow$ sinc gives sinc $\leftrightarrow$ rect.

Answer. $X(t) \leftrightarrow 2\pi x(-\omega)$.

2019–2020 • Q8(a) EEE 211 • EEE p.25–26 **MARKS 15**

Show that the Fourier transform of an impulse train is again an impulse train.

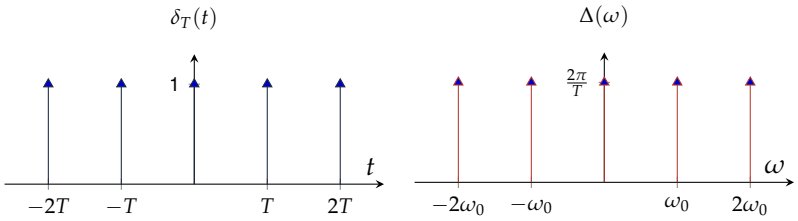
2018–2019 • Q3(a) EEE 211 • EEE p.28 **MARKS 10**

Show that the Fourier transform of an impulse train is an impulse train; sketch both.

SOLUTION

Let $\delta_T(t) = \sum_n \delta(t - nT)$, $\omega_0 = 2\pi/T$. Its Fourier-series coefficients are $a_k = \frac{1}{T}\int_{-T/2}^{T/2} \delta(t) e^{-jk\omega_0 t}\,\mathrm{d}t = \frac{1}{T}$, so $\delta_T(t) = \frac{1}{T}\sum_k e^{jk\omega_0 t}$. Using $e^{jk\omega_0 t} \leftrightarrow 2\pi\delta(\omega - k\omega_0)$:

$$\Delta(\omega) = \frac{2\pi}{T}\sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_0) = \omega_0\,\delta_{\omega_0}(\omega).$$



Spacing T in time becomes spacing $2\pi/T$ in frequency: a narrow train gives a wide spectral spacing and vice versa.

Answer. $\sum \delta(t - nT) \;\leftrightarrow\; \frac{2\pi}{T}\sum \delta(\omega - k\frac{2\pi}{T})$.

2022–2023 • Q2(a) — CSE 311 CSE 311 • TF p.6 **MARKS 20**

Prove that $g(t) * h(t) \leftrightarrow G(f)H(f)$. Show that the output of an LTI system is the convolution of the input and its impulse response.

SOLUTION

$$\mathcal{F}\{g * h\} = \iint g(\tau)h(t - \tau)e^{-j2\pi ft}\,\mathrm{d}\tau\,\mathrm{d}t = \int g(\tau)\Big[\int h(t - \tau)e^{-j2\pi ft}\,\mathrm{d}t\Big]\,\mathrm{d}\tau.$$

Inner integral with $s = t - \tau$: $e^{-j2\pi f\tau}H(f)$. Hence

$$\mathcal{F}\{g * h\} = H(f)\int g(\tau)e^{-j2\pi f\tau}\,\mathrm{d}\tau = G(f)H(f). \blacksquare$$

LTI output. Write $x(t) = \int x(\tau)\delta(t - \tau)\,\mathrm{d}\tau$. By time invariance the response to $\delta(t - \tau)$ is $h(t - \tau)$; by linearity (superposition of the integral) $y(t) = \int x(\tau)h(t - \tau)\,\mathrm{d}\tau = x * h$.

Answer. $g * h \leftrightarrow GH$; $y = x * h$ from sifting + linearity + time invariance.

09

Fourier Transform: Systems, Energy and Bandwidth

ANIK SIR • LECTURES 11–14

$H(\omega)=\mathcal{F}\{h\},\ Y=XH,$ Parseval $E=\int|x|^2\,\mathrm{d}t=\frac{1}{2\pi}\int|X(\omega)|^2\,\mathrm{d}\omega,\ e^{-at}u(t)\leftrightarrow\frac{1}{a+j\omega},\ \frac{\sin Wt}{\pi t}\leftrightarrow\operatorname{rect}\frac{\omega}{2W}.$

2023–2024 • Q1(c) CSE 219 • 22 batch p.2 MARKS 15

Find the Fourier transform of $\operatorname{rect}(t)$, and then of $\operatorname{rect}\left(\frac{t-\tau}{T}\right)\cos 2\pi f_0 t$.

SOLUTION

$\operatorname{rect}(t)=1$ for $|t|<\frac{1}{2}$: $\displaystyle\int_{-1/2}^{1/2}e^{-j2\pi ft}\,\mathrm{d}t=\frac{\sin\pi f}{\pi f}=\operatorname{sinc}(f).$

Scaling and shifting: $\operatorname{rect}\left(\frac{t-\tau}{T}\right)\leftrightarrow G(f)=T\operatorname{sinc}(fT)\,e^{-j2\pi f\tau}$. Multiplying by $\cos 2\pi f_0 t$ shifts the spectrum to $\pm f_0$ with half amplitude:

$$\mathcal{F}\left\{\operatorname{rect}\left(\frac{t-\tau}{T}\right)\cos 2\pi f_0 t\right\}=\frac{T}{2}\left[\operatorname{sinc}\left((f-f_0)T\right)e^{-j2\pi(f-f_0)\tau}+\operatorname{sinc}\left((f+f_0)T\right)e^{-j2\pi(f+f_0)\tau}\right].$$

Answer. $\operatorname{sinc}(f)$; and $\frac{T}{2}\left[\operatorname{sinc}((f-f_0)T)e^{-j2\pi(f-f_0)\tau}+\operatorname{sinc}((f+f_0)T)e^{-j2\pi(f+f_0)\tau}\right]$.

2023–2024 • Q2(b) CSE 219 • 22 batch p.2 MARKS 20

Using Parseval's theorem, evaluate $\displaystyle I(a)=\int_{-\infty}^{\infty}\frac{f^2}{\left(a^2+(2\pi f)^2\right)^2}\,\mathrm{d}f,\ a>0.$

SOLUTION

Take $y(t)=te^{-at}u(t)\leftrightarrow Y(f)=\dfrac{1}{(a+j2\pi f)^2}$, so $|Y|^2=\dfrac{1}{(a^2+4\pi^2f^2)^2}$. The derivative property gives $y'\leftrightarrow j2\pi fY$, hence

$$|j2\pi fY|^2=4\pi^2f^2|Y|^2\ \Rightarrow\ I(a)=\frac{1}{4\pi^2}\int|j2\pi fY(f)|^2\,\mathrm{d}f=\frac{1}{4\pi^2}\int_{-\infty}^{\infty}|y'(t)|^2\,\mathrm{d}t.$$

With $y'=(1-at)e^{-at}u(t)$:

$$\int_0^{\infty}(1-at)^2e^{-2at}\,\mathrm{d}t=\frac{1}{2a}-\frac{2a}{4a^2}+\frac{2a^2}{8a^3}=\frac{1}{4a}.$$

Answer. $I(a)=\dfrac{1}{4\pi^2}\cdot\dfrac{1}{4a}=\dfrac{1}{16\pi^2a}$. (Check with $a=1$: substitute $u=2\pi f$, $\frac{1}{8\pi^3}\int\frac{u^2}{(1+u^2)^2}\,\mathrm{d}u=\frac{1}{8\pi^3}\cdot\frac{\pi}{2}$.)

REPEATED ×2 **Same question in 2 papers — stacked below, solved once.**

2020–2021 • Q2(c) EEE 211 • EEE p.17 **MARKS 20**

A causal LTI system is described by $y'' + 6y' + 8y = 2x$. Find $H(\omega)$ and the response to $x(t) = te^{-2t}u(t)$.

2018–2019 • Q4(a) EEE 211 • EEE p.29 **MARKS 17**

For the causal system $y'' + 6y' + 8y = 2x$, use the Fourier transform to find $h(t)$ and the output for $x(t) = te^{-2t}u(t)$.

SOLUTION

With $s = j\omega$: $H = \dfrac{2}{s^2+6s+8} = \dfrac{2}{(s+2)(s+4)} = \dfrac{1}{s+2} - \dfrac{1}{s+4}$, so

$$h(t) = \left(e^{-2t} - e^{-4t}\right)u(t) \qquad \text{(causal, stable)}.$$

For $x = te^{-2t}u$: $X = \dfrac{1}{(s+2)^2}$ and $Y = \dfrac{2}{(s+2)^3(s+4)}$. Partial fractions (residues at $s=-2$ with $g(s) = \frac{2}{s+4}$: $g(-2) = 1$, $g'(-2) = -\frac{1}{2}$, $\frac{1}{2}g''(-2) = \frac{1}{4}$; at $s=-4$: $\frac{2}{(-2)^3} = -\frac{1}{4}$):

$$Y = \frac{1/4}{s+2} - \frac{1/2}{(s+2)^2} + \frac{1}{(s+2)^3} - \frac{1/4}{s+4}.$$

Invert with $\frac{1}{(s+a)^n} \leftrightarrow \frac{t^{n-1}}{(n-1)!}e^{-at}u(t)$:

$$y(t) = \left[\tfrac{1}{4}e^{-2t} - \tfrac{1}{2}te^{-2t} + \tfrac{1}{2}t^2e^{-2t} - \tfrac{1}{4}e^{-4t}\right]u(t).$$

Check: $y(0) = \frac{1}{4} - \frac{1}{4} = 0$ (the output is zero at $t=0$, as for a second-order system).

Answer. $H = \dfrac{2}{(j\omega+2)(j\omega+4)}$, $h = (e^{-2t} - e^{-4t})u$; $y = \left[\frac{1}{4}e^{-2t} - \frac{1}{2}te^{-2t} + \frac{1}{2}t^2e^{-2t} - \frac{1}{4}e^{-4t}\right]u(t)$.

2018–2019 • Q2(b) EEE 211 • EEE p.27–28 **MARKS 10**

The signal $x(t) = e^{-\alpha t}u(t)$ is applied to an ideal low-pass system $h(t) = \dfrac{\sin 2t}{\pi t}$. Find $Y(\omega)$. For what value of α does the output contain half the input energy?

SOLUTION

$H(\omega) = 1$ for $|\omega| < 2$ and 0 elsewhere, $X = \dfrac{1}{\alpha + j\omega}$, so $Y(\omega) = \dfrac{1}{\alpha + j\omega}$ for $|\omega| < 2$ and 0 otherwise.

Input energy: $E_x = \int_0^\infty e^{-2\alpha t}\,dt = \dfrac{1}{2\alpha}$. Output energy:

$$E_y = \frac{1}{2\pi}\int_{-2}^{2} \frac{d\omega}{\alpha^2+\omega^2} = \frac{1}{\pi\alpha}\tan^{-1}\frac{2}{\alpha}.$$

Setting $E_y = \frac{1}{2}E_x$: $\dfrac{1}{\pi\alpha}\tan^{-1}\dfrac{2}{\alpha} = \dfrac{1}{4\alpha} \Rightarrow \tan^{-1}\dfrac{2}{\alpha} = \dfrac{\pi}{4} \Rightarrow \alpha = 2$.

Answer. $Y = \frac{1}{\alpha+j\omega}$ for $|\omega| < 2$ (else 0); $\alpha = 2$.

2018–2019 • Q2(a) EEE 211 • EEE p.27 MARKS 10

Prove that $E = \int |x(t)|^2\,dt = \dfrac{1}{2\pi}\int |X(\omega)|^2\,d\omega$. Find the energy of $\operatorname{sinc}(t/2)$ (with $\operatorname{sinc} x = \frac{\sin x}{x}$).

SOLUTION

$$\int |x|^2\,dt = \int x(t)\,x^*(t)\,dt = \int x(t)\Big[\frac{1}{2\pi}\int X^*(\omega)e^{-j\omega t}\,d\omega\Big]\,dt = \frac{1}{2\pi}\int X^*(\omega)\Big[\int x(t)e^{-j\omega t}\,dt\Big]\,d\omega = \frac{1}{2\pi}\int |X|^2\,d\omega. \blacksquare$$

$\operatorname{sinc}(t/2) = \dfrac{\sin(t/2)}{t/2} = 2\pi\cdot\dfrac{\sin(t/2)}{\pi t} \leftrightarrow 2\pi\,\operatorname{rect}\dfrac{\omega}{1}$ (height 2π for $|\omega| < \frac{1}{2}$). Hence

$$E = \frac{1}{2\pi}(2\pi)^2(1) = 2\pi.$$

(If the convention $\operatorname{sinc} x = \frac{\sin \pi x}{\pi x}$ is used instead, $E = 2$.)

Answer. Parseval proved; $E = 2\pi$ for $\sin(t/2)/(t/2)$.

2023–2024 • Q6(b) EEE 211 • EEE p.4 MARKS 10

Find the Fourier transform of $\dfrac{\sin(at)}{\pi t}$ and, by Parseval's relation, the energy of the signal.

SOLUTION

$\dfrac{\sin at}{\pi t} \leftrightarrow \operatorname{rect}\dfrac{\omega}{2a}$ (1 for $|\omega| < a$, otherwise 0).

$$E = \frac{1}{2\pi}\int_{-a}^{a} 1\,d\omega = \frac{a}{\pi}.$$

Answer. $X = 1$ for $|\omega| < a$; $E = a/\pi$.

2020–2021 • Q2(b) EEE 211 • EEE p.17 MARKS 7

Find the 3-dB bandwidth of $x(t) = e^{-3t}u(t)$.

SOLUTION

$X = \dfrac{1}{3 + j\omega}$, $|X|^2 = \dfrac{1}{9+\omega^2}$. The 3-dB point is where $|X|^2$ is half its peak value $\frac{1}{9}$: $9 + \omega^2 = 18 \Rightarrow \omega = 3$.

Answer. $B = 3$ rad/s $\left(= \frac{3}{2\pi} \approx 0.477\text{ Hz}\right)$.

2019–2020 • Q6(a) EEE 211 • EEE p.25 MARKS 15

Find the $1/e$ duration of $x_1(t) = Ae^{-t/T}u(t)$ and of $x_1(3t)$ and $x_1(t/2)$.

SOLUTION

The $1/e$ duration is the time at which the signal falls to $1/e$ of its peak: $Ae^{-t/T} = A/e \Rightarrow t = T$.

signal	expression	$1/e$ duration
$x_1(t)$	$Ae^{-t/T}$	T
$x_1(3t)$	$Ae^{-3t/T}$	$T/3$
$x_1(t/2)$	$Ae^{-t/(2T)}$	$2T$

Compressing in time by 3 divides the duration by 3 (and widens the bandwidth by 3).

Answer. $T,\ T/3,\ 2T$.

2021–2022 • Q6(b) EEE 211 • EEE p.15 MARKS 15

For $h(t) = e^{2t}u(-t)$, find the 3-dB bandwidth, the $1/e$ duration, and their product.

SOLUTION

$H(\omega) = \int_{-\infty}^{0} e^{(2-j\omega)t}\,\mathrm{d}t = \dfrac{1}{2-j\omega}$, so $|H|^2 = \dfrac{1}{4+\omega^2}$; half of the peak $\frac{1}{4}$ at $\omega = 2$.
The signal e^{2t} ($t<0$) equals $1/e$ of its peak at $t = -\frac{1}{2}$, so the duration is $\frac{1}{2}$ s.

Answer. $B_{3dB} = 2$ rad/s; duration $\frac{1}{2}$ s; product $= 1$.

2020–2021 • Q2(a) EEE 211 • EEE p.17 MARKS 8

Find the power spectral density of the impulse train $x(t) = \sum_n \delta(t - nT)$.

SOLUTION

x is periodic with $a_k = 1/T$ for all k ($\omega_0 = 2\pi/T$). The power spectral density of a periodic signal is a line spectrum, $S(\omega) = 2\pi\sum_k |a_k|^2\delta(\omega - k\omega_0)$:

$$S_x(\omega) = \frac{2\pi}{T^2}\sum_{k=-\infty}^{\infty}\delta\Big(\omega - \frac{2\pi k}{T}\Big).$$

Each line carries power $1/T^2$ (the area $\frac{1}{2\pi}\int$ of one impulse); the total is infinite, consistent with an impulse train having infinite power.

Answer. $S_x(\omega) = \dfrac{2\pi}{T^2}\sum_k \delta(\omega - k\omega_0)$.

2022–2023 • Q4(a) EEE 211 • EEE p.9 MARKS 15

An AM signal $s(t) = m(t)\cos\omega_c t$ is received by a synchronous demodulator whose local oscillator has a phase error θ: $\cos(\omega_c t + \theta)$, followed by a low-pass filter with gain 2. Find the output.

SOLUTION

$$s(t)\cos(\omega_c t + \theta) = m(t)\cos\omega_c t\cos(\omega_c t + \theta) = \tfrac{1}{2}m(t)\big[\cos\theta + \cos(2\omega_c t + \theta)\big].$$

The low-pass filter removes the $2\omega_c$ term and doubles the rest: $y(t) = m(t)\cos\theta$.

Answer. $y(t) = m(t)\cos\theta$. Full output for $\theta = 0$; zero for $\theta = 90^\circ$ (phase error attenuates, or kills, the signal).

REPEATED ×4 **Same question in 4 papers — stacked below, solved once.**

2022–2023 • Q2(b) — CSE 311 CSE 311 • TF p.6 MARKS 15

Distinguish a distortionless system from an all-pass system; how can you test for distortionless transmission?

2021–2022 • Q2(a) — CSE 311 CSE 311 • TF p.8 MARKS 20

All-pass versus distortionless systems; frequency response; why are ideal filters unrealizable?

2020–2021 • Q4(a) — CSE 311 CSE 311 • TF p.12 MARKS 20

Characteristics of a distortionless system and their physical meaning.

2018–2019 • Q1(b) — CSE 311 CSE 311 • TF p.18 MARKS 14

Distortionless transmission; is an all-pass system distortionless?

SOLUTION

Distortionless: the output is a scaled, delayed copy of the input, $y(t) = Kx(t - t_d)$. Then

$$H(\omega) = Ke^{-j\omega t_d}: \qquad |H(\omega)| = K \text{ (constant gain)}, \qquad \angle H = -\omega t_d \text{ (linear phase)}.$$

Physical meaning: every frequency is amplified equally (no amplitude distortion) and delayed by the same t_d (group delay $-\frac{\mathrm{d}\angle H}{\mathrm{d}\omega} = t_d$ constant; no phase/dispersion distortion).

All-pass: $|H(\omega)| = 1$ for all ω but the phase is arbitrary. It passes all frequencies with equal gain, yet if the phase is not linear the components are delayed unequally and the waveform changes. So an all-pass system is distortionless *only if* its phase is linear ($H = e^{-j\omega t_d}$); in general, no.

Test: measure $|H|$ and $\angle H$ (or the group delay) over the signal band; both must be flat/linear. Equivalently, send a pulse and check that only its size and position change.

Ideal filters: an ideal low-pass has $h(t) = \dfrac{\sin Wt}{\pi t}$, which is nonzero for $t < 0$: non-causal, so it cannot be built (it would need the input before it arrives). Practical filters approximate the brick-wall shape with a transition band and a delay.

Answer. Distortionless: $H = Ke^{-j\omega t_d}$ (flat magnitude, linear phase). All-pass: $|H| = 1$ only; distortionless iff phase is linear. Ideal filters have non-causal h.

2021–2022 • Q5(a) EEE 211 • EEE p.14 MARKS 18

Derive the Fourier transform as the limiting case of the Fourier series as the period $T \to \infty$.

SOLUTION

Let $x(t)$ be a finite-duration signal and $x_T(t)$ its periodic repetition with period T ($\omega_0 = 2\pi/T$). The Fourier-series coefficients are

$$a_k = \frac{1}{T}\int_{-T/2}^{T/2} x_T(t)e^{-jk\omega_0 t}\,\mathrm{d}t = \frac{1}{T}\int_{-\infty}^{\infty} x(t)e^{-jk\omega_0 t}\,\mathrm{d}t = \frac{1}{T}X(k\omega_0),$$

so $x_T(t) = \displaystyle\sum_k \frac{X(k\omega_0)}{T}e^{jk\omega_0 t} = \frac{1}{2\pi}\sum_k X(k\omega_0)e^{jk\omega_0 t}\,\omega_0$. As $T \to \infty$: $\omega_0 \to \mathrm{d}\omega$, $k\omega_0 \to \omega$ and the sum becomes an integral,

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)e^{j\omega t}\,\mathrm{d}\omega, \qquad X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}\,\mathrm{d}t.$$

Answer. $a_k = \frac{1}{T} X(k\omega_0)$; as $T \to \infty$ the harmonic spacing $\omega_0 \to \mathrm{d}\omega$ and the series becomes the inverse Fourier transform.

2018–2019 • Q4(a) — CSE 311 CSE 311 • TF p.18 MARKS 18

State the frequency-shifting property. How is it used in frequency-division multiplexing (FDM)?

SOLUTION

If $x(t) \leftrightarrow X(\omega)$ then $x(t)e^{j\omega_0 t} \leftrightarrow X(\omega - \omega_0)$, so

$$x(t)\cos\omega_0 t \;\leftrightarrow\; \tfrac{1}{2}\big[X(\omega-\omega_0) + X(\omega+\omega_0)\big].$$

Multiplying by a carrier moves the baseband spectrum up to $\pm\omega_0$. In FDM each message $m_i(t)$ is multiplied by a different carrier ω_i, spaced by more than the message bandwidth so the shifted spectra do not overlap, then all are added and sent on one channel. The receiver picks one band with a band-pass filter and multiplies by the same carrier ($\cos^2$ gives back the baseband) followed by a low-pass filter.

Answer. $x(t)e^{j\omega_0 t} \leftrightarrow X(\omega - \omega_0)$; FDM stacks messages on non-overlapping carriers.

10

Sampling and Reconstruction

ASHRAF SIR • SAMPLING

Impulse sampling: $x_p(t) = x(t)\sum\delta(t-nT) \Rightarrow X_p(\omega) = \frac{1}{T}\sum_k X(\omega - k\omega_s)$, $\omega_s = 2\pi/T$. No aliasing iff $\omega_s > 2\omega_M$.

2022–2023 • Q1(c) CSE 219 • TF p.1 MARKS 15

Let $x(t)$ be band-limited ($X(\omega) = 0$ for $|\omega| > \omega_M$). Show that if $\omega_s > 2\omega_M$, $x(t)$ is uniquely determined by its samples $x(nT)$.

SOLUTION

Sample with the impulse train $p(t) = \sum_n \delta(t-nT)$: $x_p(t) = \sum_n x(nT)\delta(t-nT)$. Since $P(\omega) = \omega_s\sum_k \delta(\omega - k\omega_s)$,

$$X_p(\omega) = \frac{1}{2\pi}X * P = \frac{1}{T}\sum_{k=-\infty}^{\infty} X(\omega - k\omega_s).$$

These are copies of X centred at multiples of ω_s, each of width $2\omega_M$. If $\omega_s - \omega_M > \omega_M$, i.e. $\omega_s > 2\omega_M$, the copies do not overlap, so the copy at $k = 0$ is intact, equal to $X(\omega)/T$.
An ideal low-pass filter with gain T and cutoff ω_c, $\omega_M < \omega_c < \omega_s - \omega_M$, returns exactly $X(\omega)$. Its impulse response $h(t) = \dfrac{T\sin\omega_c t}{\pi t}$ gives the interpolation formula

$$x(t) = \sum_n x(nT)\,\frac{\sin\omega_c(t-nT)}{\omega_c(t-nT)}\cdot\frac{T\omega_c}{\pi} \quad \left(\omega_c = \tfrac{\omega_s}{2} :\ x(t) = \sum_n x(nT)\,\mathrm{sinc}\,\tfrac{\pi(t-nT)}{T}\right).$$

Because X is recovered uniquely from the samples, so is $x(t)$.

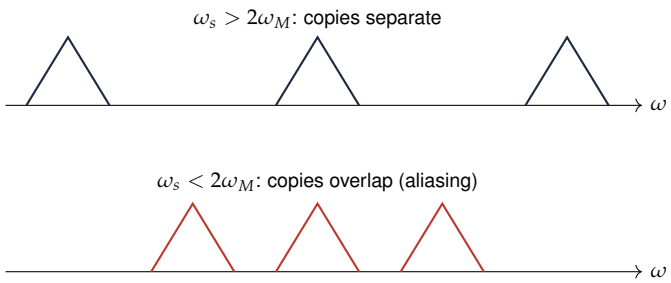
Answer. No overlap when $\omega_s > 2\omega_M$; the $k = 0$ copy is isolated by a low-pass filter, so the samples determine $x(t)$ uniquely.

2023–2024 • Q7(c) CSE 219 • 22 batch p.4 MARKS 10

State the Nyquist–Shannon sampling theorem. Illustrate $X_p(j\omega)$ when it is satisfied and when it is violated, and explain why aliasing cannot be undone after sampling.

SOLUTION

Theorem. A signal band-limited to ω_M can be recovered exactly from samples taken at rate $\omega_s > 2\omega_M$ (the Nyquist rate is $2\omega_M$).



Aliasing is irreversible because in the overlap region the sum $X(\omega) + X(\omega - \omega_s)$ is all that survives; many different pairs of spectra produce the same sum, and a frequency $\omega_0 + \omega_s$ produces exactly the same samples as ω_0. Nothing in the samples tells them apart, so no filter can restore the original.

Answer. Sample at $\omega_s > 2\omega_M$. Below that, shifted copies overlap and the added spectra cannot be separated.

2022–2023 • Q4(b) CSE 219 • TF p.2 MARKS 10

Let $x[n]$ be a signal whose DTFT satisfies $X(e^{j\omega}) = 0$ for $\frac{2\pi}{9} \leq |\omega| \leq \pi$. What is the lowest rate at which $x[n]$ can be sampled (decimated) without aliasing?

SOLUTION

The spectrum is non-zero only for $|\omega| < \omega_M = \frac{2\pi}{9}$. Keeping every M-th sample, $y[n] = x[Mn]$, puts copies of the spectrum (stretched by M) at spacing 2π; the spectrum of y is $\frac{1}{M}\sum_k X\left(e^{j(\omega - 2\pi k)/M}\right)$, whose occupied width is $2M\omega_M$. No overlap requires

$$2M\omega_M \leq 2\pi \;\Rightarrow\; M \leq \frac{\pi}{\omega_M} = \frac{9}{2}.$$

The largest integer is $M = 4$.

Answer. Keep every 4th sample ($M = 4$): the lowest rate is $\frac{1}{4}$ of the original sampling rate.

2023–2024 • Q5(c) CSE 219 • 22 batch p.3 MARKS 10

Compare ideal (sinc) reconstruction, zero-order hold and linear interpolation. Which would you choose for a real-time DAC, and why?

SOLUTION

	ideal (sinc)	zero-order hold	linear interpolation
$h(t)$	$\mathrm{sinc}(\pi t/T)$	rectangle of width T	triangle of width $2T$
$\lvert H(\omega)\rvert$	ideal brick wall	$T\,\lvert\mathrm{sinc}(\omega T/2)\rvert$ (droop, images remain)	$T\,\mathrm{sinc}^2(\omega T/2)$ (better image rejection, more droop)
causal?	no, infinite	yes	no (needs the next sample; causal if delayed by T)
exact?	yes (band-limited)	staircase, error	piecewise-linear, smaller error

Real-time DAC: a zero-order hold, followed by an analog low-pass (reconstruction) filter and, if needed, a $1/\,\mathrm{sinc}$ droop compensator. It is causal, needs only the current sample and no look-ahead, and is simple to implement;

the sinc is non-causal and linear interpolation requires the next sample (one-sample delay).

Answer. Choose the zero-order hold plus an analog smoothing filter (causal, cheap); ideal sinc is unrealisable and linear interpolation needs look-ahead.

2020–2021 • Q1(b) — CSE 311 CSE 311 • TF p.12 MARKS 10

Can the past samples of a band-limited signal foretell its future?

SOLUTION

In principle, yes: a band-limited signal is fully described by its samples, $x(t) = \sum x(nT)\,\mathrm{sinc}\,\frac{\pi(t-nT)}{T}$, and such a signal is analytic (infinitely smooth), so a complete record of its past determines the future. In practice, no: the formula needs infinitely many samples, the interpolation weights decay only like $1/t$, and any noise or truncation makes extrapolation hopelessly unstable.

Answer. Theoretically yes (band-limited $\Rightarrow$ analytic); practically no, because of noise, finite records and ill-conditioning.

REPEATED ×2 **Same question in 2 papers — stacked below, solved once.**

2020–2021 • Q3(a) — CSE 311 CSE 311 • TF p.12 MARKS 24

Why can practical signals not be fully recovered after sampling? Explain aliasing and how to minimise its effect.

2018–2019 • Q3(b) — CSE 311 CSE 311 • TF p.18 MARKS 20

What causes aliasing, what problems does it create and how is it compensated?

SOLUTION

Cause. Practical signals are time-limited, hence never strictly band-limited; their spectra have tails beyond $\omega_s/2$. Also, the reconstruction filter is not ideal. When $\omega_s < 2\omega_M$ (or because of the tails) the shifted copies of the spectrum overlap, and high-frequency content appears as a lower frequency: $f \to |f - kf_s|$.
Problem. The overlap adds energy to the wanted band and cannot be removed afterwards (the same samples describe many signals); it causes distortion and spurious tones (e.g. the wheel that seems to turn backwards).
Remedy.

- Pass the signal through an **anti-aliasing low-pass filter** (cutoff $\leq f_s/2$) before sampling.
- Sample above the Nyquist rate (oversampling) so that a filter with a gradual transition band suffices.
- Use a good reconstruction filter after the DAC.

Answer. Aliasing = overlap of spectral copies when $f_s < 2f_M$ (or from signal tails); minimise with an anti-aliasing filter and oversampling.

REPEATED ×2 **Same question in 2 papers — stacked below, solved once.**

2021–2022 • Q3(b) — CSE 311 CSE 311 • TF p.8 MARKS 15

The Nyquist-rate samples of a band-limited signal $g(t)$ are all zero except $g(0) = A$. Find $g(t)$.

2018–2019 • Q1(a) — CSE 311 CSE 311 • TF p.18 MARKS 21

A signal $g(t)$ is band-limited to 50 Hz and its samples, 0.01 s apart, are all 0 except $g(0) = 1$. Find $g(t)$ and $G(f)$. Also state the Fourier transform of $\delta(t)$.

SOLUTION

Interpolation formula: $g(t) = \sum_n g(nT)\,\operatorname{sinc}\dfrac{\pi(t-nT)}{T}$, with only $n = 0$ non-zero:

$$g(t) = A\,\frac{\sin(\pi t/T)}{\pi t/T} = A\,\operatorname{sinc}\left(\frac{\pi t}{T}\right), \qquad T = \frac{1}{2B}.$$

For $B = 50$ Hz, $T = 0.01$ s (exactly the Nyquist interval) and $A = 1$:

$$g(t) = \frac{\sin 100\pi t}{100\pi t}, \qquad G(f) = 0.01\,\operatorname{rect}\left(\frac{f}{100}\right)\ (= 0.01 \text{ for } |f| < 50 \text{ Hz, else } 0).$$

Also $\delta(t) \leftrightarrow 1$.

Answer. $g(t) = A\,\operatorname{sinc}(2\pi Bt)$; for 50 Hz, $g(t) = \frac{\sin 100\pi t}{100\pi t}$, $G(f) = 0.01\,\operatorname{rect}(f/100)$.

2018–2019 • Q3(b) EEE 211 • EEE p.28 MARKS 10

Find the Fourier transform of $x(t) = \sin 1000\pi t + \cos 1200\pi t$. Find the sampling period such that no aliasing occurs.

SOLUTION

$$X(\omega) = \frac{\pi}{j}\big[\delta(\omega - 1000\pi) - \delta(\omega + 1000\pi)\big] + \pi\big[\delta(\omega - 1200\pi) + \delta(\omega + 1200\pi)\big].$$

The highest frequency is $\omega_M = 1200\pi$ rad/s (600 Hz), so the Nyquist rate is $2\omega_M = 2400\pi$ rad/s (1200 Hz):

$$T < \frac{2\pi}{2\omega_M} = \frac{1}{1200}\ \text{s} \approx 0.833\ \text{ms}.$$

Answer. Impulses at $\pm 1000\pi$, $\pm 1200\pi$; sample faster than 1200 Hz, i.e. $T < \frac{1}{1200}$ s.

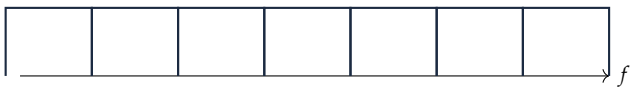
2019–2020 • Q3 — CSE 311 CSE 311 • TF p.15 MARKS 35

Find the Fourier transforms of $2B\,\operatorname{sinc}(2\pi Bt)$ and $B\,\operatorname{sinc}^2(\pi Bt)$. Sketch the spectrum of each when sampled at 1.5 times the Nyquist rate, and write the general expression for the sampled spectrum.

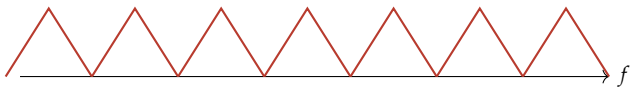
SOLUTION

With $\operatorname{sinc} x = \frac{\sin x}{x}$: $2B\,\operatorname{sinc}(2\pi Bt) = \dfrac{\sin 2\pi Bt}{\pi t} \leftrightarrow \operatorname{rect}\left(\dfrac{f}{2B}\right)$ (height 1, $|f| < B$), and $B\,\operatorname{sinc}^2(\pi Bt) \leftrightarrow \Lambda\left(\frac{f}{B}\right) = \max(0, 1 - |f|/B)$.
Both are band-limited to B (Nyquist rate $2B$). At $f_s = 1.5(2B) = 3B$ the copies are centred at multiples of $3B$:

$$G_s(f) = f_s \sum_{n=-\infty}^{\infty} G(f - nf_s) \quad \text{with ideal impulse sampling at } T = \tfrac{1}{f_s}.$$



rect: copies at $0, \pm 3B, \pm 6B, \ldots$ with gaps of width B



triangle: copies at multiples of $3B$, no overlap

Answer. $\mathrm{rect}(f/2B)$ and $\Lambda(f/B)$; sampled spectra are copies repeated every $3B$ (gaps of width B), scaled by f_s.

2019–2020 • Q7(b) EEE 211 • EEE p.25 MARKS 20

Let $x(t) = \dfrac{\sin(\omega_B t/2)}{\pi t} \sum_n \delta\left(t - \frac{n\pi}{\omega_B}\right)$ be applied to an LTI system with $h(t) = \dfrac{\sin(3\omega_B t/2)}{\pi t}$. Find the output $y(t)$.

SOLUTION

Let $x_0(t) = \dfrac{\sin(\omega_B t/2)}{\pi t} \leftrightarrow X_0 = \mathrm{rect}\,\dfrac{\omega}{\omega_B}$ (1 for $|\omega| < \omega_B/2$). The impulse train has $T = \pi/\omega_B$, so $\omega_s = 2\pi/T = 2\omega_B$, and

$$X(\omega) = \frac{\omega_B}{\pi} \sum_k X_0(\omega - 2k\omega_B).$$

The copies sit at $0, \pm 2\omega_B, \ldots$ with half-width $\omega_B/2$ (no aliasing: $\omega_s = 2\omega_B > 2 \cdot \frac{\omega_B}{2}$). The system is an ideal low-pass filter with cutoff $\frac{3}{2}\omega_B$, which passes the $k = 0$ copy ($|\omega| < \omega_B/2$) and rejects the copies at $\pm 2\omega_B$, which start at $1.5\omega_B$ (the edge, of measure zero). Hence

$$Y(\omega) = \frac{\omega_B}{\pi} X_0(\omega), \qquad y(t) = \frac{\omega_B}{\pi} \cdot \frac{\sin(\omega_B t/2)}{\pi t}.$$

Answer. $y(t) = \dfrac{\omega_B}{\pi^2} \dfrac{\sin(\omega_B t/2)}{t}$.

11

DFT, FFT and Fast Algorithms

ASHRAF SIR • DFT AND FFT

$X[k]=\sum_{n=0}^{N-1}x[n]W_N^{kn}$, $W_N=e^{-j2\pi/N}$, $x[n]=\frac{1}{N}\sum_k X[k]W_N^{-kn}$. Linear convolution of lengths L,M via the DFT needs $N\ge L+M-1$.

2023–2024 • Q5(a) CSE 219 • 22 batch p.3 MARKS 15

Compute the 8-point DFT of $x[n]=\{1,2,3,4,3,2\}$ using the radix-2 decimation-in-time FFT, showing all butterflies.

SOLUTION

Zero-pad to $N=8$: $x=\{1,2,3,4,3,2,0,0\}$. Bit-reversed input order $x[0],x[4],x[2],x[6],x[1],x[5],x[3],x[7]=1,3,3,0,2,2,4,0$. Twiddles: $W_8^0=1$, $W_8^1=\frac{1-j}{\sqrt{2}}$, $W_8^2=-j$, $W_8^3=\frac{-1-j}{\sqrt{2}}$.

Stage 1 (pairs, $W_2^0=1$): $(1,3)\to(4,-2)$; $(3,0)\to(3,3)$; $(2,2)\to(4,0)$; $(4,0)\to(4,4)$.

Stage 2 (blocks of 4, $W_4^0=1$, $W_4^1=-j$):

$$A=[4,-2,3,3]\to A_0=4+3=7,\ A_1=-2-3j,\ A_2=4-3=1,\ A_3=-2+3j,$$
$$B=[4,0,4,4]\to B_0=8,\ B_1=-4j,\ B_2=0,\ B_3=4j.$$

Stage 3 ($X[k]=A_k+W_8^kB_k$, $X[k+4]=A_k-W_8^kB_k$):

k	$X[k]$	$X[k+4]$
0	$7+8=15$	$7-8=-1$
1	$(-2-3j)+W_8^1(-4j)=-4.828-5.828j$	$0.828-0.172j$
2	$1+(-j)(0)=1$	1
3	$(-2+3j)+W_8^3(4j)=0.828+0.172j$	$-4.828+5.828j$

$$X=\{15,\ -4.828-5.828j,\ 1,\ 0.828+0.172j,\ -1,\ 0.828-0.172j,\ 1,\ -4.828+5.828j\}.$$

The result is conjugate-symmetric, $X[8-k]=X^*[k]$, as expected for real input.

Answer. $X[k]=\{15,\ -4.828-5.828j,\ 1,\ 0.828+0.172j,\ -1,\ 0.828-0.172j,\ 1,\ -4.828+5.828j\}$.

2023–2024 • Q5(b) CSE 219 • 22 batch p.3 MARKS 10

Find the linear convolution of $x=\{1,2\}$ and $h=\{1,1\}$ using the DFT.

SOLUTION

The result has length $2+2-1=3$, so use $N=4$ (zero-padding).

$$X=\{3,\ 1-2j,\ -1,\ 1+2j\},\qquad H=\{2,\ 1-j,\ 0,\ 1+j\}.$$

$$Y=XH=\{6,\ -1-3j,\ 0,\ -1+3j\}.$$

Inverse DFT, $y[n]=\frac{1}{4}\sum_k Y[k]j^{kn}$: $y[0]=\frac{6-1-3j-1+3j}{4}=1$, $y[1]=\frac{12}{4}=3$, $y[2]=\frac{8}{4}=2$, $y[3]=0$.

Answer. $y[n]=\{1,3,2\}$ (direct check: $(1+2z^{-1})(1+z^{-1})=1+3z^{-1}+2z^{-2}$).

2023–2024 • Q7(a) CSE 219 • 22 batch p.4 MARKS 15

Compute the DFT of $x=\{1,2,3,4,9,0,7,8\}$ using the 2×4 Cooley–Tukey (matrix) method: (i) DFT of the columns, (ii) twiddle-factor multiplication, (iii) DFT of the rows.

SOLUTION

Write $n=4n_1+n_2$ ($n_1=0,1$; $n_2=0..3$) and $k=k_1+2k_2$ ($k_1=0,1$; $k_2=0..3$).

$$\text{matrix of } x:\ \begin{bmatrix}1 & 2 & 3 & 4\\ 9 & 0 & 7 & 8\end{bmatrix}.$$

(i) 2-point DFT of each column: $\begin{bmatrix}10 & 2 & 10 & 12\\ -8 & 2 & -4 & -4\end{bmatrix}$.

(ii) Twiddles $W_8^{k_1n_2}$: row $k_1=0$ unchanged; row $k_1=1$ multiplied by $W_8^{n_2}$, $W_8=\frac{1-j}{\sqrt{2}}$:

$$[-8,\ 2W_8,\ -4W_8^2,\ -4W_8^3]=[-8,\ 1.414-1.414j,\ 4j,\ 2.828+2.828j].$$

(iii) 4-point DFT of each row: row 0 gives $X[0],X[2],X[4],X[6]$; row 1 gives $X[1],X[3],X[5],X[7]$:

row 0: $\mathrm{DFT}_4[10,2,10,12]=[34,\ 10j,\ 6,\ -10j]$,
row 1: $\mathrm{DFT}_4[-8,\ 1.414-1.414j,\ 4j,\ 2.828+2.828j]=[-3.757+5.414j,\ -12.243-2.586j,\ -12.243+2.586j,\ -3.757-5$

Interleaving:

$$X=\{34,\ -3.757+5.414j,\ 10j,\ -12.243-2.586j,\ 6,\ -12.243+2.586j,\ -10j,\ -3.757-5.414j\}.$$

Answer. $X[k]$ as above ($X[0]=34$ is the sum of the samples; $X[k]$ conjugate-symmetric).

2023–2024 • Q7(b) CSE 219 • 22 batch p.4 MARKS 10

Derive the chirp-z (Bluestein) algorithm for computing the DFT. Why must the FFT size satisfy $M\geq 2N-1$?

SOLUTION

Use $nk=\frac{1}{2}\left[n^2+k^2-(k-n)^2\right]$ in $X[k]=\sum_{n=0}^{N-1}x[n]W^{nk}$, $W=e^{-j2\pi/N}$:

$$X[k]=W^{k^2/2}\sum_{n=0}^{N-1}\underbrace{x[n]W^{n^2/2}}_{a[n]}\ \underbrace{W^{-(k-n)^2/2}}_{b[k-n]}=W^{k^2/2}\,(a*b)[k].$$

So the DFT is: (1) multiply x by the chirp $W^{n^2/2}$; (2) convolve with the chirp $b[m]=W^{-m^2/2}$; (3) multiply by $W^{k^2/2}$. The convolution is done with FFTs of any convenient size (e.g. a power of two), which is why Bluestein works for any N (including primes) in $O(N\log N)$.

Why $M \geq 2N-1$. The index $k-n$ runs from $-(N-1)$ to $N-1$, so b has $2N-1$ non-zero points. The linear convolution of a (length N) with b must not wrap around circularly: we need $M \geq N+(2N-1)-1$ for full linear convolution, but only outputs $k=0..N-1$ are used and those are uncontaminated once $M \geq 2N-1$ (the wrapped terms land outside the N wanted samples).

Answer. $X[k] = W^{k^2/2}\sum x[n]W^{n^2/2}W^{-(k-n)^2/2}$: chirp multiply, convolve, chirp multiply; zero-pad to $M \geq 2N-1$ so circular wrap-around does not corrupt $X[0..N-1]$.

2022–2023 • Q4(c) CSE 219 • TF p.2 MARKS 15

Compare the computational complexity of the direct DFT with the radix-2 FFT.

SOLUTION

Direct DFT: each of N outputs needs N complex multiplications and $N-1$ additions, so N^2 multiplications, $N(N-1)$ additions: $O(N^2)$.
Radix-2 FFT ($N=2^m$): $\log_2 N$ stages, each with $N/2$ butterflies (1 complex multiplication, 2 additions). So

$$\tfrac{N}{2}\log_2 N \text{ multiplications}, \qquad N\log_2 N \text{ additions} : \; O(N\log N).$$

N	N^2	$\frac{N}{2}\log_2 N$	speed-up
8	64	12	5.3
64	4096	192	21
1024	1 048 576	5120	205

The saving grows with N: the FFT reuses sub-results by splitting the sum into even and odd samples.

Answer. Direct: N^2 multiplications. FFT: $\frac{N}{2}\log_2 N$, i.e. about $205\times$ fewer for $N=1024$.

12

Laplace Transform: ROC, Properties and Inverse

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$X(s) = \int x(t)e^{-st}\,\mathrm{d}t$. Right-sided ⇒ ROC is a right half-plane; left-sided ⇒ left half-plane; two-sided ⇒ a vertical strip. Poles are never inside the ROC.

2021–2022 • Q7(a) EEE 211 • EEE p.15 MARKS 12

Find the Laplace transform and ROC of (i) $e^{-at}u(t)$ for $a > 0$ and $a < 0$, (ii) $-e^{-at}u(-t)$ for $a < 0$.

SOLUTION

(i) $\displaystyle\int_0^{\infty} e^{-(s+a)t}\,\mathrm{d}t = \frac{1}{s+a}$, valid for $\mathrm{Re}\{s\} > -a$. For $a > 0$ the ROC $\mathrm{Re}\{s\} > -a$ contains the $j\omega$-axis (stable, Fourier transform exists). For $a < 0$ the ROC $\mathrm{Re}\{s\} > |a|$ excludes the $j\omega$-axis (growing exponential; no Fourier transform).

(ii) $\displaystyle -\int_{-\infty}^{0} e^{-(s+a)t}\,\mathrm{d}t = \frac{1}{s+a}$, valid for $\mathrm{Re}\{s\} < -a$. For $a < 0$ this half-plane ($\mathrm{Re}\,s < |a|$) contains the $j\omega$-axis.

Same algebraic expression, different ROC ⇒ different signal: the ROC is part of the answer.

Answer. (i) $\frac{1}{s+a}$, $\mathrm{Re}\,s > -a$. (ii) $\frac{1}{s+a}$, $\mathrm{Re}\,s < -a$.

2022–2023 • Q1(b) CSE 219 • TF p.1 MARKS 10

For $x(t) = e^{-at}$, $0 < t < T$ (zero otherwise), is $s = -a$ in the ROC? (Note the apparent pole there.)

SOLUTION

$$X(s) = \int_0^{T} e^{-(s+a)t}\,\mathrm{d}t = \frac{1 - e^{-(s+a)T}}{s+a}.$$

At $s = -a$ the numerator and denominator both vanish, so the singularity is removable: using $1 - e^{-\epsilon T} \approx \epsilon T$, $X(-a) = T$ (indeed $\int_0^T 1\,\mathrm{d}t = T$). The integral converges for every s (finite-duration signal), so the ROC is the whole s-plane and $s = -a$ is in it. The zero of the numerator cancels the pole: no pole at $s = -a$.

Answer. Yes: $X(s) = \frac{1-e^{-(s+a)T}}{s+a}$ has a removable singularity at $-a$ ($X(-a) = T$); ROC = all s.

2022–2023 • Q2(a) CSE 219 • TF p.1 MARKS 10

State and prove the time-domain and s-domain differentiation properties of the Laplace transform.

SOLUTION

Time differentiation: $\mathcal{L}\{x'(t)\} = sX(s) - x(0^-)$. *Proof* (integration by parts): $\int_{0^-}^{\infty} x'e^{-st}\,\mathrm{d}t = \left[xe^{-st}\right]_{0^-}^{\infty} + s\int_{0^-}^{\infty} xe^{-st}\,\mathrm{d}t = -x(0^-) + sX(s)$ (the boundary term vanishes at ∞ inside the ROC). Applying twice: $\mathcal{L}\{x''\} = s^2X - sx(0^-) - x'(0^-)$.

s-domain differentiation: $\mathcal{L}\{tx(t)\} = -\dfrac{\mathrm{d}X(s)}{\mathrm{d}s}$. *Proof:* $\dfrac{\mathrm{d}}{\mathrm{d}s}\int x(t)e^{-st}\,\mathrm{d}t = \int(-t)x(t)e^{-st}\,\mathrm{d}t = -\mathcal{L}\{tx(t)\}$.

Answer. $x' \leftrightarrow sX - x(0^-);\ tx(t) \leftrightarrow -X'(s)$.

2022–2023 • Q2(b) CSE 219 • TF p.1 **MARKS 10**

Find the inverse Laplace transform of $X(s) = \dfrac{s+2}{s^2+7s+12}$ with ROC $-4 < \mathrm{Re}\{s\} < -3$.

SOLUTION

$X(s) = \dfrac{s+2}{(s+3)(s+4)} = \dfrac{A}{s+3} + \dfrac{B}{s+4}$ with $A = \dfrac{-1}{1} = -1$, $B = \dfrac{-2}{-1} = 2$:

$$X(s) = -\frac{1}{s+3} + \frac{2}{s+4}.$$

The ROC lies to the *left* of the pole $s = -3$ (left-sided term) and to the *right* of $s = -4$ (right-sided term):

$$-\frac{1}{s+3},\ \mathrm{Re}\,s < -3 \longrightarrow e^{-3t}u(-t); \qquad \frac{2}{s+4},\ \mathrm{Re}\,s > -4 \longrightarrow 2e^{-4t}u(t).$$

Answer. $x(t) = 2e^{-4t}u(t) + e^{-3t}u(-t)$.

2022–2023 • Q2(c) CSE 219 • TF p.1 **MARKS 15**

Find the Laplace transform and ROC of $x(t) = e^{-b|t-2|}$, $b > 0$.

SOLUTION

Let $w(t) = e^{-b|t|} = e^{-bt}u(t) + e^{bt}u(-t)$. Then

$$W(s) = \frac{1}{s+b} - \frac{1}{s-b} = \frac{2b}{b^2 - s^2}, \qquad -b < \mathrm{Re}\{s\} < b.$$

Since $x(t) = w(t-2)$, the time-shift property gives

$$X(s) = e^{-2s}\,\frac{2b}{b^2 - s^2}, \qquad -b < \mathrm{Re}\{s\} < b \text{ (shift does not change the ROC).}$$

Answer. $X(s) = \dfrac{2b\,e^{-2s}}{b^2 - s^2}$, ROC: $-b < \mathrm{Re}\,s < b$.

2023–2024 • Q6(b) CSE 219 • 22 batch p.3 **MARKS 13**

Find the Laplace transform and ROC of $x(t) = |t|e^{-2|t|} - e^{-t}\cos 3t\,u(t) + t^2e^{-4t}u(t)$ and sketch the ROC.

SOLUTION

Term 1: $|t|e^{-2|t|} = te^{-2t}u(t) + (-t)e^{2t}u(-t)$. The first part is $\frac{1}{(s+2)^2}$ for $\mathrm{Re}\, s > -2$. For the second, $e^{2t}u(-t) \leftrightarrow -\frac{1}{s-2}$ ($\mathrm{Re}\, s < 2$) and $tx \leftrightarrow -X'$ give $te^{2t}u(-t) \leftrightarrow -\frac{1}{(s-2)^2}$, so $(-t)e^{2t}u(-t) \leftrightarrow \frac{1}{(s-2)^2}$.

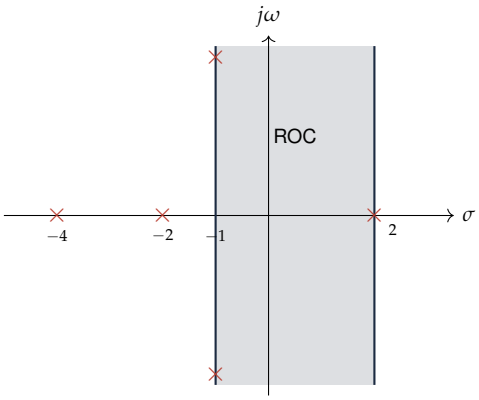
$$X_1 = \frac{1}{(s+2)^2} + \frac{1}{(s-2)^2}, \qquad -2 < \mathrm{Re}\, s < 2.$$

Term 2: $e^{-t}\cos 3t\, u(t) \leftrightarrow \dfrac{s+1}{(s+1)^2+9}$, $\mathrm{Re}\, s > -1$.

Term 3: $t^2 e^{-4t}u(t) \leftrightarrow \dfrac{2}{(s+4)^3}$, $\mathrm{Re}\, s > -4$.

$$X(s) = \frac{1}{(s+2)^2} + \frac{1}{(s-2)^2} - \frac{s+1}{(s+1)^2+9} + \frac{2}{(s+4)^3}.$$

ROC = intersection $= \{-1 < \mathrm{Re}\, s < 2\}$.



(Poles: double at -2, double at 2, $-1 \pm 3j$, triple at -4; the strip lies between the right-sided boundary -1 and the left-sided boundary 2.)

Answer. ROC $-1 < \mathrm{Re}\{s\} < 2$ with $X(s)$ above.

2023–2024 • Q6(c) CSE 219 • 22 batch p.3 MARKS 10

Can a causal system be unstable? Can a stable system be non-causal? Explain with the ROC.

SOLUTION

Causal $\Leftrightarrow$ ROC is a right half-plane $\mathrm{Re}\, s > \sigma_{\max}$. Stable $\Leftrightarrow$ ROC contains the $j\omega$-axis. They are independent conditions.

- *Causal but unstable:* $H(s) = \dfrac{1}{s-1}$, ROC $\mathrm{Re}\, s > 1$, $h = e^t u(t)$. Right half-plane, but it does not include the $j\omega$-axis (pole in the right half of the plane).
- *Stable but non-causal:* $H(s) = \dfrac{1}{s-1}$, ROC $\mathrm{Re}\, s < 1$, $h = -e^t u(-t)$. ROC contains the $j\omega$-axis ($\int |h| = 1 < \infty$) but it is left-sided.

For a rational $H(s)$, causal *and* stable $\Leftrightarrow$ all poles in the left half-plane.

Answer. Yes to both: $\frac{1}{s-1}$ with ROC $\mathrm{Re}\, s > 1$ (causal, unstable) or $\mathrm{Re}\, s < 1$ (stable, non-causal).

2022–2023 • Q4(a) CSE 219 • TF p.2 MARKS 10

What is the relation between the Laplace transform and the Fourier transform?

SOLUTION

With $s = \sigma + j\omega$: $X(s) = \int \left[x(t)e^{-\sigma t}\right]e^{-j\omega t}\,\mathrm{d}t = \mathcal{F}\{x(t)e^{-\sigma t}\}$. The Laplace transform is the Fourier transform of x after multiplication by a decaying exponential, which makes more signals transformable. If the ROC contains the $j\omega$-axis ($\sigma = 0$),

$$X(j\omega) = X(s)\big|_{s=j\omega}.$$

If the ROC does not include the axis, the Fourier transform does not exist (in the ordinary sense), although the Laplace transform does; e.g. $u(t) \to \frac{1}{s}$ but $\mathcal{F}\{u\} = \pi\delta(\omega) + \frac{1}{j\omega}$.

Answer. $X(j\omega) = X(s)|_{s=j\omega}$ provided the ROC includes the $j\omega$-axis.

2021–2022 • Q7(b) EEE 211 • EEE p.15 MARKS 12

Using the convolution property, find the inverse Laplace transform of $\dfrac{1}{(s-a)^3}$ (right-sided).

SOLUTION

$\dfrac{1}{s-a} \leftrightarrow e^{at}u(t)$. Convolution in time is multiplication in s:

$$\frac{1}{(s-a)^2} \leftrightarrow \int_0^t e^{a\tau}e^{a(t-\tau)}\,\mathrm{d}\tau = te^{at}u(t), \qquad \frac{1}{(s-a)^3} \leftrightarrow \int_0^t \tau e^{a\tau}e^{a(t-\tau)}\,\mathrm{d}\tau = \frac{t^2}{2}e^{at}u(t).$$

Answer. $\dfrac{t^2}{2}e^{at}u(t)$, ROC $\operatorname{Re} s > a$.

2018–2019 • Q5(a) EEE 211 • EEE p.29 MARKS 15

A real signal $x(t)$ has Laplace transform $X(s)$ with two poles, no finite zeros, a pole at $s = -1 + j$, such that $e^{2t}x(t)$ is not absolutely integrable, and $X(0) = 5$. Find $X(s)$ and its ROC.

SOLUTION

x real $\Rightarrow$ poles come in conjugate pairs: $-1 \pm j$. Two poles and no zeros:

$$X(s) = \frac{K}{(s+1-j)(s+1+j)} = \frac{K}{s^2+2s+2}, \qquad X(0) = \frac{K}{2} = 5 \Rightarrow K = 10.$$

$e^{2t}x(t)$ is not absolutely integrable $\Leftrightarrow \sigma = -2$ (the value that makes $e^{-\sigma t} = e^{2t}$) is outside the ROC. Both poles lie on $\operatorname{Re} s = -1$, so the ROC is either $\operatorname{Re} s > -1$ or $\operatorname{Re} s < -1$; since $\sigma = -2$ must be excluded, the ROC is $\operatorname{Re} s > -1$.

Answer. $X(s) = \dfrac{10}{s^2+2s+2}$, ROC: $\operatorname{Re}\{s\} > -1$ (so $x(t) = 10e^{-t}\sin t\, u(t)$).

13

Laplace Transform: Differential Equations, Circuits and Feedback

ASHRAF SIR • LAPLACE IN SYSTEMS

$\mathcal{L}\{y'\} = sY - y(0^-)$, $\mathcal{L}\{y''\} = s^2Y - sy(0^-) - y'(0^-)$. Total response = zero-input (initial conditions) + zero-state (input). Feedback: $T = \dfrac{G}{1+GH}$. Stable $\Leftrightarrow$ all closed-loop poles in the open left half-plane.

2023–2024 • Q6(a) CSE 219 • 22 batch p.3 MARKS 12

Solve $y'' + 7y' + 12y = x' + 2x$ with $x(t) = u(t)$ and zero initial conditions by the Laplace transform.

SOLUTION

$$(s^2 + 7s + 12)Y = (s+2)X, \quad X = \frac{1}{s} \;\Rightarrow\; Y = \frac{s+2}{s(s+3)(s+4)}.$$

Partial fractions: $\dfrac{A}{s} + \dfrac{B}{s+3} + \dfrac{C}{s+4}$ with $A = \dfrac{2}{12} = \dfrac{1}{6}$, $B = \dfrac{-1}{(-3)(1)} = \dfrac{1}{3}$, $C = \dfrac{-2}{(-4)(-1)} = -\dfrac{1}{2}$.

$$y(t) = \left[\tfrac{1}{6} + \tfrac{1}{3}e^{-3t} - \tfrac{1}{2}e^{-4t}\right]u(t).$$

Check: $y(0) = \frac{1}{6} + \frac{1}{3} - \frac{1}{2} = 0$ and $y(\infty) = \frac{2}{12} = H(0)\cdot 1$.

Answer. $y(t) = \left(\frac{1}{6} + \frac{1}{3}e^{-3t} - \frac{1}{2}e^{-4t}\right)u(t)$.

2023–2024 • Q8(b) EEE 211 • EEE p.6 MARKS 18

For $y'' + 5y' + 6y = x' + x$ with $y(0^-) = 2$, $y'(0^-) = 1$ and $x(t) = e^{-4t}u(t)$, find the zero-input response (ZIR) and zero-state response (ZSR) by the Laplace transform.

SOLUTION

$$(s^2 + 5s + 6)Y - \left[sy(0^-) + y'(0^-) + 5y(0^-)\right] = (s+1)X - x(0^-).$$

With $x(0^-) = 0$ and $X = \frac{1}{s+4}$:

$$Y(s) = \underbrace{\frac{2s+11}{(s+2)(s+3)}}_{\text{ZIR}} + \underbrace{\frac{s+1}{(s+2)(s+3)(s+4)}}_{\text{ZSR}}.$$

ZIR: $\dfrac{7}{s+2}-\dfrac{5}{s+3}$. ZSR: $-\dfrac{1/2}{s+2}+\dfrac{2}{s+3}-\dfrac{3/2}{s+4}$.

$$y_{zi}(t)=\big(7e^{-2t}-5e^{-3t}\big)u(t),$$
$$y_{zs}(t)=\big(-\tfrac{1}{2}e^{-2t}+2e^{-3t}-\tfrac{3}{2}e^{-4t}\big)u(t),$$
$$y(t)=\big(\tfrac{13}{2}e^{-2t}-3e^{-3t}-\tfrac{3}{2}e^{-4t}\big)u(t).$$

Check: $y_{zi}(0)=2$, $y'_{zi}(0)=-14+15=1$ (ok).

Answer. ZIR $=7e^{-2t}-5e^{-3t}$; ZSR $=-\frac{1}{2}e^{-2t}+2e^{-3t}-\frac{3}{2}e^{-4t}$ (for $t\ge 0$).

2021–2022 • Q7(c) EEE 211 • EEE p.15 MARKS 11

Solve $y'+2y=e^{-3t}u(t)$ with $y(0^-)=1$ using the Laplace transform.

SOLUTION

$$sY-1+2Y=\frac{1}{s+3}\Rightarrow Y=\frac{1}{s+2}+\frac{1}{(s+2)(s+3)}=\frac{1}{s+2}+\frac{1}{s+2}-\frac{1}{s+3}=\frac{2}{s+2}-\frac{1}{s+3}.$$

Answer. $y(t)=\big(2e^{-2t}-e^{-3t}\big)u(t)$; $y(0)=1$ (ok).

2018–2019 • Q8(a) EEE 211 • EEE p.31 MARKS 15

For $y''+3y'+Ky=x''$, find the values of K for which the system is BIBO stable. For $K=2$, find the zero-input response with $y(0^+)=0$, $y'(0^+)=1$.

SOLUTION

$H(s)=\dfrac{s^2}{s^2+3s+K}$. The poles are the roots of s^2+3s+K. For a second-order polynomial with positive leading coefficient, both roots lie in the left half-plane iff all coefficients are positive:

$$3>0,\qquad K>0\ \Rightarrow\ K>0.$$

(For $0<K\le\frac{9}{4}$ the poles are real and negative, for $K>\frac{9}{4}$ complex with real part $-\frac{3}{2}$.) At $K=2$ the characteristic equation $s^2+3s+2=(s+1)(s+2)$, so

$$y_{zi}=Ae^{-t}+Be^{-2t},\quad y(0)=A+B=0,\ y'(0)=-A-2B=1\Rightarrow B=-1,\ A=1.$$

Answer. BIBO stable iff $K>0$. For $K=2$: $y_{zi}(t)=e^{-t}-e^{-2t}$, $t\ge 0$.

2023–2024 • Q5(b) EEE 211 • EEE p.4 MARKS 15

A plant $G(s)=\dfrac{1}{s+2}$ is controlled by $C(s)=\dfrac{s+1}{s}$ in a unity-feedback loop with input $x(t)=Au(t)$. Does the output settle to the input value? What happens if the plant is $\dfrac{1}{s-2}$?

SOLUTION

Open loop $L = CG = \dfrac{s+1}{s(s+2)}$; closed loop

$$T(s) = \frac{L}{1+L} = \frac{s+1}{s^2+2s+s+1} = \frac{s+1}{s^2+3s+1}.$$

Poles: $s = \frac{-3\pm\sqrt{5}}{2} = -0.38, -2.62$ — both in the left half-plane, so T is stable and the final value theorem applies: $y(\infty) = \lim_{s\to 0} s \cdot T(s)\frac{A}{s} = T(0)A = A$. Yes, the integrator in the controller gives zero steady-state error.
For $G = \dfrac{1}{s-2}$: $T = \dfrac{s+1}{s(s-2)+s+1} = \dfrac{s+1}{s^2-s+1}$, poles $\frac{1}{2} \pm j\frac{\sqrt{3}}{2}$ in the right half-plane: the output oscillates with growing amplitude, so there is no final value (the final value theorem cannot be used for an unstable system).

Answer. Stable loop: $y(\infty) = A$ (since $T(0) = 1$). With $\frac{1}{s-2}$ the closed-loop poles are $\frac{1}{2} \pm j0.866$: unstable, no steady state.

2020–2021 • Q3(a) EEE 211 • EEE p.18 MARKS 15

Two systems $H_1(s) = \dfrac{1}{s^2+bs+(b-3)}$ and $H_2(s) = -2$ are connected in a feedback loop (H_1 forward, H_2 in the feedback path). Find the overall $H(s)$ and the values of b for which it is stable.

NOTE The figure is read as standard negative feedback, $H = \dfrac{H_1}{1+H_1H_2}$. If the diagram uses a positive summing junction, replace $1 + H_1H_2$ with $1 - H_1H_2$.

SOLUTION

$$H = \frac{H_1}{1+H_1H_2} = \frac{1}{(s^2+bs+b-3)-2} = \frac{1}{s^2+bs+(b-5)}.$$

Stable iff both coefficients are positive: $b > 0$ and $b - 5 > 0$, i.e. $b > 5$.

Answer. $H(s) = \dfrac{1}{s^2+bs+b-5}$; stable for $b > 5$.

2019–2020 • Q3 EEE 211 • EEE p.23 MARKS 35

A multipath channel is $y(t) = x(t) + a\,x(t-T)$. Find $H(s)$. Show that the inverse system can be built as a feedback system, find its impulse response, and state when it is causal and stable.

SOLUTION

Channel. $H(s) = 1 + ae^{-sT}$, $h(t) = \delta(t) + a\delta(t-T)$.

Inverse system. $H_{inv}(s) = \dfrac{1}{1+ae^{-sT}}$. This is a feedback loop with forward path 1 and feedback path ae^{-sT} (a delay T with gain a) and *negative* feedback: $w(t) = x(t) - a\,w(t-T)$.

Impulse response. Expand for $|ae^{-sT}| < 1$: $\dfrac{1}{1+ae^{-sT}} = \sum_{k=0}^{\infty}(-a)^k e^{-ksT}$, so

$$h_{inv}(t) = \sum_{k=0}^{\infty}(-a)^k\delta(t-kT).$$

Causality and stability. All impulses are at $t \geq 0$: causal (ROC a right half-plane, $\mathrm{Re}\, s > \frac{\ln|a|}{T}$). Absolute integrability: $\sum_k |a|^k < \infty \iff |a| < 1$. The ROC then includes the $j\omega$-axis.

Answer. $H = 1 + ae^{-sT}$; $h_{inv} = \sum_{k\ge 0}(-a)^k \delta(t - kT)$ (feedback of $-a\,e^{-sT}$); causal, and stable iff $|a| < 1$.

2018–2019 • Q5(b) EEE 211 • EEE p.29–30 MARKS 20

A current source $i_s(t) = 2$ A for $t \le 0$ and $4e^{-2t}$ A for $t \ge 0$ drives a parallel combination of $R = 2\,\Omega$ and $L = 2$ H. Find the inductor current $i_L(t)$ for $t \ge 0$ using the Laplace transform.

SOLUTION

Initial condition. For $t < 0$ the source is constant, so the inductor is a short circuit and carries the whole current: $i_L(0^-) = 2$ A.

Node equation. With $v = L\frac{\mathrm{d}i_L}{\mathrm{d}t}$ across both elements: $i_s = \dfrac{v}{R} + i_L = \dfrac{L}{R}i_L' + i_L = i_L' + i_L$ (since $L/R = 1$). Transform:

$$I_S = sI_L - i_L(0^-) + I_L = (s+1)I_L - 2, \qquad I_S = \frac{4}{s+2}.$$

$$I_L = \frac{I_S + 2}{s+1} = \frac{4}{(s+1)(s+2)} + \frac{2}{s+1} = \frac{6}{s+1} - \frac{4}{s+2}.$$

Answer. $i_L(t) = 6e^{-t} - 4e^{-2t}$ A, $t \ge 0$ (check: $i_L(0) = 2$ A (ok)).

14

The z-Transform

ASHRAF SIR • Z-TRANSFORM

$X(z) = \sum_n x[n]z^{-n}$. Right-sided $\Rightarrow$ ROC outside the outermost pole; left-sided $\Rightarrow$ inside the innermost pole; two-sided $\Rightarrow$ an annulus. Causal and stable $\Leftrightarrow$ all poles inside the unit circle.

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Find the z-transform and ROC of $x[n] = a^n u[n]$.

SOLUTION

$$X(z) = \sum_{n=0}^{\infty} (az^{-1})^n = \frac{1}{1 - az^{-1}} = \frac{z}{z - a}, \qquad |az^{-1}| < 1 \iff |z| > |a|.$$

Pole at $z = a$, zero at $z = 0$; the ROC is the exterior of the circle through the pole. It contains the unit circle iff $|a| < 1$.

Answer. $X(z) = \dfrac{1}{1 - az^{-1}}$, ROC $|z| > |a|$.

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For $x[n] = a^n$, $0 \le n \le N - 1$ (zero otherwise), show that the ROC excludes $z = 0$.

SOLUTION

$$X(z) = \sum_{n=0}^{N-1} a^n z^{-n} = \frac{1 - a^N z^{-N}}{1 - az^{-1}} = 1 + az^{-1} + \cdots + a^{N-1} z^{-(N-1)}.$$

The sum is finite, so it converges for every z where each term is finite. The terms $z^{-1}, \ldots, z^{-(N-1)}$ are infinite at $z = 0$ (poles of order up to $N - 1$ there), hence $z = 0$ must be excluded. The apparent pole at $z = a$ cancels (numerator vanishes there), so

$$\text{ROC: } 0 < |z| \le \infty \ \ (\text{all } z \ne 0;\ \text{for } N = 1 \text{ the point } z = 0 \text{ is included}).$$

Answer. Finite, right-sided with $n \ge 0$ terms: ROC is the whole plane except $z = 0$.

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Derive the contour-integral formula for the inverse z-transform.

SOLUTION

Start from $X(z) = \sum_n x[n]z^{-n}$ in its ROC. Multiply by z^{k-1} and integrate around a counter-clockwise closed contour C inside the ROC encircling the origin:

$$\oint_C X(z)z^{k-1}\,\mathrm{d}z = \sum_n x[n] \oint_C z^{k-n-1}\,\mathrm{d}z.$$

By Cauchy's integral theorem, $\oint_C z^{m-1}\,\mathrm{d}z = 2\pi j$ for $m = 0$ and 0 for $m \neq 0$. Only $n = k$ survives:

$$\oint_C X(z)z^{k-1}\,\mathrm{d}z = 2\pi j\, x[k] \;\Rightarrow\; x[n] = \frac{1}{2\pi j}\oint_C X(z)z^{n-1}\,\mathrm{d}z.$$

In practice this is evaluated by residues at the poles inside C.

Answer. $x[n] = \dfrac{1}{2\pi j}\oint_C X(z)z^{n-1}\,\mathrm{d}z.$

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Find the inverse z-transform of $X(z) = \log\left(1 + az^{-1}\right)$, $|z| > |a|$.

SOLUTION

For $|u| < 1$, $\log(1+u) = \sum_{n=1}^{\infty} \dfrac{(-1)^{n+1}}{n}u^n$. With $u = az^{-1}$, which satisfies $|u| < 1$ exactly when $|z| > |a|$,

$$X(z) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}a^n}{n}z^{-n}.$$

Comparing with $\sum x[n]z^{-n}$:

$$x[n] = \frac{(-1)^{n+1}a^n}{n}\,u[n-1].$$

Answer. $x[n] = \dfrac{(-1)^{n+1}a^n}{n}u[n-1].$

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Find all possible ROCs of $X(z) = \dfrac{1 + 2z^{-1} - z^{-2}}{1 - \frac{11}{6}z^{-1} + z^{-2} - \frac{1}{6}z^{-3}}$.

SOLUTION

Multiply by z^3: denominator $z^3 - \frac{11}{6}z^2 + z - \frac{1}{6} = (z-1)(z-\frac{1}{2})(z-\frac{1}{3})$ (sum of roots $\frac{11}{6}$, product $\frac{1}{6}$). Poles: $1,\ \frac{1}{2},\ \frac{1}{3}$. Numerator: $z(z^2 + 2z - 1)$ gives zeros at 0 and $-1 \pm \sqrt{2}$ (no cancellation).
The ROC is bounded by pole circles and contains no pole, so there are four possibilities:

ROC	sequence type	unit circle?
$\lvert z\rvert > 1$	right-sided (causal)	no (unstable)
$\frac{1}{2} < \lvert z\rvert < 1$	two-sided	no
$\frac{1}{3} < \lvert z\rvert < \frac{1}{2}$	two-sided	no
$\lvert z\rvert < \frac{1}{3}$	left-sided	no

(There is a pole on $|z| = 1$, so no ROC contains the unit circle: the DTFT does not exist as an ordinary function.)

Answer. Four ROCs: $|z| > 1$, $\frac{1}{2} < |z| < 1$, $\frac{1}{3} < |z| < \frac{1}{2}$, $|z| < \frac{1}{3}$.

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A system has poles at $z = \frac{1}{3}, \frac{2}{3}, 2$. State the ROC when the sequence is (i) right-sided, (ii) its Fourier transform converges, (iii) its Fourier transform diverges, (iv) left-sided.

SOLUTION

Pole magnitudes: $\frac{1}{3} < \frac{2}{3} < 2$.

(i) Right-sided: outside the outermost pole, $|z| > 2$.

(ii) Fourier transform converges $\Leftrightarrow$ the ROC contains $|z| = 1$: $\frac{2}{3} < |z| < 2$ (two-sided, not causal).

(iii) Diverges: any ROC without the unit circle: $|z| > 2$, $\frac{1}{3} < |z| < \frac{2}{3}$, or $|z| < \frac{1}{3}$.

(iv) Left-sided: inside the innermost pole, $|z| < \frac{1}{3}$.

Answer. (i) $|z| > 2$; (ii) $\frac{2}{3} < |z| < 2$; (iii) $|z| > 2$, $\frac{1}{3} < |z| < \frac{2}{3}$ or $|z| < \frac{1}{3}$; (iv) $|z| < \frac{1}{3}$.

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Find the z-transform of $x[n] = |n|\left(\frac{1}{2}\right)^{|n|}$ and draw its pole-zero plot.

SOLUTION

Split: $x[n] = n(\frac{1}{2})^n u[n] + (-n)(\frac{1}{2})^{-n} u[-n-1]$. Using $na^n u[n] \leftrightarrow \dfrac{az^{-1}}{(1-az^{-1})^2}$, $|z| > |a|$:

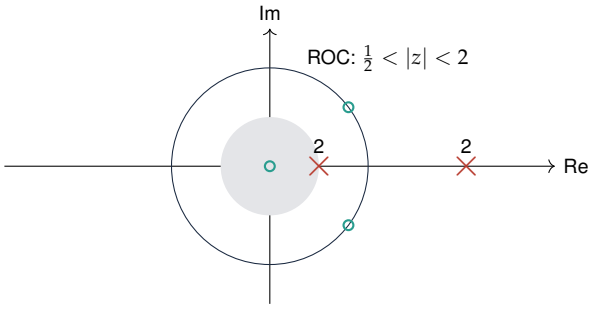
$$X_1(z) = \frac{\frac{1}{2}z^{-1}}{(1-\frac{1}{2}z^{-1})^2} = \frac{\frac{1}{2}z}{(z-\frac{1}{2})^2}, \quad |z| > \tfrac{1}{2}.$$

The left part is X_1 with $z \to z^{-1}$ (time reversal; the $n = 0$ term is zero):

$$X_2(z) = X_1(1/z) = \frac{\frac{1}{2}z}{(1-\frac{1}{2}z)^2} = \frac{2z}{(z-2)^2}, \quad |z| < 2.$$

$$X(z) = \frac{\frac{1}{2}z}{(z-\frac{1}{2})^2} + \frac{2z}{(z-2)^2}, \qquad \tfrac{1}{2} < |z| < 2.$$

Numerator: $z\left[\frac{1}{2}(z-2)^2 + 2(z-\frac{1}{2})^2\right] = \frac{5}{2}z\left(z^2 - \frac{8}{5}z + 1\right)$. Zeros: $z = 0$ and $z = 0.8 \pm 0.6j$ (on the unit circle). Poles: double at $\frac{1}{2}$ and double at 2.



Answer. $X(z) = \dfrac{\frac{1}{2}z}{(z-\frac{1}{2})^2} + \dfrac{2z}{(z-2)^2}$, ROC $\frac{1}{2} < |z| < 2$; zeros $0,\ 0.8 \pm 0.6j$; double poles at $\frac{1}{2}$ and 2.

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Compare the Laplace and z-transforms. Why is the z-transform used for discrete-time signals?

SOLUTION

Both are generalisations of the Fourier transform with a convergence factor; they are related by $z = e^{sT}$.

	Laplace	z-transform
signal	continuous $x(t)$	discrete $x[n]$
variable	$s = \sigma + j\omega$	$z = re^{j\omega}$
Fourier axis	$j\omega$-axis	unit circle $\lvert z\rvert = 1$
stability	poles in left half-plane	poles inside unit circle
derivative / delay	s / e^{-sT}	delay z^{-1} (algebraic)

For sequences, $\sum x[n]e^{-sTn}$ is a function of e^{sT} only; z turns it into an ordinary power series (rational functions for difference equations), where a unit delay is simply multiplication by z^{-1}.

Answer. $z = e^{sT}$ maps the s-plane's left half-plane onto the interior of the unit circle; z gives rational transforms for difference equations.